

MatrixMultiplicationBias

January 12, 2026

1 Triton Kernel Profiling on AMD GPUs

1.1 Get the System Hardware Info

```
[1]: !rocminfo
```

```
ROcK module version 6.16.6 is loaded
```

```
=====
```

```
HSA System Attributes
```

```
=====
```

```
Runtime Version:          1.18
Runtime Ext Version:      1.11
System Timestamp Freq.:   1000.000000MHz
Sig. Max Wait Duration:   18446744073709551615 (0xFFFFFFFFFFFFFFFF) (timestamp
count)
Machine Model:            LARGE
System Endianness:        LITTLE
Mwaitx:                   DISABLED
XNACK enabled:            NO
DMAbuf Support:           YES
VMM Support:              YES
```

```
=====
```

```
HSA Agents
```

```
=====
```

```
*****
```

```
Agent 1
```

```
*****
```

```
  Name:                    Intel(R) Xeon(R) Platinum 8470
  Uuid:                    CPU-XX
  Marketing Name:          Intel(R) Xeon(R) Platinum 8470
  Vendor Name:             CPU
  Feature:                 None specified
  Profile:                 FULL_PROFILE
  Float Round Mode:        NEAR
  Max Queue Number:        0(0x0)
  Queue Min Size:          0(0x0)
  Queue Max Size:          0(0x0)
```

```

Queue Type:          MULTI
Node:                0
Device Type:         CPU
Cache Info:
  L1:                 32768(0x8000) KB
Chip ID:             0(0x0)
ASIC Revision:       0(0x0)
Cacheline Size:      64(0x40)
Max Clock Freq. (MHz): 0
BDFID:               0
Internal Node ID:    0
Compute Unit:        13
SIMDs per CU:        0
Shader Engines:      0
Shader Arrs. per Eng.: 0
WatchPts on Addr. Ranges:1
Memory Properties:
Features:             None
Pool Info:
  Pool 1
    Segment:          GLOBAL; FLAGS: FINE GRAINED
    Size:              230893524(0xdc327d4) KB
    Allocatable:       TRUE
    Alloc Granule:     4KB
    Alloc Recommended Granule:4KB
    Alloc Alignment:   4KB
    Accessible by all: TRUE
  Pool 2
    Segment:          GLOBAL; FLAGS: EXTENDED FINE GRAINED
    Size:              230893524(0xdc327d4) KB
    Allocatable:       TRUE
    Alloc Granule:     4KB
    Alloc Recommended Granule:4KB
    Alloc Alignment:   4KB
    Accessible by all: TRUE
  Pool 3
    Segment:          GLOBAL; FLAGS: KERNARG, FINE GRAINED
    Size:              230893524(0xdc327d4) KB
    Allocatable:       TRUE
    Alloc Granule:     4KB
    Alloc Recommended Granule:4KB
    Alloc Alignment:   4KB
    Accessible by all: TRUE
  Pool 4
    Segment:          GLOBAL; FLAGS: COARSE GRAINED
    Size:              230893524(0xdc327d4) KB
    Allocatable:       TRUE
    Alloc Granule:     4KB

```

```

        Alloc Recommended Granule:4KB
        Alloc Alignment:          4KB
        Accessible by all:        TRUE
    ISA Info:
*****
Agent 2
*****
    Name:                gfx942
    Uuid:                 GPU-1c03b713ca52c40c
    Marketing Name:       AMD Instinct MI300X VF
    Vendor Name:          AMD
    Feature:               KERNEL_DISPATCH
    Profile:               BASE_PROFILE
    Float Round Mode:      NEAR
    Max Queue Number:      128(0x80)
    Queue Min Size:        64(0x40)
    Queue Max Size:        131072(0x20000)
    Queue Type:            MULTI
    Node:                  1
    Device Type:           GPU
    Cache Info:
        L1:                32(0x20) KB
        L2:                4096(0x1000) KB
        L3:                262144(0x40000) KB
    Chip ID:               29877(0x74b5)
    ASIC Revision:         1(0x1)
    Cacheline Size:        128(0x80)
    Max Clock Freq. (MHz): 2100
    BDFID:                 65280
    Internal Node ID:       1
    Compute Unit:           304
    SIMDs per CU:           4
    Shader Engines:         32
    Shader Arrs. per Eng.:  1
    WatchPts on Addr. Ranges:4
    Coherent Host Access:   FALSE
    Memory Properties:
    Features:               KERNEL_DISPATCH
    Fast F16 Operation:     TRUE
    Wavefront Size:         64(0x40)
    Workgroup Max Size:     1024(0x400)
    Workgroup Max Size per Dimension:
        x                   1024(0x400)
        y                   1024(0x400)
        z                   1024(0x400)
    Max Waves Per CU:       32(0x20)
    Max Work-item Per CU:   2048(0x800)
    Grid Max Size:          4294967295(0xffffffff)

```

Grid Max Size per Dimension:

x	2147483647(0x7fffffff)
y	65535(0xffff)
z	65535(0xffff)

Max fbarriers/Workgrp: 32

Packet Processor uCode:: 191

SDMA engine uCode:: 25

IOMMU Support:: None

Pool Info:

Pool 1

Segment:	GLOBAL; FLAGS: COARSE GRAINED
Size:	200998912(0xbfb0000) KB
Allocatable:	TRUE
Alloc Granule:	4KB
Alloc Recommended Granule:	2048KB
Alloc Alignment:	4KB
Accessible by all:	FALSE

Pool 2

Segment:	GLOBAL; FLAGS: EXTENDED FINE GRAINED
Size:	200998912(0xbfb0000) KB
Allocatable:	TRUE
Alloc Granule:	4KB
Alloc Recommended Granule:	2048KB
Alloc Alignment:	4KB
Accessible by all:	FALSE

Pool 3

Segment:	GLOBAL; FLAGS: FINE GRAINED
Size:	200998912(0xbfb0000) KB
Allocatable:	TRUE
Alloc Granule:	4KB
Alloc Recommended Granule:	2048KB
Alloc Alignment:	4KB
Accessible by all:	FALSE

Pool 4

Segment:	GROUP
Size:	64(0x40) KB
Allocatable:	FALSE
Alloc Granule:	0KB
Alloc Recommended Granule:	0KB
Alloc Alignment:	0KB
Accessible by all:	FALSE

ISA Info:

ISA 1

Name:	amdgcncn-amd-amdhscn--gfx942:sramecc+:xnack-
Machine Models:	HSA_MACHINE_MODEL_LARGE
Profiles:	HSA_PROFILE_BASE
Default Rounding Mode:	NEAR
Default Rounding Mode:	NEAR

```

Fast f16:                TRUE
Workgroup Max Size:      1024(0x400)
Workgroup Max Size per Dimension:
  x                      1024(0x400)
  y                      1024(0x400)
  z                      1024(0x400)
Grid Max Size:           4294967295(0xffffffff)
Grid Max Size per Dimension:
  x                      2147483647(0x7fffffff)
  y                      65535(0xffff)
  z                      65535(0xffff)
FBarrier Max Size:       32
ISA 2
Name:                    amdgcN-amd-amdhsa--gfx9-4-generic:sramecc+:xnack-
Machine Models:          HSA_MACHINE_MODEL_LARGE
Profiles:                 HSA_PROFILE_BASE
Default Rounding Mode:    NEAR
Default Rounding Mode:    NEAR
Fast f16:                TRUE
Workgroup Max Size:      1024(0x400)
Workgroup Max Size per Dimension:
  x                      1024(0x400)
  y                      1024(0x400)
  z                      1024(0x400)
Grid Max Size:           4294967295(0xffffffff)
Grid Max Size per Dimension:
  x                      2147483647(0x7fffffff)
  y                      65535(0xffff)
  z                      65535(0xffff)
FBarrier Max Size:       32
*** Done ***

```

1.2 Check profiling environment

```
[2]: !rocprof-compute --version
```

```

-----
rocprofiler-compute version: 3.2.3 (release)
Git revision:      1d722c14
-----

```

1.3 Non-Fusion MatMulBias Kernel

```
[3]: %%writefile nonfusion_matmulbias.py
import argparse
import torch

import triton

```

```

import triton.language as tl
import triton.profiler as proton

def _matmul_launch_metadata(grid, kernel, args):
    ret = {}
    M, N, K, WS = args["M"], args["N"], args["K"], args.get("WARP_SPECIALIZE",
↪False)
    BM, BN, BK = args["BLOCK_SIZE_M"], args["BLOCK_SIZE_N"],
↪args["BLOCK_SIZE_K"]
    ws_str = "_ws" if WS else ""
    ret["name"] = f"{kernel.name}{ws_str} [M={M}, N={N}, K={K}] [BM={BM},
↪BN={BN} BK={BK}]"
    if "output_ptr" in args:
        bytes_per_elem = args["output_ptr"].element_size()
    else:
        bytes_per_elem = 2
    ret[f"flops{bytes_per_elem * 8}"] = 2. * M * N * K
    ret["bytes"] = bytes_per_elem * (M * K + N * K + M * N)
    return ret

def _bias_launch_metadata(grid, kernel, args):
    ret = {}
    M, N, WS = args["M"], args["N"], args.get("WARP_SPECIALIZE", False)
    BM, BN = args["BLOCK_SIZE_M"], args["BLOCK_SIZE_N"]
    ws_str = "_ws" if WS else ""
    ret["name"] = f"{kernel.name}{ws_str} [M={M}, N={N}] [BM={BM}, BN={BN}]"
    if "output_ptr" in args:
        bytes_per_elem = args["output_ptr"].element_size()
    else:
        bytes_per_elem = 2
    ret[f"flops{bytes_per_elem * 8}"] = 2. * M * N
    ret["bytes"] = bytes_per_elem * (M + N + M * N)
    return ret

def matmul_autotune_config(pre_hook=None):
    return [
        triton.Config({'BLOCK_SIZE_M': BM, 'BLOCK_SIZE_N': BN, "BLOCK_SIZE_K":
↪BK, "GROUP_SIZE_M": 8}, num_stages=s,
        num_warps=w, pre_hook=pre_hook)
        for BM in [128]
        for BN in [128, 256]
        for BK in [64, 128]
        for s in ([3, 4, 5])
        for w in [4, 8]
    ]

```

```

def bias_autotune_config(pre_hook=None):
    return [
        triton.Config({'BLOCK_SIZE_M': BM, 'BLOCK_SIZE_N': BN, "GROUP_SIZE_M": 8}, num_stages=s,
        ↪}, num_warps=w, pre_hook=pre_hook)
        for BM in [128]
        for BN in [128, 256]
        for s in ([3, 4, 5])
        for w in [4, 8]
    ]

# MatMul kernel
@triton.autotune(
    configs=matmul_autotune_config(),
    key=['M', 'N', 'K'],
)
@triton.jit(launch_metadata=_matmul_launch_metadata)
def matmul_kernel(
    a_ptr, b_ptr, c_ptr,
    M, N, K,
    stride_am, stride_ak,
    stride_bk, stride_bn,
    stride_cm, stride_cn,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, BLOCK_SIZE_K: 8,
    ↪tl.constexpr,
    GROUP_SIZE_M: tl.constexpr
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)
    pid_n = (pid % num_pid_in_group) // group_size_m

    tl.assume(pid_m >= 0)
    tl.assume(pid_n >= 0)
    tl.assume(stride_am > 0)
    tl.assume(stride_ak > 0)
    tl.assume(stride_bn > 0)
    tl.assume(stride_bk > 0)
    tl.assume(stride_cm > 0)
    tl.assume(stride_cn > 0)

    offs_am = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M

```

```

    offs_bn = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
    offs_k = tl.arange(0, BLOCK_SIZE_K)
    a_ptrs = a_ptr + (offs_am[:, None] * stride_am + offs_k[None, :] *
↪stride_ak)
    b_ptrs = b_ptr + (offs_k[:, None] * stride_bk + offs_bn[None, :] *
↪stride_bn)

    accumulator = tl.zeros((BLOCK_SIZE_M, BLOCK_SIZE_N), dtype=tl.float32)
    for k in range(0, tl.cdiv(K, BLOCK_SIZE_K)):
        a = tl.load(a_ptrs, mask=offs_k[None, :] < K - k * BLOCK_SIZE_K,
↪other=0.0)
        b = tl.load(b_ptrs, mask=offs_k[:, None] < K - k * BLOCK_SIZE_K,
↪other=0.0)
        accumulator = tl.dot(a, b, accumulator)
        a_ptrs += BLOCK_SIZE_K * stride_ak
        b_ptrs += BLOCK_SIZE_K * stride_bk
    c = accumulator.to(tl.float16)

    offs_cm = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
    offs_cn = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
    c_ptrs = c_ptr + stride_cm * offs_cm[:, None] + stride_cn * offs_cn[None, :]
    c_mask = (offs_cm[:, None] < M) & (offs_cn[None, :] < N)
    tl.store(c_ptrs, c, mask=c_mask)

# Bias kernel
@triton.autotune(
    configs=bias_autotune_config(),
    key=['M', 'K'],
)
@triton.jit(launch_metadata=_bias_launch_metadata)
def bias_kernel(
    a_ptr,
    bias_ptr,
    output_ptr,
    M, N,
    stride_m, stride_n,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, GROUP_SIZE_M:
↪tl.constexpr,
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)

```



```

pid_n = (pid % num_pid_in_group) // group_size_m

offs_m = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M
offs_n = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
a_ptrs = a_ptr + (offs_m[:, None] * stride_m + offs_n[None, :] * stride_n)

a = tl.load(a_ptrs)

bias = tl.load(bias_ptr + offs_n, mask=offs_n < N, other=0.0)

output = a + bias[None, :]

offs_m = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
offs_n = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
output_ptrs = output_ptr + stride_m * offs_m[:, None] + stride_n * ␣
↪offs_n[None, :]
output_mask = (offs_m[:, None] < M) & (offs_n[None, :] < N)
tl.store(output_ptrs, output, mask=output_mask)

# MatMulBias kernel wrapper function
def matmulbias(a: torch.Tensor, b: torch.Tensor, bias: torch.Tensor):
    assert a.shape[1] == b.shape[0], "Incompatible dimensions"
    assert a.is_contiguous(), "Matrix A must be contiguous"
    M, K = a.shape
    K, N = b.shape
    assert bias.shape[0] == N, "BIAS has incompatible dimensions"
    c = torch.empty((M, N), device=a.device, dtype=torch.float16)
    matmul_grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.
↪cdiv(N, META['BLOCK_SIZE_N']), )
    matmul_kernel[matmul_grid](
        a, b, c,
        M, N, K,
        a.stride(0), a.stride(1),
        b.stride(0), b.stride(1),
        c.stride(0), c.stride(1),
    )

    o = torch.empty((M, N), device=c.device, dtype=torch.float16)
    bias_grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.
↪cdiv(N, META['BLOCK_SIZE_N']), )
    bias_kernel[bias_grid](
        c, bias, o,
        M, N,
        c.stride(0), c.stride(1),
    )

```

```

return o

def get_rtol():
    target = triton.runtime.driver.active.get_current_target()
    if target.backend == "hip":
        if target.arch == "gfx90a":
            return 1e-2
        elif target.arch == "gfx942":
            return 1e-2
    else:
        return 0

def show_profile(profile_name):
    import triton.profiler.viewer as proton_viewer
    metric_names = ["time/ms", "tflop16/s"]
    file_name = f"{profile_name}.hatchet"
    tree, metrics = proton_viewer.parse(metric_names, file_name)

    print(f"Proton profile results for {profile_name}")
    proton_viewer.print_tree(tree, metrics)

if __name__ == "__main__":
    parser = argparse.ArgumentParser()
    parser.add_argument("--profile", action="store_true")
    parser.add_argument("--verify", action="store_true")
    args = parser.parse_args()

    # Test matrices
    torch.manual_seed(0)
    M = 1024
    N = 1024
    a = torch.randn((M, N), device='cuda', dtype=torch.float16)
    b = torch.randn((N, M), device='cuda', dtype=torch.float16)
    bias = torch.randn(N, device='cuda', dtype=torch.float16)
    # bias = torch.zeros(N, device='cuda', dtype=torch.float16)

    if args.profile:
        print("Profiling the nonfusion matmulbias kernel ...")
        proton.start("nonfusion_matmulbias", hook="triton")
    else:
        print("Running the nonfusion matmulbias kernel ...")
        nonfusion_triton_output = matmulbias(a, b, bias)
    if args.verify:
        proton.finalize()
    print(f"nonfusion_triton_output: {nonfusion_triton_output}")

```

```

if args.profile:
    show_profile("nonfusion_matmulbias")

if args.verify:
    torch_output = torch.addmm(bias, a, b)
    print(f"torch_output: {torch_output}")
    print("Verifying triton results with torch ...")
    triton.testing.assert_close(nonfusion_triton_output, torch_output,
    ↪atol=1e-2, rtol=get_rtol())
    print("OK")

```

Overwriting nonfusion_matmulbias.py

1.3.1 Pre-Build the Non-Fusion MatMulBias Kernel

```
[4]: !python nonfusion_matmulbias.py --verify
```

```

Running the nonfusion matmulbias kernel ...
nonfusion_triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,
8.8203,  6.4609],
      [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
      [13.8125, -6.4609,  0.5146, ..., 48.0000, 25.1562, -27.5156],
      ...,
      [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
      [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
      [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
      device='cuda:0', dtype=torch.float16)
torch_output: tensor([[ 2.7480, 43.6562, 30.9531, ..., 26.2500,  8.8203,
6.4609],
      [ 7.1602, -21.3906, 17.7969, ..., -16.8125, 29.6406, -48.3125],
      [13.8125, -6.4609,  0.5142, ..., 48.0312, 25.1406, -27.5156],
      ...,
      [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
      [ 9.6094, -52.2500, 43.0000, ..., -45.8125, -49.3125, 32.7188],
      [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1562]],
      device='cuda:0', dtype=torch.float16)
Verifying triton results with torch ...
OK

```

1.3.2 Profile the Non-Fusion MatMulBias Kernel

Using Triton Proton to generate the report

```
[5]: !python nonfusion_matmulbias.py --profile
```

```

Profiling the nonfusion matmulbias kernel ...
nonfusion_triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,
8.8203,  6.4609],
      [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],

```



```

https://github.com/ROCm/amdsmi
INFO Rocprofiler-Compute version: 3.2.3
INFO Profiler choice: rocprofiler-sdk
INFO Path: /workspace/user/workloads/nonfusion_matmulbias/MI300X_A1
INFO Target: MI300X_A1
INFO Command: python nonfusion_matmulbias.py
INFO Kernel Selection: None
INFO Dispatch Selection: None
INFO Hardware Blocks: []
INFO Report Sections: []
INFO
INFO ~~~~~
INFO Collecting Performance Counters
INFO ~~~~~
INFO
INFO [Run 1/12] [Approximate profiling time left: pending first
measurement...]
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:01:25.227616
139800787537472 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.179592 sec
INFO      |-> [rocprofiler-sdk] W20260112 19:01:25.227730
139800787537472 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec
INFO      |-> [rocprofiler-sdk] W20260112 19:01:26.510456
139800787537472 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)
INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:01:30.326758
139800787537472 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      5.099028 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:01:30.340155
139800787537472 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/182_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:01:30.360703
139800787537472 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/182_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:01:32.748100
139800787537472 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/182_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:01:32.751051
139800787537472 simple_timer.cpp:55] [rocprofv3] output generation ::
2.418662 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:01:32.752147
139800787537472 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.419827 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12] [Approximate profiling time left: 2 minutes and 40
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:01:41.252780
140404982701632 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.165516 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:01:41.252890
140404982701632 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:01:42.528637
140404982701632 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...

```

```

INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260112 19:01:45.371925

140404982701632 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.119036 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:01:45.385094

140404982701632 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/208_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:01:45.405906

140404982701632 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/208_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:01:47.190738

140404982701632 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/208_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:01:47.193599

140404982701632 simple_timer.cpp:55] [rocprofv3] output generation ::
1.816162 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:01:47.194687

140404982701632 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.817316 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12] [Approximate profiling time left: 2 minutes and 7
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:01:53.609245
140020396166720 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178113 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:01:53.609353
140020396166720 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:01:54.884943
140020396166720 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260112 19:01:57.858233
140020396166720 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.248881 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:01:57.871715
140020396166720 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/229_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:01:57.892158
140020396166720 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/229_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:00.212006
140020396166720 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/229_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:02:00.214890
140020396166720 simple_timer.cpp:55] [rocprofv3] output generation ::
2.350901 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:00.215981
140020396166720 simple_timer.cpp:55] [rocprofv3] tool finalization ::
2.352063 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 4/12] [Approximate profiling time left: 1 minute and 53
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:02:07.858964
140620350250560 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.166368 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:07.859069
140620350250560 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:09.136378
140620350250560 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...

```

```

INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260112 19:02:12.009616

140620350250560 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.150547 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:02:12.022951
140620350250560 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/250_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:12.044645
140620350250560 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/250_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:13.967108
140620350250560 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/250_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:02:13.970010
140620350250560 simple_timer.cpp:55] [rocprofv3] output generation ::
1.954774 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:13.971100
140620350250560 simple_timer.cpp:55] [rocprofv3] tool finalization ::
1.955931 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12] [Approximate profiling time left: 1 minute and 36
seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/nonfu
sion_matmulbias/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:02:20.249028
140055684128320 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128131 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:20.249129
140055684128320 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:21.520773
140055684128320 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:02:24.294457
140055684128320 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.045328 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:02:24.307751
140055684128320 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/272_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:24.330320
140055684128320 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/272_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:24.463229
140055684128320 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/272_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:02:24.466065
140055684128320 simple_timer.cpp:55] [rocprofv3] output generation ::
0.166015 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:24.467074
140055684128320 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.167116 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 6/12] [Approximate profiling time left: 1 minute and 13
seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_0.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:02:26.132054
139991775702592 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128112 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:26.132157
139991775702592 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:27.403289
139991775702592 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...

```

```

INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260112 19:02:30.174532

139991775702592 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.042374 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:02:30.187861

139991775702592 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/292_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:30.210406

139991775702592 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/292_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:30.329546

139991775702592 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/292_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:02:30.332384

139991775702592 simple_timer.cpp:55] [rocprofv3] output generation ::
0.152241 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:30.333361

139991775702592 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.153290 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12][Approximate profiling time left: 55 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_1.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:02:31.958908
140122325388864 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.129005 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:31.959013
140122325388864 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:33.229526
140122325388864 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:02:36.002614
140122325388864 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.043601 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:02:36.015951
140122325388864 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/313_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:36.038602
140122325388864 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/313_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:36.160746
140122325388864 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/313_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:02:36.163665
140122325388864 simple_timer.cpp:55] [rocprofv3] output generation ::
0.155435 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:36.164650
140122325388864 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.156492 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 8/12] [Approximate profiling time left: 41 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_2.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:02:37.826589
140227165208128 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.130180 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:37.826703
140227165208128 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:39.104501
140227165208128 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[

```

```

2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260112 19:02:41.872511

140227165208128 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.045809 sec

INFO    |-> [rocprofiler-sdk] E20260112 19:02:41.885901

140227165208128 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/333_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260112 19:02:41.908577

140227165208128 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/333_counter_collectio
n.csv

INFO    |-> [rocprofiler-sdk] E20260112 19:02:42.030311

140227165208128 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/333_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260112 19:02:42.033141

140227165208128 simple_timer.cpp:55] [rocprofv3] output generation ::
0.155013 sec

INFO    |-> [rocprofiler-sdk] W20260112 19:02:42.034156

140227165208128 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.156102 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 9/12][Approximate profiling time left: 29 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_3.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260112 19:02:43.688269
140223665352256 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128638 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:43.688376
140223665352256 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:44.963292
140223665352256 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:02:47.765577
140223665352256 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.077201 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:02:47.778933
140223665352256 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/354_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:47.805316
140223665352256 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/354_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:02:47.948864
140223665352256 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/354_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:02:47.951735
140223665352256 simple_timer.cpp:55] [rocprofv3] output generation ::
0.180530 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:47.952725
140223665352256 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.181590 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 10/12] [Approximate profiling time left: 18 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_4.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:02:49.694696
140092912528960 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127635 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:49.694797
140092912528960 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:50.962549
140092912528960 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[

```

```

2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260112 19:02:53.819254

140092912528960 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.124457 sec

INFO    |-> [rocprofiler-sdk] E20260112 19:02:53.832797

140092912528960 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/375_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260112 19:02:53.868723

140092912528960 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/375_counter_collectio
n.csv

INFO    |-> [rocprofiler-sdk] E20260112 19:02:54.074719

140092912528960 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/375_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260112 19:02:54.077745

140092912528960 simple_timer.cpp:55] [rocprofv3] output generation ::
0.252701 sec

INFO    |-> [rocprofiler-sdk] W20260112 19:02:54.078694

140092912528960 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.253722 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 11/12] [Approximate profiling time left: 9 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_5.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:02:56.063743
140299422707264 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128927 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:56.063857
140299422707264 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:02:57.375897
140299422707264 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:03:00.238343
140299422707264 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.174487 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:03:00.251904
140299422707264 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/395_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:03:00.287684
140299422707264 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/395_counter_collectio
n.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:03:00.491612
140299422707264 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/395_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:03:00.494650
140299422707264 simple_timer.cpp:55] [rocprofv3] output generation ::
0.250513 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:03:00.495615
140299422707264 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.251550 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 12/12] [Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1/perfmon/pmc_perf_6.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:03:02.478505
139914782920256 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127872 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:03:02.478607
139914782920256 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:03:03.745738
139914782920256 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the nonfusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] nonfusion_triton_output: tensor([[

```

```

2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260112 19:03:06.601146

139914782920256 simple_timer.cpp:55] [rocprofv3] 'python
nonfusion_matmulbias.py' ::      4.122539 sec

INFO    |-> [rocprofiler-sdk] E20260112 19:03:06.614885

139914782920256 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/416_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260112 19:03:06.660788

139914782920256 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/416_counter_collectio
n.csv

INFO    |-> [rocprofiler-sdk] E20260112 19:03:06.802429

139914782920256 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/nonfusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/416_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260112 19:03:06.805490

139914782920256 simple_timer.cpp:55] [rocprofv3] output generation ::
0.198298 sec

INFO    |-> [rocprofiler-sdk] W20260112 19:03:06.806483

139914782920256 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.199381 sec

INFO    |-> [rocprofiler-sdk]
WARNING Detected differing Workgroup_Size values while joining
pmc_perf.csv
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/nonfusion_matmulbias/MI300X_A1
INFO [roofline] No roofline data found. Generating...

```

Empirical Roofline Calculation

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Total detected GPU devices: 1

GPU Device 0 (gfx942) with 304 CUs: Profiling...

99% [|||||]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.1 ms, mean:4205.2 GB/sec, stdev=8.5 GB/sec

99% [|||||]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:2611340115968 bytes, duration:413.4 ms, mean:6286.8 GB/sec, stdev=18.1
GB/sec

99% [|||||]
L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:1632087572480 bytes, duration:63.5 ms, mean:25719.6 GB/sec, stdev=55.0
GB/sec

99% [|||||]
L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:127506841600 bytes, duration:3.8 ms, mean:33631.8 GB/sec, stdev=207.3
GB/sec

99% [|||||]
LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:79691776000 bytes, duration:1.3 ms, mean:61974.8 GB/sec, stdev=682.1
GB/sec

99% [|||||]
Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:82976.7 GFLOPS,
stdev=243.5 GFLOPS

99% [|||||]
Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39568.0 GFLOPS,
stdev=78.5 GFLOPS

99% [|||||]
Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:91.524 ms, mean:5791.3 GFLOPS,
stdev=9.8 GFLOPS

99% [|||||]
Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:265214230528, duration:2.019 ms, mean:131535.0 GFLOPS,
stdev=1334.6 GFLOPS

99% [|||||]
Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:122406567936, duration:1.2 ms, mean:105744.2 GFLOPS,
stdev=1638.6 GFLOPS

99% [|||||]
Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82829.7 GOPS,
stdev=139.6 GOPS

99% [|||||]

```

Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:265214230528, duration:3.7 ms, mean:72135.1 GOPS,
stdev=731.0 GOPS
 99% [||||| ]
Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:122406567936, duration:6.2 ms, mean:19820.0 GOPS,
stdev=61.5 GOPS
100% [||||| ]
Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [||||| ]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:10200547328000, duration:5.3 ms, mean:1920435.0 GFLOPS,
stdev=14616.6 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:5100273664000, duration:5.4 ms, mean:941115.1 GFLOPS,
stdev=3468.6 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:2550136832000, duration:4.9 ms, mean:525202.6 GFLOPS,
stdev=3077.2 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1275068416000, duration:9.1 ms, mean:140691.2 GFLOPS,
stdev=537.4 GFLOPS
 99% [||||| ]
Peak MFMA FLOPs (F64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:637534208000, duration:4.4 ms, mean:146202.9 GFLOPS,
stdev=861.4 GFLOPS
 99% [||||| ]
Peak MFMA IOPs (I8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:5100273664000, duration:5.1 ms, mean:1003608.9 GOPS,
stdev=4615.9 GOPS

```

INFO [roofline] Empirical Roofline PDFs saved!

1.3.3 Analyze the Non-Fusion MatMulBias Kernel

Using Triton Proton to analyze the report

```
[7]: !proton-viewer -m flops16,time/s -f full nonfusion_matmulbias.hatchet
```

```
3590238240768.000 0.998 ROOT
```

```
nan 0.868 _ZN2at6native29vectorized_elementwise_kernelILi4EN
S0_11FillFunctorIiEESt5arrayIPcLm1EEEEviT0_T1_
```


Legend (Metric: flops16 (inc) Min: 25478299648.00 Max: 3590238240768.00)

```
name User code      Only in left graph
Only in right graph
```

Using ROCm Systems Compute to analyze the report

```
[8]: !rocprof-compute analyze -p workloads/nonfusion_matmulbias/MI300X_A1
```

[illegible]

```
INFO Analysis mode = cli
INFO [analysis] deriving rocprofiler-compute metrics...
WARNING PC sampling: can not detect pc sampling method without /workspa
ce/user/workloads/nonfusion_matmulbias/MI300X_A1/ps_file_pc_sampling_host_trap.c
sv
```

0. Top Stats

0.1 Top Kernels

Kernel_Name			Count	Sum(ns)
Mean(ns)	Median(ns)	Pct		

0	void at::native::vectorized_elementwise_	1404.00	59575021.12
42432.35	42342.75 85.66		
	kernel<4, at::native::FillFunctor<int>,		
	std::array<char*, 1ul> >(int, at::nat...		
1	bias_kernel	1594.00	5755918.81
3610.99	3471.25 8.28		
2	matmul_kernel	152.00	3846198.53
25303.94	24692.70 5.53		
3	__amd_rocclr_copyBuffer	89.00	268785.00
3020.06	2986.59 0.39		
4	void at::native::(anonymous namespace)::	3.00	18722.91
6240.97	7286.66 0.03		
	distribution_elementwise_grid_stride_ker		
	nel<float, 4, at::native::templates::...		
5	void at::native::(anonymous namespace)::	7.00	16406.69
2343.81	2194.94 0.02		
	CatArrayBatchedCopy_contig<at::native::(		
	anonymous namespace)::OpaqueType<2u>,...		
6	void at::native::vectorized_elementwise_	3.00	10276.75
3425.58	3417.75 0.01		
	kernel<4, at::native::compare_scalar_ker		
	nel<double>(at::TensorIteratorBase&, ...		
7	void at::native::vectorized_elementwise_	2.00	6163.75
3081.88	3081.88 0.01		
	kernel<8, at::native::AUnaryFunctor<c10:		

```

:Half, c10::Half, bool, at::native::(...)

8 void at::native::vectorized_elementwise_      2.00      5222.00
2611.00      2611.00      0.01
kernel<8, at::native::BinaryFunctor<c10:

:Half, c10::Half, bool, at::native::(...)

9 void at::native::vectorized_elementwise_      2.00      5191.50
2595.75      2595.75      0.01
kernel<8, at::native::AbsFunctor<c10::Ha

lf>, std::array<char*, 2ul> >(int, at...
```

0.2 Dispatch List

Dispatch_ID	Kernel_Name	GPU_ID
0	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
1	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0

```

2          2 void at::native::(anonymous
namespace)::distribution_elementwise_grid_stride_ker          0
          nel<float, 4,
at::native::templates::cuda::normal_and_transform<c10::Half, float
          , at::CUDAGeneratorImpl*,
at::native::templates::cuda::normal_kernel<at::CUDAGen
          eratorImpl*>(at::TensorBase const&, double, double,
at::CUDAGeneratorImpl*)::...

```

```

3          3 matmul_kernel
0

```

```

4          4 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

```

```

5          5 matmul_kernel
0

```

```

6          6 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

```

```

7          7 matmul_kernel
0

```

```

8          8 void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
          std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
          1>)

```

```

9          9 matmul_kernel
0

```

1. System Info

	Info
workload_name	nonfusion_matmulbias
command	python nonfusion_matmulbias.py
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Mon Jan 12 19:01:21 2026 (UTC)
version	3
hostname	f57b369e0635
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-164-generic
amd_gpu_kernel_version	nan
cpu_memory	230893524
gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942

gpu_chip_id	29877
gpu_l1	32
gpu_l2	4096
cu_per_gpu	304
simd_per_cu	4
se_per_gpu	32
wave_size	64
workgroup_max_size	1024
max_waves_per_cu	32
max_sclk	2100
max_mclk	1300
cur_sclk	2100
cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160
pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak
Pct of Peak				

2.1.0 0.12	VALU FLOPs	95.48	Gflop/s	81715.2
2.1.1 0.77	VALU IOPs	629.19	Giop/s	81715.2
2.1.2 2614886.4 0.0	MFMA FLOPs (F8)	0.0	Gflop/s	
2.1.3 1307443.2 0.0	MFMA FLOPs (BF16)	0.0	Gflop/s	
2.1.4 1307443.2 0.31	MFMA FLOPs (F16)	3994.13	Gflop/s	
2.1.5 163430.4 0.0	MFMA FLOPs (F32)	0.0	Gflop/s	
2.1.6 163430.4 0.0	MFMA FLOPs (F64)	0.0	Gflop/s	
2.1.7	MFMA FLOPs (F6F4)		Gflop/s	
2.1.8 2614886.4 0.0	MFMA IOPs (Int8)	0.0	Giop/s	
2.1.9 37.83	Active CUs	115.0	Cus	304.0
2.1.10 1.57	SALU Utilization	1.57	Pct	100.0
2.1.11 1.61	VALU Utilization	1.61	Pct	100.0

2.1.12 0.23	MFMA Utilization	0.23	Pct	100.0
2.1.13 0.4	VMEM Utilization	0.4	Pct	100.0
2.1.14 0.28	Branch Utilization	0.28	Pct	100.0
2.1.15 99.35	VALU Active Threads	63.58	Threads	64.0
2.1.16 5.18	IPC	0.26	Instr/cycle	5.0
2.1.17 74.96	Wavefront Occupancy	7291.76	Wavefronts	9728.0
2.1.18 0.31	Theoretical LDS Bandwidth	251.72	Gb/s	81715.2
2.1.19 2.24	LDS Bank Conflicts/Access	0.72	Conflicts/access	32.0
2.1.20 17.27	vL1D Cache Hit Rate	17.27	Pct	100.0
2.1.21 8.37	vL1D Cache BW	6842.65	Gb/s	81715.2
2.1.22 43.01	L2 Cache Hit Rate	43.01	Pct	100.0
2.1.23 18.66	L2 Cache BW	6421.33	Gb/s	34406.4

2.1.24 6.13	L2-Fabric Read BW	326.63	Gb/s	5324.8
2.1.25 56.46	L2-Fabric Write BW	3006.17	Gb/s	5324.8
2.1.26	L2-Fabric Read Latency	423.6	Cycles	
2.1.27	L2-Fabric Write Latency	391.33	Cycles	
2.1.28 98.12	sL1D Cache Hit Rate	98.12	Pct	100.0
2.1.29 0.2	sL1D Cache BW	42.84	Gb/s	21504.0
2.1.30 99.74	L1I Hit Rate	99.74	Pct	100.0
2.1.31 2.21	L1I BW	474.17	Gb/s	21504.0
2.1.32	L1I Fetch Latency		Cycles	

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff	Instr Dispatch	Exec
LDS	L2 Cache	

SALU: 413405

CUs	Req	Util	Active
	: 9455	: 0 %	
		xGMI/PCIE	

Wave	Instr Buf	
5: 304	<-----	11

SMEM:

Lat	:
28163	
N/A	Hit : 43
%	

RVGPRseq: 38.0

Wave N-1	Instr Buf	VALU:
371116		Vector L1 Cache
Request		

SGPRs : 37

MFMA:

Rd	Hit	%	Rd
6089	: 169518	: 17 %	

$\begin{array}{c} \text{---} \\ > \\ \text{---} \end{array}$
 $\begin{array}{c} \text{---} \\ < \\ \text{---} \end{array}$
 $\begin{array}{c} \text{---} \\ < \\ \text{---} \end{array}$

Wt : 7275218 Lat : N/A Wr : 1821829 Wr :
1821829 Rd :
15185

```
>          <----- Latency (cycles)      Rd:
15180
```

```

----->          <-----
>          ----->  Rd      :    424      Wr:
1817262

```

```
-----> Atomic:      0
                                         0
```

63
---> Rd :
69

←

43

```

|
----->                                0 <
-----
>                                |
|
Lat : N/A Atomic: 0 Wr :
N/A |
v
^
<-----
>

| Workgroups:

|
0 Instr L1 Cache
  GMI
|
|
|
|
| Fetch :
215331 Hit : 98 % Req
: 2639

-----
-----<-----

Lat : 1143.0 cycles

```

4. Roofline

4.1 Roofline

140691.2
140691 GFLOP/s

36379.2

9406.7

2432.3
C C

628.9 61975 GB/s

33632 GB/s

162.6

42.1 4205 GB/s

0.01 0.18 3.16 56.23 1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)
5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct
5.1.1	CPF Stall	2.17	0.06	7.30	Pct
5.1.2	CPF-L2 Utilization	23.48	0.96	50.80	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	2.43	0.06	10.93	Pct

5.2 Packet Processor

Metric_ID	Metric	Avg	Min	Max	Unit
5.2.0	CPC SYNC FIFO Full Rate				Pct
5.2.1	CPC CANE Stall Rate				Pct
5.2.2	CPC ADC Utilization				Pct
5.2.3	CPC Utilization	100.0	100.0	100.0	Pct

5.2.4	CPC Stall Rate	23.93	7.45	52.42	Pct
5.2.5	CPC Packet Decoding Utilization	56.61	24.47	135.11	Pct
5.2.6	CPC-Workgroup Manager Utilization	29.8	0.51	70.69	Pct
5.2.7	CPC-L2 Utilization	0.67	0.17	1.34	Pct
5.2.8	CPC-UTCL1 Stall	0.42	0.11	5.37	Pct
5.2.9	CPC-UTCL2 Utilization	0.41	0.04	7.21	Pct

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	100.0	100.0	100.0	Pct
6.1.2	Scheduler-Pipe Utilization	0.0	0.0	0.0	Pct

6.1.3	Scheduler-Pipe Wave Utilization				Pct
6.1.4	Workgroup Manager Utilization	52.24	5.96	82.95	Pct
6.1.5	Shader Engine Utilization	47.1	0.24	94.08	Pct
6.1.6	SIMD Utilization	35.48	0.01	93.25	Pct
6.1.7 Workgroups	Dispatched Workgroups	0.0	0.0	0.0	
6.1.8 Wavefronts	Dispatched Wavefronts	0.0	0.0	0.0	
6.1.9 Cycles/wave	VGPR Writes				
6.1.10 Cycles/wave	SGPR Writes				

6.2 Workgroup Manager - Resource Allocation

Metric_ID Unit	Metric	Avg	Min	Max
6.2.0 Pct	Not-scheduled Rate (Workgroup Manager)	0.0	0.0	0.0
6.2.1 Pct	Not-scheduled Rate (Scheduler-Pipe)	0.0	0.0	0.0
6.2.2	Scheduler-Pipe FIFO Full Rate			

Pct

6.2.3	Scheduler-Pipe Stall Rate	0.0	0.0	0.0
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Pct

6.2.4	Scratch Stall Rate	0.0	0.0	0.0
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Pct

6.2.5	Insufficient SIMD Waveslots	0.0	0.0	0.0
-------	-----------------------------	-----	-----	-----

Pct

6.2.6	Insufficient SIMD VGPRs	0.0	0.0	0.0
-------	-------------------------	-----	-----	-----

Pct

6.2.7	Insufficient SIMD SGPRs	0.0	0.0	0.0
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Pct

6.2.8	Insufficient CU LDS	0.0	0.0	0.0
-------	---------------------	-----	-----	-----

Pct

6.2.9	Insufficient CU Barriers	0.0	0.0	0.0
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Pct

6.2.10	Reached CU Workgroup Limit	0.0	0.0	0.0
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Pct

6.2.11	Reached CU Wavefront Limit	0.0	0.0	0.0
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Pct

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID	Metric	Avg	Min	Max	Unit
-----------	--------	-----	-----	-----	------

7.1.0	Grid Size	1810567.77	32.00	4194304.00	Work items
7.1.1	Workgroup Size	316.81	32.00	1024.00	Work items
7.1.2	Total Wavefronts	0.00	0.00	0.00	
7.1.3	Saved Wavefronts	0.00	0.00	0.00	
7.1.4	Restored Wavefronts	0.00	0.00	0.00	
7.1.5	VGPRs	38.39	8.00	104.00	
7.1.6	AGPRs	2.91	0.00	192.00	
7.1.7	SGPRs	36.80	16.00	112.00	
7.1.8	LDS Allocation	9.39	0.00	29184.00	Bytes
7.1.9	Scratch Allocation	0.00	0.00	0.00	Bytes/workitem

7.2 Wavefront Runtime Stats

Metric_ID	Metric	Avg	Min
Max Unit			

7.2.0	Kernel Time (Nanosec)	21255.03	1620.25
65706.00	Ns		
7.2.1	Kernel Time (Cycles)	59931.43	17034.12
136810.50	Cycle		
7.2.2	Instructions per wavefront	323.25	19.50
3555.00	Instr/wavefront		
7.2.3	Wave Cycles	297295311.37	4512.00
949742956.00	Cycles per kernel		
7.2.4	Dependency Wait Cycles	65604119.59	3824.00
203139208.00	Cycles per kernel		
7.2.5	Issue Wait Cycles	227175885.06	8.00
736585300.00	Cycles per kernel		
7.2.6	Active Cycles	4129825.53	296.00
16351660.00	Cycles per kernel		
7.2.7	Wavefront Occupancy	7291.76	4.22
40190.17	Wavefronts		

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0	VALU	365027.17	20.00	2731520.00	Instr per kernel
10.1.1	VMEM	116343.26	2.00	262144.00	Instr per kernel
10.1.2	LDS	9455.38	0.00	212992.00	Instr per kernel
10.1.3	MFMA	6088.92	0.00	131072.00	Instr per kernel

10.1.4	SALU	413405.03	24.00	924160.00	Instr per kernel
10.1.5	SMEM	28163.01	0.00	68096.00	Instr per kernel
10.1.6	Branch	56693.61	0.00	131072.00	Instr per kernel

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.2.0	INT32	198105.83	9.00	393216.00	Instr per kernel
10.2.1	INT64	28600.72	0.00	668160.00	Instr per kernel
10.2.2	F16-ADD	3990.85	0.00	8192.00	Instr per kernel
10.2.3	F16-MUL	0.00	0.00	0.00	Instr per kernel
10.2.4	F16-FMA	10.02	0.00	16384.00	Instr per kernel
10.2.5	F16-Trans	0.00	0.00	0.00	Instr per kernel
10.2.6	F32-ADD	23.80	0.00	38912.00	Instr per kernel
10.2.7	F32-MUL	725.97	0.00	123392.00	Instr per kernel
10.2.8	F32-FMA	71.41	0.00	116736.00	Instr per kernel
10.2.9	F32-Trans	690.26	0.00	65024.00	Instr per kernel
10.2.10	F64-ADD	0.00	0.00	0.00	Instr per kernel
10.2.11	F64-MUL	0.00	0.00	1.00	Instr per kernel
10.2.12	F64-FMA	0.00	0.00	6.00	Instr per kernel
10.2.13	F64-Trans	0.00	0.00	1.00	Instr per kernel
10.2.14	Conversion	2097.79	0.00	58368.00	Instr per kernel

10.3 VMEM Instr Mix

Metric_ID	Metric	Avg	Min	Max
Unit				

10.3.0	Global/Generic Instr	112494.93	0.0	262144.0
Instr per kernel				
10.3.1	Global/Generic Read	0.07	0.0	6.0
Instr per kernel				
10.3.2	Global/Generic Write	112494.86	0.0	262144.0
Instr per kernel				
10.3.3	Global/Generic Atomic	0.0	0.0	0.0
Instr per kernel				
10.3.4	Spill/Stack Instr	3848.33	0.0	36864.0
Instr per kernel				
10.3.5	Spill/Stack Coalesceable Instr			
Instr per kernel				
10.3.6	Spill/Stack Read	2660.34	0.0	32768.0
Instr per kernel				
10.3.7	Spill/Stack Write	1187.99	0.0	4096.0
Instr per kernel				
10.3.8	Spill/Stack Atomic	0.0	0.0	0.0
Instr per kernel				

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0	MFMA-I8	0.0	0.0	0.0	Instr per kernel
10.4.1	MFMA-F8	0.0	0.0	0.0	Instr per kernel
10.4.2	MFMA-F16	6088.92	0.0	131072.0	Instr per kernel
10.4.3	MFMA-BF16	0.0	0.0	0.0	Instr per kernel

10.4.4	MFMA-F32	0.0	0.0	0.0	Instr per kernel
10.4.5	MFMA-F64	0.0	0.0	0.0	Instr per kernel
10.4.6	MFMA-F6F4				Instr per kernel

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
11.1.0	VALU FLOPs	95.48	Gflop	81715.2	0.12
11.1.1	VALU IOPs	629.19	Giop	81715.2	0.77
11.1.2	MFMA FLOPs (F8)	0.0	Gflop	2614886.4	0.0
11.1.3	MFMA FLOPs (BF16)	0.0	Gflop	1307443.2	0.0
11.1.4	MFMA FLOPs (F16)	3994.13	Gflop	1307443.2	0.31
11.1.5	MFMA FLOPs (F32)	0.0	Gflop	163430.4	0.0
11.1.6	MFMA FLOPs (F64)	0.0	Gflop	163430.4	0.0
11.1.7	MFMA FLOPs (F6F4)		Gflop		

11.1.8	MFMA IOPs (INT8)	0.0	Giop	2614886.4	0.0
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11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.26	0.03	1.28	Instr/cycle
11.2.1	IPC (Issued)	0.98	0.85	1.0	Instr/cycle
11.2.2	SALU Utilization	1.57	0.0	11.71	Pct
11.2.3	VALU Utilization	1.61	0.0	34.72	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.4	0.0	0.86	Pct
11.2.6	Branch Utilization	0.28	0.0	1.61	Pct
11.2.7	VALU Active Threads	63.58	28.46	64.0	Threads
11.2.8	MFMA Utilization	0.23	0.0	5.75	Pct

11.2.9	MFMA Instr Cycles	32.0	32.0	32.0	Cycles/instr
11.2.10	VMEM Latency				Cycles
11.2.11	SMEM Latency				Cycles
11.3 Arithmetic Operations					
Metric_ID	Metric	Avg	Min	Max	Unit
11.3.0 kernel	FLOPs (Total)	100118853.75	0.0	2147745792.0	Ops per
11.3.1 kernel	IOPs (Total)	14509218.95	960.0	59179008.0	Ops per
11.3.2 kernel	F8 OPs	0.0	0.0	0.0	Ops per
11.3.3 kernel	F16 OPs	100017550.71	0.0	2147483648.0	Ops per
11.3.4 kernel	BF16 OPs	0.0	0.0	0.0	Ops per
11.3.5 kernel	F32 OPs	101302.77	0.0	29491200.0	Ops per
11.3.6 kernel	F64 OPs	0.27	0.0	896.0	Ops per
11.3.7	F6F4 OPs				Ops per

kernel

11.3.8	INT8 OPs	0.0	0.0	0.0	Ops per
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kernel

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	0.23	Pct of peak
12.1.1	Access Rate	0.15	Pct of peak
12.1.2	Theoretical Bandwidth (% of Peak)	0.31	Pct of peak
12.1.3	Bank Conflict Rate	2.24	Pct of peak

12.2 LDS Stats

Metric_ID	Metric	Avg	Min	Max
12.2.0	LDS Instrs	9455.38	0.0	212992.0

Instr per kernel

12.2.1 LDS LOAD
Instr per kernel

12.2.2 LDS STORE
Instr per kernel

12.2.3 LDS ATOMIC
Instr per kernel

12.2.4 LDS LOAD Bandwidth
Gbps

12.2.5	LDS STORE Bandwidth Gbps			
12.2.6	LDS ATOMIC Bandwidth Gbps			
12.2.7	Theoretical Bandwidth Bytes per kernel	6065772.52	0.0	134217728.0
12.2.8	LDS Latency Cycles			
12.2.9	Bank Conflicts/Access Conflicts/access	0.72	0.0	1.0
12.2.10	Index Accesses Cycles per kernel	47668.43	0.0	1048576.0
12.2.11	Atomic Return Cycles Cycles per kernel	0.0	0.0	0.0
12.2.12	Bank Conflict Cycles per kernel	279.58	0.0	2304.0
12.2.13	Addr Conflict Cycles per kernel	0.01	0.0	24.0
12.2.14	Unaligned Stall Cycles per kernel	0.0	0.0	0.0
12.2.15	Mem Violations Accesses per kernel	0.0	0.0	0.0
12.2.16	LDS Command FIFO Full Rate Cycles per kernel			

12.2.17 LDS Data FIFO Full Rate
Cycles per kernel

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	2.21	Pct of peak
13.1.1	Cache Hit Rate	98.24	Pct of peak
13.1.2	L1I-L2 Bandwidth	0.04	Pct of peak

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
13.2.0 per kernel	Req	215331.02	18.0	778240.0	Req
13.2.1 per kernel	Hits	213763.4	14.0	766147.0	Hits
13.2.2 Misses per kernel	Misses - Non Duplicated	99.22	1.0	480.0	
13.2.3 Misses per kernel	Misses - Duplicated	1440.89	1.0	18130.0	
13.2.4	Cache Hit Rate	98.24	77.78	99.66	Pct
13.2.5	Instruction Fetch Latency				

Cycles

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
13.3.0	L1I-L2 Bandwidth	168885.09	1024.00	645120.00	Bytes per kernel

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.20	Pct of peak
14.1.1	Cache Hit Rate	91.06	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0	Req	28168.97	0.00	77824.00	Req per kernel
14.2.1	Hits	26120.95	0.00	67904.00	Req per kernel
14.2.2	Misses - Non Duplicated	68.89	0.00	320.00	Req per kernel
14.2.3	Misses- Duplicated	1980.20	0.00	9600.00	Req per kernel

kernel

14.2.4	Cache Hit Rate	91.06	0.00	93.69	Pct
14.2.5 kernel	Read Req (Total)	28163.01	0.00	68096.00	Req per
14.2.6 kernel	Atomic Req	0.00	0.00	0.00	Req per
14.2.7 kernel	Read Req (1 DWord)	17.95	0.00	29184.00	Req per
14.2.8 kernel	Read Req (2 DWord)	11.95	0.00	19456.00	Req per
14.2.9 kernel	Read Req (4 DWord)	28133.11	0.00	65536.00	Req per
14.2.10 kernel	Read Req (8 DWord)	0.01	0.00	8.00	Req per
14.2.11 kernel	Read Req (16 DWord)	0.00	0.00	0.00	Req per

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	4409.10	0.00	20480.00	Bytes per kernel
14.3.1	Read Req	68.89	0.00	320.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	0.00	0.00	0.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID Max	Metric Unit	Avg	Min
15.1.0 90.88	Address Processing Unit Busy Pct	33.23	0.0
15.1.1 6.64	Address Stall Pct	0.37	0.0
15.1.2 69.01	Data Stall Pct	26.53	0.0
15.1.3 0.0	Data-Processor → Address Stall Pct	0.0	0.0
15.1.4	Sequencer → TA Address Stall Cycles per kernel		
15.1.5	Sequencer → TA Command Stall Cycles per kernel		
15.1.6	Sequencer → TA Data Stall Cycles per kernel		
15.1.7 262144.0	Total Instructions Instructions per kernel	116343.26	2.0
15.1.8 262144.0	Global/Generic Instructions Instructions per kernel	112494.93	0.0

15.1.9	Global/Generic Read Instructions	0.07	0.0
6.0	Instructions per kernel		
15.1.10	Global/Generic Read Instructions for LDS		
	Instructions per kernel		
15.1.11	Global/Generic Write Instructions	112494.86	0.0
262144.0	Instructions per kernel		
15.1.12	Global/Generic Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.13	Spill/Stack Instructions	3848.33	0.0
36864.0	Instructions per kernel		
15.1.14	Spill/Stack Read Instructions	2660.34	0.0
32768.0	Instructions per kernel		
15.1.15	Spill/Stack Read Instructions for LDS		
	Instructions per kernel		
15.1.16	Spill/Stack Write Instructions	1187.99	0.0
4096.0	Instructions per kernel		
15.1.17	Spill/Stack Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.18	Spill/Stack Total Cycles	61386.95	0.0
589824.0	Cycles per kernel		
15.1.19	Spill/Stack Coalesced Read	62.12	0.0
512.0	Cycles per kernel		
15.1.20	Spill/Stack Coalesced Write	0.0	0.0
0.0	Cycles per kernel		

15.2 Data-Return Path

Metric_ID Unit	Metric	Avg	Min	Max
15.2.0 310.86	Data-Return Busy Pct	34.1	0.0	
15.2.1 79.55	Cache RAM → Data-Return Stall Pct	33.28	0.0	
15.2.2 Pct	Workgroup manager → Data-Return Stall	0.0	0.0	0.0
15.2.3 16384.0	Coalescable Instructions Instructions per kernel	25.68	0.0	
15.2.4 32768.0	Read Instructions Instructions per kernel	2660.41	0.0	
15.2.5 262144.0	Write Instructions Instructions per kernel	113682.85	1.0	
15.2.6	Atomic Instructions Instructions per kernel	0.0	0.0	0.0
15.2.7	Write Ack Instructions Instructions per kernel			

INFO Not showing table with empty column(s): 16.6 L1D Addr
Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
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16.1.0	Hit rate	17.27	Pct of peak
16.1.1	Bandwidth	8.37	Pct of peak
16.1.2	Utilization	44.15	Pct of peak
16.1.3	Coalescing	27.62	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3	Max
16.2.0 94.98	Stalled on L2 Data	0.02	3.75	6.1	93.05	
16.2.1 82.88	Stalled on L2 Req	0.0	0.0	0.0	76.85	
16.2.2	Stalled on Address					
16.2.3	Stalled on Data					
16.2.4	Stalled on Latency FIFO					
16.2.5	Stalled on Request FIFO					
16.2.6	Stalled on Read Return					
16.2.7	Tag RAM Stall (Read)	0.0	0.0	0.0	0.0	0.0

16.2.8	Tag RAM Stall (Write)	0.0	0.0	0.0	0.0	1.4
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16.2.9	Tag RAM Stall (Atomic)	0.0	0.0	0.0	0.0	0.0
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16.3 L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
16.3.0 per kernel	Total Req	7444736.16	32.0	16777216.0	Req
16.3.1 per kernel	Read Req	169517.73	0.0	2097152.0	Req
16.3.2 per kernel	Write Req	7275218.43	16.0	16777216.0	Req
16.3.3 per kernel	Atomic Req	0.0	0.0	0.0	Req
16.3.4 Bytes per kernel	Cache BW	239400569.35	256.0	536870912.0	
16.3.5	Cache Hit Rate	17.27	-29.63	84.09	Pct
16.3.6 per kernel	Cache Accesses	1870316.95	2.0	4194304.0	Req
16.3.7 per kernel	Cache Hits	28262.55	-8.0	393216.0	Req
16.3.8	Invalidations	157.05	1.0	304.0	Req

per kernel

16.3.9	L1-L2 BW	119185921.06	192.0	268435456.0
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Bytes per kernel

16.3.10	Tag RAM 0 Req				Req
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per kernel

16.3.11	Tag RAM 1 Req				Req
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per kernel

16.3.12	Tag RAM 2 Req				Req
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per kernel

16.3.13	Tag RAM 3 Req				Req
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per kernel

16.3.14	L1-L2 Read	20225.62	0.0	262144.0	Req
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per kernel

16.3.15	L1-L2 Write	1821828.78	1.0	4194304.0	Req
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per kernel

16.3.16	L1-L2 Atomic	0.0	0.0	0.0	Req
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per kernel

16.3.17	L1 Access Latency				
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Cycles

16.3.18	L1-L2 Read Latency				
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Cycles

16.3.19	L1-L2 Write Latency				
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Cycles

16.4 L1D - L2 Transactions

Metric_ID	Metric	Xfer	Coherency	Avg	Min
Max	Unit				
16.4.0	NC - Read	Read	NC	0.00	0.00
0.00	Req per kernel				
16.4.1	UC - Read	Read	UC	0.00	0.00
0.00	Req per kernel				
16.4.2	CC - Read	Read	CC	0.00	0.00
0.00	Req per kernel				
16.4.3	RW - Read	Read	RW	20225.39	0.00
262144.00	Req per kernel				
16.4.4	RW - Write	Write	RW	1821828.73	0.00
4194304.00	Req per kernel				
16.4.5	NC - Write	Write	NC	0.00	0.00
0.00	Req per kernel				
16.4.6	UC - Write	Write	UC	0.05	0.00
6.00	Req per kernel				
16.4.7	CC - Write	Write	CC	0.00	0.00
0.00	Req per kernel				
16.4.8	NC - Atomic	Atomic	NC	0.00	0.00
0.00	Req per kernel				
16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
0.00	Req per kernel				
16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
0.00	Req per kernel				

16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
0.00	Req per kernel				

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min	Max
Units				
16.5.0	Req	1870316.95	2.0	4194304.0
Req per kernel				
16.5.1	Inflight Req			
Req per kernel				
16.5.2	Hit Ratio	92.79	0.0	100.0
Pct				
16.5.3	Hits	1721231.5	0.0	3888317.0
Req per kernel				
16.5.4	Translation Misses	7094.28	0.0	16384.0
Req per kernel				
16.5.5	Misses under Translation Miss			
Req per kernel				
16.5.6	Permission Misses	0.05	0.0	128.0
Req per kernel				

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	50.30	Pct
17.1.1	Bandwidth	18.66	Pct
17.1.2	Hit Rate	43.01	Pct
17.1.3	L2-Fabric Read BW	326.63	Gb/s
17.1.4	L2-Fabric Write and Atomic BW	3006.17	Gb/s
17.1.5	HBM Bandwidth	5324.80	Gb/s

17.2 L2 - Fabric Transactions

Metric_ID	Metric	Avg	Min	Max
Unit				
17.2.0	Read BW	1925075.09	1536.0	
18954240.0	Bytes per kernel			
17.2.1	HBM Read Traffic	98.89	64.24	
143.33	Pct			
17.2.2	Remote Read Traffic	1.54	0.0	
35.76	Pct			
17.2.3	Uncached Read Traffic	3.72	0.0	
144.46	Pct			
17.2.4	Write and Atomic BW	116304771.46	32.0	
268435456.0	Bytes per kernel			
17.2.5	HBM Write and Atomic Traffic	97.28	0.0	
100.0	Pct			
17.2.6	Remote Write and Atomic Traffic	2.72	0.0	
100.0	Pct			

17.2.7 Pct	Atomic Traffic	0.0	0.0	0.0
17.2.8 100.0	Uncached Write and Atomic Traffic Pct	2.72	0.0	
17.2.9 1144.28	Read Latency Cycles	423.6	150.72	
17.2.10 1058.34	Write and Atomic Latency Cycles	391.33	264.0	
17.2.11 Cycles	Atomic Latency			
17.2.12 Pct	Read Stall			
17.2.13 Pct	Write Stall			

17.3 L2 Cache Accesses

Metric_ID Unit	Metric	Avg	Min	Max
17.3.0 Bytes per kernel	Bandwidth	236133195.54	3072.0	537406336.0
17.3.1 Bytes per kernel	Read Bandwidth			
17.3.2 Bytes per kernel	Write Bandwidth			

17.3.3 Atomic Bandwidth
Bytes per kernel

17.3.4 per kernel	Req	1844790.59	24.0	4198487.0	Req
17.3.5 per kernel	Read Req	22961.34	23.0	270048.0	Req
17.3.6 per kernel	Write Req	1821828.78	1.0	4194304.0	Req
17.3.7 per kernel	Atomic Req	0.0	0.0	0.0	Req
17.3.8 per kernel	Streaming Req	0.0	0.0	3.0	Req
17.3.9 per kernel	Bypassss Req				Req
17.3.10 per kernel	Probe Req	0.0	0.0	2.0	Req
17.3.11 per kernel	Input Buffer Req				Req
17.3.12	Cache Hit	43.01	20.0	61.4	Pct
17.3.13 Hits per kernel	Hits	921151.34	10.0	2101048.0	
17.3.14 Misses per kernel	Misses	923676.93	13.0	2097443.0	

17.3.15	Writeback	908714.0	5.0	2097208.0	
Cachelines per kernel					
17.3.16	Writeback (Internal)	843635.76	0.0	1966087.0	
Cachelines per kernel					
17.3.17	Writeback (vL1D Req)	64995.27	0.0	131090.0	
Cachelines per kernel					
17.3.18	Evict (Internal)	787441.25	0.0	1835120.0	
Cachelines per kernel					
17.3.19	Evict (vL1D Req)	0.0	0.0	0.0	
Cachelines per kernel					
17.3.20	NC Req	0.0	0.0	0.0	Req
per kernel					
17.3.21	UC Req	7.29	0.0	515.0	Req
per kernel					
17.3.22	CC Req	0.0	0.0	0.0	Req
per kernel					
17.3.23	RW Req	1844821.32	24.0	4198304.0	Req
per kernel					

17.5 L2 - Fabric Interface Stalls

Metric_ID	Metric				Type
Transaction	Avg	Min	Max	Unit	
17.5.0	Read - PCIe Stall				PCIe Stall
	Pct				Read
17.5.1	Read - Infinity Fabric Stall				Infinity Fabric Stall
					Read

Pct				
17.5.2	Read - HBM Stall Pct		HBM Stall	Read
17.5.3	Write - PCIe Stall Pct		PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct		Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct		HBM Stall	Write
17.5.6	Write - Credit Starvation 0.0 0.0 0.0 Pct		Credit Starvation	Write

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID Unit	Metric	Avg	Min	Max
17.6.0 Req per kernel	Read (32B)	0.0	0.0	0.0
17.6.1 1248.0 Req per kernel	Read (64B)	291.05	22.0	
17.6.2 147456.0 Req per kernel	Read (128B)	14894.13	0.0	
17.6.3 1030.0 Req per kernel	Read (Uncached)	14.47	0.0	
17.6.4 148688.0 Req per kernel	HBM Read	15180.34	23.0	

17.6.5	Remote Read	6.22	0.0	255.0
Req per kernel				
17.6.6	Read Bandwidth - PCIe			
Bytes per kernel				
17.6.7	Read Bandwidth - Infinity Fabric			
Bytes per kernel				
17.6.8	Read Bandwidth - HBM			
Bytes per kernel				
17.6.9	Write and Atomic (32B)	0.06	0.0	6.0
Req per kernel				
17.6.10	Write and Atomic (Uncached)	0.05	0.0	6.0
Req per kernel				
17.6.11	Write and Atomic (64B)	1817262.02	0.0	
4194304.0	Req per kernel			
17.6.12	HBM Write and Atomic	1817262.03	0.0	
4194304.0	Req per kernel			
17.6.13	Remote Write and Atomic	0.05	0.0	6.0
Req per kernel				
17.6.14	Write Bandwidth - PCIe			
Bytes per kernel				
17.6.15	Write Bandwidth - Infinity Fabric			
Bytes per kernel				
17.6.16	Write Bandwidth - HBM			
Bytes per kernel				

17.6.17	Atomic	0.0	0.0	0.0
Req per kernel				

17.6.18	Atomic - HBM
Req per kernel	

17.6.19	Atomic Bandwidth - PCIe
Bytes per kernel	

17.6.20	Atomic Bandwidth - Infinity Fabric
Bytes per kernel	

17.6.21	Atomic Bandwidth - HBM
Bytes per kernel	

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	42.62	8.75	0.00	61.41	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	0.00	33.16	37.89	50.12	61.31
18.2.1	1.00	0.00	33.16	37.98	50.12	61.31
18.2.2	2.00	0.00	33.16	37.98	50.12	61.31
18.2.3	3.00	0.00	33.16	37.98	50.12	61.31

18.2.4	4.00	0.00	33.16	37.73	50.00	61.31
18.2.5	5.00	33.07	33.16	37.98	50.00	61.31
18.2.6	6.00	0.00	33.16	37.56	50.00	61.31
18.2.7	7.00	0.00	33.16	37.98	50.12	61.31
18.2.8	8.00	0.00	35.82	37.98	50.12	66.67
18.2.9	9.00	0.00	35.82	37.98	50.00	100.00
18.2.10	10.00	0.00	35.50	37.98	50.00	61.31
18.2.11	11.00	0.00	35.50	37.98	50.00	61.31
18.2.12	12.00	0.00	35.50	37.98	50.00	63.03
18.2.13	13.00	0.00	35.41	37.89	50.00	65.33
18.2.14	14.00	0.00	35.50	37.98	50.00	61.31
18.2.15	15.00	0.00	35.50	37.98	50.00	61.31
18.2.16	16.00	0.00	35.82	37.98	50.12	66.67
18.2.17	17.00	0.00	35.82	37.98	50.00	100.00
18.2.18	18.00	0.00	35.50	37.98	50.00	61.31
18.2.19	19.00	0.00	35.50	37.98	50.00	61.31
18.2.20	20.00	0.00	35.50	37.98	50.00	63.03
18.2.21	21.00	0.00	35.41	37.89	50.00	65.33
18.2.22	22.00	0.00	35.50	37.98	50.00	61.31
18.2.23	23.00	0.00	35.50	37.98	50.00	61.31
18.2.24	24.00	0.00	33.16	37.56	50.12	61.31
18.2.25	25.00	0.00	33.16	37.56	50.12	61.31
18.2.26	26.00	0.00	33.16	37.56	50.12	61.31
18.2.27	27.00	0.00	33.16	37.56	50.12	61.31

18.2.28	28.00	0.00	33.16	37.56	50.00	61.31
18.2.29	29.00	33.07	33.16	37.56	50.00	61.31
18.2.30	30.00	0.00	33.16	37.56	50.00	61.31
18.2.31	31.00	0.00	33.16	37.56	50.12	61.31
18.2.32	32.00	0.00	33.16	37.56	50.12	61.31
18.2.33	33.00	0.00	33.16	37.56	50.12	61.31
18.2.34	34.00	0.00	33.16	37.56	50.12	61.31
18.2.35	35.00	0.00	33.16	37.56	50.12	61.31
18.2.36	36.00	0.00	33.16	37.56	50.00	61.31
18.2.37	37.00	33.07	33.16	37.56	50.00	61.31
18.2.38	38.00	0.00	33.16	37.56	50.00	61.31
18.2.39	39.00	0.00	33.16	37.56	50.12	61.31
18.2.40	40.00	0.00	35.82	37.98	50.12	66.67
18.2.41	41.00	0.00	35.82	37.98	50.00	100.00
18.2.42	42.00	0.00	35.50	37.98	50.00	61.31
18.2.43	43.00	0.00	35.50	37.98	50.00	61.31
18.2.44	44.00	0.00	35.50	37.98	50.00	63.03
18.2.45	45.00	0.00	35.41	37.98	50.00	65.33
18.2.46	46.00	0.00	35.50	37.98	50.00	61.31
18.2.47	47.00	0.00	35.50	37.98	50.00	61.31
18.2.48	48.00	0.00	35.82	37.98	50.12	66.67
18.2.49	49.00	0.00	35.82	37.98	50.00	100.00
18.2.50	50.00	0.00	35.50	37.98	50.00	61.31
18.2.51	51.00	0.00	35.50	37.98	50.00	61.31

18.2.52	52.00	0.00	35.50	37.98	50.00	63.03
18.2.53	53.00	0.00	35.41	37.89	50.00	65.33
18.2.54	54.00	0.00	35.50	37.98	50.00	61.31
18.2.55	55.00	0.00	35.50	37.98	50.00	61.31
18.2.56	56.00	0.00	33.16	37.56	50.12	61.31
18.2.57	57.00	0.00	33.16	37.56	50.12	61.31
18.2.58	58.00	0.00	33.16	37.56	50.12	61.31
18.2.59	59.00	0.00	33.16	37.56	50.12	61.31
18.2.60	60.00	0.00	33.16	37.56	50.00	61.31
18.2.61	61.00	33.07	33.16	37.56	50.00	61.31
18.2.62	62.00	0.00	33.16	37.56	50.00	61.31
18.2.63	63.00	0.00	33.16	37.56	50.12	61.31
18.2.64	64.00	0.00	35.82	37.98	50.12	61.31
18.2.65	65.00	0.00	35.82	37.98	50.00	61.31
18.2.66	66.00	0.00	35.50	37.98	50.00	61.31
18.2.67	67.00	0.00	35.50	37.98	50.00	61.31
18.2.68	68.00	0.00	35.50	37.98	50.00	63.03
18.2.69	69.00	0.00	35.41	37.89	50.00	65.33
18.2.70	70.00	32.90	35.50	37.98	50.00	61.31
18.2.71	71.00	32.90	35.50	37.98	50.00	61.31
18.2.72	72.00	0.00	33.16	37.56	50.12	61.31
18.2.73	73.00	33.07	33.16	37.56	50.12	61.31
18.2.74	74.00	0.00	33.16	37.56	50.12	61.31
18.2.75	75.00	0.00	33.16	37.56	50.12	61.31

18.2.76	76.00	0.00	33.16	37.56	50.00	61.31
18.2.77	77.00	33.07	33.16	37.56	50.00	61.31
18.2.78	78.00	0.00	33.16	37.56	50.00	61.31
18.2.79	79.00	0.00	33.16	37.56	50.12	61.31
18.2.80	80.00	0.00	33.16	37.98	50.12	61.31
18.2.81	81.00	33.07	33.16	38.03	50.12	61.31
18.2.82	82.00	0.00	33.16	37.98	50.12	61.31
18.2.83	83.00	0.00	33.16	37.98	50.12	61.31
18.2.84	84.00	0.00	33.16	37.98	50.00	61.31
18.2.85	85.00	33.07	33.16	38.07	50.00	61.31
18.2.86	86.00	0.00	33.16	37.98	50.00	61.31
18.2.87	87.00	0.00	33.16	37.98	50.12	61.31
18.2.88	88.00	0.00	35.82	37.98	50.12	61.31
18.2.89	89.00	0.00	35.82	37.98	50.00	61.31
18.2.90	90.00	0.00	35.50	37.98	50.00	61.31
18.2.91	91.00	0.00	35.50	37.98	50.00	61.31
18.2.92	92.00	0.00	35.50	37.98	50.00	63.03
18.2.93	93.00	0.00	35.41	37.94	50.00	65.33
18.2.94	94.00	32.90	35.50	37.98	50.00	61.31
18.2.95	95.00	32.90	35.50	37.98	50.00	61.31
18.2.96	96.00	0.00	35.96	50.00	50.12	100.00
18.2.97	97.00	0.00	35.96	50.00	50.00	100.00
18.2.98	98.00	0.00	35.50	50.00	50.00	85.71
18.2.99	99.00	0.00	35.50	37.38	50.00	100.00

18.2.100	100.00	0.00	35.50	37.38	50.00	100.00
18.2.101	101.00	0.00	35.41	37.38	50.00	100.00
18.2.102	102.00	0.00	35.50	50.00	50.00	100.00
18.2.103	103.00	0.00	35.50	50.00	50.00	100.00
18.2.104	104.00	0.00	33.16	37.38	50.12	75.00
18.2.105	105.00	0.00	33.16	37.77	50.12	100.00
18.2.106	106.00	0.00	33.16	37.56	50.12	100.00
18.2.107	107.00	0.00	33.16	37.56	50.12	100.00
18.2.108	108.00	0.00	33.16	37.89	50.00	100.00
18.2.109	109.00	0.00	33.16	37.98	50.00	100.00
18.2.110	110.00	0.00	33.16	37.56	50.00	100.00
18.2.111	111.00	0.00	33.16	37.77	50.12	100.00
18.2.112	112.00	0.00	33.16	37.56	50.12	61.31
18.2.113	113.00	0.00	33.16	37.56	50.12	61.31
18.2.114	114.00	0.00	33.16	37.56	50.12	61.31
18.2.115	115.00	0.00	33.16	37.56	50.12	61.31
18.2.116	116.00	0.00	33.16	37.56	50.00	61.31
18.2.117	117.00	0.00	33.16	37.56	50.00	61.31
18.2.118	118.00	0.00	33.16	37.56	50.00	61.31
18.2.119	119.00	0.00	33.16	37.56	50.12	61.31
18.2.120	120.00	0.00	35.82	37.98	50.12	100.00
18.2.121	121.00	0.00	35.82	37.98	50.00	100.00
18.2.122	122.00	0.00	35.50	37.98	50.00	61.31
18.2.123	123.00	0.00	35.50	37.98	50.00	61.31

18.2.124	124.00	0.00	35.50	37.98	50.00	63.03
18.2.125	125.00	0.00	35.50	37.89	50.00	65.33
18.2.126	126.00	0.00	35.50	37.98	50.00	61.31
18.2.127	127.00	0.00	35.50	37.98	50.00	61.31

18.3 L2 Requests (per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.3.0	0.00	0.00	400.00	412.00	32768.00	32789.00
18.3.1	1.00	0.00	400.00	412.00	32768.00	32789.00
18.3.2	2.00	0.00	400.00	412.00	32848.00	32869.00
18.3.3	3.00	0.00	400.00	416.00	32768.00	32789.00
18.3.4	4.00	0.00	400.00	412.00	32848.00	32869.00
18.3.5	5.00	0.00	400.00	412.00	32848.00	32869.00
18.3.6	6.00	0.00	400.00	416.00	32848.00	32869.00
18.3.7	7.00	0.00	400.00	416.00	32848.00	32869.00
18.3.8	8.00	0.00	388.00	414.00	32768.00	32790.00
18.3.9	9.00	0.00	388.00	414.00	32768.00	32789.00
18.3.10	10.00	0.00	388.00	414.00	32768.00	32789.00
18.3.11	11.00	0.00	388.00	414.00	32768.00	32789.00
18.3.12	12.00	0.00	388.00	414.00	32768.00	32790.00
18.3.13	13.00	0.00	388.00	414.00	32848.00	32869.00
18.3.14	14.00	0.00	387.00	414.00	32768.00	32789.00
18.3.15	15.00	0.00	387.00	414.00	32768.00	32789.00
18.3.16	16.00	0.00	388.00	414.00	32768.00	32790.00
18.3.17	17.00	0.00	388.00	414.00	32768.00	32789.00

18.3.18	18.00	0.00	388.00	414.00	32768.00	32789.00
18.3.19	19.00	0.00	388.00	414.00	32768.00	32789.00
18.3.20	20.00	0.00	387.00	414.00	32768.00	32790.00
18.3.21	21.00	0.00	387.00	414.00	32848.00	32869.00
18.3.22	22.00	0.00	387.00	414.00	32768.00	32789.00
18.3.23	23.00	0.00	387.00	414.00	32768.00	32789.00
18.3.24	24.00	0.00	400.00	412.00	32768.00	32789.00
18.3.25	25.00	0.00	400.00	412.00	32768.00	32789.00
18.3.26	26.00	0.00	400.00	412.00	32848.00	32869.00
18.3.27	27.00	0.00	400.00	416.00	32768.00	32789.00
18.3.28	28.00	0.00	400.00	412.00	32848.00	32869.00
18.3.29	29.00	0.00	400.00	412.00	32848.00	32869.00
18.3.30	30.00	0.00	400.00	416.00	32848.00	32869.00
18.3.31	31.00	0.00	400.00	416.00	32848.00	32869.00
18.3.32	32.00	0.00	400.00	412.00	32768.00	32789.00
18.3.33	33.00	0.00	400.00	412.00	32768.00	32789.00
18.3.34	34.00	0.00	400.00	412.00	32848.00	32869.00
18.3.35	35.00	0.00	400.00	416.00	32768.00	32789.00
18.3.36	36.00	0.00	400.00	412.00	32848.00	32869.00
18.3.37	37.00	0.00	400.00	412.00	32848.00	32869.00
18.3.38	38.00	0.00	400.00	416.00	32848.00	32869.00
18.3.39	39.00	0.00	400.00	416.00	32848.00	32869.00
18.3.40	40.00	0.00	388.00	414.00	32768.00	32790.00
18.3.41	41.00	0.00	387.00	414.00	32768.00	32789.00

18.3.42	42.00	0.00	388.00	414.00	32768.00	32789.00
18.3.43	43.00	0.00	387.00	414.00	32768.00	32789.00
18.3.44	44.00	0.00	387.00	414.00	32768.00	32790.00
18.3.45	45.00	0.00	387.00	414.00	32848.00	32869.00
18.3.46	46.00	0.00	387.00	414.00	32768.00	32789.00
18.3.47	47.00	0.00	387.00	414.00	32768.00	32789.00
18.3.48	48.00	0.00	388.00	414.00	32768.00	32790.00
18.3.49	49.00	0.00	387.00	414.00	32768.00	32789.00
18.3.50	50.00	0.00	387.00	414.00	32768.00	32789.00
18.3.51	51.00	0.00	387.00	414.00	32768.00	32789.00
18.3.52	52.00	0.00	387.00	414.00	32768.00	32790.00
18.3.53	53.00	0.00	386.00	414.00	32848.00	32869.00
18.3.54	54.00	0.00	386.75	414.00	32768.00	32789.00
18.3.55	55.00	0.00	387.00	414.00	32768.00	32789.00
18.3.56	56.00	0.00	400.00	412.00	32768.00	32789.00
18.3.57	57.00	0.00	400.00	412.00	32768.00	32789.00
18.3.58	58.00	0.00	400.00	412.00	32848.00	32869.00
18.3.59	59.00	0.00	400.00	416.00	32768.00	32789.00
18.3.60	60.00	0.00	400.00	412.00	32848.00	32869.00
18.3.61	61.00	0.00	400.00	412.00	32848.00	32869.00
18.3.62	62.00	0.00	400.00	416.00	32848.00	32869.00
18.3.63	63.00	0.00	400.00	416.00	32848.00	32869.00
18.3.64	64.00	0.00	388.00	414.00	32768.00	32790.00
18.3.65	65.00	0.00	387.00	414.00	32768.00	32789.00

18.3.66	66.00	0.00	388.00	414.00	32768.00	32789.00
18.3.67	67.00	0.00	387.00	414.00	32768.00	32789.00
18.3.68	68.00	0.00	387.00	414.00	32768.00	32790.00
18.3.69	69.00	0.00	387.00	414.00	32848.00	32869.00
18.3.70	70.00	0.00	387.00	414.00	32768.00	32789.00
18.3.71	71.00	0.00	387.00	414.00	32768.00	32789.00
18.3.72	72.00	0.00	400.00	412.00	32768.00	32789.00
18.3.73	73.00	0.00	400.00	412.00	32768.00	32789.00
18.3.74	74.00	0.00	400.00	412.00	32848.00	32869.00
18.3.75	75.00	0.00	400.00	416.00	32768.00	32789.00
18.3.76	76.00	0.00	400.00	412.00	32848.00	32869.00
18.3.77	77.00	0.00	400.00	412.00	32848.00	32869.00
18.3.78	78.00	0.00	400.00	416.00	32848.00	32869.00
18.3.79	79.00	0.00	400.00	416.00	32848.00	32869.00
18.3.80	80.00	0.00	400.00	412.00	32768.00	32789.00
18.3.81	81.00	0.00	400.00	412.00	32768.00	32789.00
18.3.82	82.00	0.00	400.00	412.00	32848.00	32869.00
18.3.83	83.00	0.00	400.00	416.00	32768.00	32789.00
18.3.84	84.00	0.00	400.00	412.00	32848.00	32869.00
18.3.85	85.00	0.00	400.00	412.00	32848.00	32869.00
18.3.86	86.00	0.00	400.00	416.00	32848.00	32869.00
18.3.87	87.00	0.00	400.00	416.00	32848.00	32869.00
18.3.88	88.00	0.00	388.00	414.00	32768.00	32790.00
18.3.89	89.00	0.00	387.00	414.00	32768.00	32789.00

18.3.90	90.00	0.00	388.00	414.00	32768.00	32789.00
18.3.91	91.00	0.00	387.00	414.00	32768.00	32789.00
18.3.92	92.00	0.00	387.00	414.00	32768.00	32790.00
18.3.93	93.00	0.00	387.00	414.00	32848.00	32869.00
18.3.94	94.00	0.00	387.00	414.00	32768.00	32789.00
18.3.95	95.00	0.00	387.00	414.00	32768.00	32789.00
18.3.96	96.00	0.00	387.00	414.00	32768.00	32790.00
18.3.97	97.00	0.00	388.00	414.00	32768.00	32789.00
18.3.98	98.00	0.00	387.00	414.00	32768.00	32789.00
18.3.99	99.00	0.00	388.00	414.00	32768.00	32789.00
18.3.100	100.00	0.00	387.00	414.00	32768.00	32790.00
18.3.101	101.00	0.00	387.00	414.00	32848.00	32869.00
18.3.102	102.00	0.00	386.00	414.00	32768.00	32789.00
18.3.103	103.00	0.00	387.00	414.00	32768.00	32789.00
18.3.104	104.00	0.00	400.00	412.00	32768.00	32789.00
18.3.105	105.00	0.00	400.00	412.00	32768.00	32789.00
18.3.106	106.00	0.00	400.00	412.00	32848.00	32869.00
18.3.107	107.00	0.00	400.00	416.00	32768.00	32789.00
18.3.108	108.00	0.00	400.00	412.00	32848.00	32869.00
18.3.109	109.00	0.00	400.00	412.00	32848.00	32869.00
18.3.110	110.00	0.00	400.00	416.00	32848.00	32869.00
18.3.111	111.00	0.00	400.00	416.00	32848.00	32869.00
18.3.112	112.00	0.00	400.00	412.00	32768.00	32789.00
18.3.113	113.00	0.00	400.00	412.00	32768.00	32789.00

18.3.114	114.00	0.00	400.00	412.00	32848.00	32869.00
18.3.115	115.00	0.00	400.00	416.00	32768.00	32789.00
18.3.116	116.00	0.00	400.00	412.00	32848.00	32869.00
18.3.117	117.00	0.00	400.00	412.00	32848.00	32869.00
18.3.118	118.00	0.00	400.00	416.00	32848.00	32869.00
18.3.119	119.00	0.00	400.00	416.00	32848.00	32869.00
18.3.120	120.00	0.00	387.00	414.00	32768.00	32790.00
18.3.121	121.00	0.00	388.00	414.00	32768.00	32789.00
18.3.122	122.00	0.00	387.00	414.00	32768.00	32789.00
18.3.123	123.00	0.00	388.00	414.00	32768.00	32789.00
18.3.124	124.00	0.00	387.00	414.00	32768.00	32790.00
18.3.125	125.00	0.00	387.00	414.00	32848.00	32869.00
18.3.126	126.00	0.00	387.00	414.00	32768.00	32789.00
18.3.127	127.00	0.00	387.00	414.00	32768.00	32789.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	169.06	14233.04	0.00
18.4.1	1.00	168.78	14233.04	0.00
18.4.2	2.00	169.43	14233.04	0.00
18.4.3	3.00	203.75	14233.04	0.00
18.4.4	4.00	201.97	14233.04	0.00
18.4.5	5.00	201.65	14233.04	0.00
18.4.6	6.00	202.36	14233.04	0.00
18.4.7	7.00	201.65	14233.04	0.00

18.4.8	8.00	164.39	14233.04	0.00
18.4.9	9.00	164.29	14233.04	0.00
18.4.10	10.00	164.47	14233.04	0.00
18.4.11	11.00	164.39	14233.04	0.00
18.4.12	12.00	198.77	14233.04	0.00
18.4.13	13.00	164.70	14233.04	0.00
18.4.14	14.00	164.22	14233.04	0.00
18.4.15	15.00	164.32	14233.04	0.00
18.4.16	16.00	164.38	14233.04	0.00
18.4.17	17.00	164.28	14233.04	0.00
18.4.18	18.00	164.46	14233.04	0.00
18.4.19	19.00	164.38	14233.04	0.00
18.4.20	20.00	198.75	14233.04	0.00
18.4.21	21.00	164.68	14233.04	0.00
18.4.22	22.00	164.22	14233.04	0.00
18.4.23	23.00	164.31	14233.04	0.00
18.4.24	24.00	169.12	14233.04	0.00
18.4.25	25.00	168.84	14233.04	0.00
18.4.26	26.00	169.49	14233.04	0.00
18.4.27	27.00	203.82	14233.04	0.00
18.4.28	28.00	202.01	14233.04	0.00
18.4.29	29.00	201.69	14233.04	0.00
18.4.30	30.00	202.41	14233.04	0.00
18.4.31	31.00	201.69	14233.04	0.00

18.4.32	32.00	169.13	14233.04	0.00
18.4.33	33.00	168.84	14233.04	0.00
18.4.34	34.00	169.49	14233.04	0.00
18.4.35	35.00	203.82	14233.04	0.00
18.4.36	36.00	202.02	14233.04	0.00
18.4.37	37.00	201.70	14233.04	0.00
18.4.38	38.00	202.42	14233.04	0.00
18.4.39	39.00	201.70	14233.04	0.00
18.4.40	40.00	164.37	14233.04	0.00
18.4.41	41.00	164.31	14233.04	0.00
18.4.42	42.00	164.48	14233.04	0.00
18.4.43	43.00	164.43	14233.04	0.00
18.4.44	44.00	198.77	14233.04	0.00
18.4.45	45.00	164.70	14233.04	0.00
18.4.46	46.00	164.24	14233.04	0.00
18.4.47	47.00	164.34	14233.04	0.00
18.4.48	48.00	164.34	14233.04	0.00
18.4.49	49.00	164.28	14233.04	0.00
18.4.50	50.00	164.44	14233.04	0.00
18.4.51	51.00	164.39	14233.04	0.00
18.4.52	52.00	198.73	14233.04	0.00
18.4.53	53.00	164.66	14233.04	0.00
18.4.54	54.00	164.22	14233.04	0.00
18.4.55	55.00	164.31	14233.04	0.00

18.4.56	56.00	169.12	14233.04	0.00
18.4.57	57.00	168.83	14233.04	0.00
18.4.58	58.00	169.48	14233.04	0.00
18.4.59	59.00	203.81	14233.04	0.00
18.4.60	60.00	201.99	14233.04	0.00
18.4.61	61.00	201.67	14233.04	0.00
18.4.62	62.00	202.40	14233.04	0.00
18.4.63	63.00	201.67	14233.04	0.00
18.4.64	64.00	164.35	14233.04	0.00
18.4.65	65.00	164.30	14233.04	0.00
18.4.66	66.00	164.46	14233.04	0.00
18.4.67	67.00	164.41	14233.04	0.00
18.4.68	68.00	198.75	14233.04	0.00
18.4.69	69.00	164.68	14233.04	0.00
18.4.70	70.00	164.22	14233.04	0.00
18.4.71	71.00	164.32	14233.04	0.00
18.4.72	72.00	169.09	14233.04	0.00
18.4.73	73.00	168.80	14233.04	0.00
18.4.74	74.00	169.45	14233.04	0.00
18.4.75	75.00	203.78	14233.04	0.00
18.4.76	76.00	201.98	14233.04	0.00
18.4.77	77.00	201.66	14233.04	0.00
18.4.78	78.00	202.38	14233.04	0.00
18.4.79	79.00	201.66	14233.04	0.00

18.4.80	80.00	169.04	14233.04	0.00
18.4.81	81.00	168.75	14233.04	0.00
18.4.82	82.00	169.41	14233.04	0.00
18.4.83	83.00	203.73	14233.04	0.00
18.4.84	84.00	201.92	14233.04	0.00
18.4.85	85.00	201.60	14233.04	0.00
18.4.86	86.00	202.32	14233.04	0.00
18.4.87	87.00	201.60	14233.04	0.00
18.4.88	88.00	164.28	14233.04	0.00
18.4.89	89.00	164.24	14233.04	0.00
18.4.90	90.00	164.38	14233.04	0.00
18.4.91	91.00	164.33	14233.04	0.00
18.4.92	92.00	198.68	14233.04	0.00
18.4.93	93.00	164.61	14233.04	0.00
18.4.94	94.00	164.15	14233.04	0.00
18.4.95	95.00	164.24	14233.04	0.00
18.4.96	96.00	164.49	14233.04	0.00
18.4.97	97.00	164.42	14233.04	0.00
18.4.98	98.00	164.62	14233.05	0.00
18.4.99	99.00	164.54	14233.04	0.00
18.4.100	100.00	198.92	14233.04	0.00
18.4.101	101.00	164.83	14233.04	0.00
18.4.102	102.00	164.27	14233.04	0.00
18.4.103	103.00	164.36	14233.04	0.00

18.4.104	104.00	169.12	14233.04	0.00
18.4.105	105.00	168.82	14233.04	0.00
18.4.106	106.00	169.48	14233.07	0.00
18.4.107	107.00	203.80	14233.04	0.00
18.4.108	108.00	201.97	14233.04	0.00
18.4.109	109.00	201.66	14233.04	0.00
18.4.110	110.00	202.39	14233.04	0.00
18.4.111	111.00	201.66	14233.04	0.00
18.4.112	112.00	169.09	14233.04	0.00
18.4.113	113.00	168.80	14233.04	0.00
18.4.114	114.00	169.45	14233.04	0.00
18.4.115	115.00	203.78	14233.04	0.00
18.4.116	116.00	202.01	14233.04	0.00
18.4.117	117.00	201.69	14233.04	0.00
18.4.118	118.00	202.39	14233.04	0.00
18.4.119	119.00	201.69	14233.04	0.00
18.4.120	120.00	164.31	14233.04	0.00
18.4.121	121.00	164.22	14233.04	0.00
18.4.122	122.00	164.40	14233.04	0.00
18.4.123	123.00	164.32	14233.04	0.00
18.4.124	124.00	198.69	14233.04	0.00
18.4.125	125.00	164.62	14233.04	0.00
18.4.126	126.00	164.16	14233.04	0.00
18.4.127	127.00	164.26	14233.04	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID L2-Fabric Atomic	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
18.5.0 0.00	0.00	117.58	14197.36
18.5.1 0.00	1.00	117.69	14197.36
18.5.2 0.00	2.00	119.44	14197.36
18.5.3 0.00	3.00	117.88	14197.36
18.5.4 0.00	4.00	119.50	14197.36
18.5.5 0.00	5.00	119.49	14197.36
18.5.6 0.00	6.00	119.94	14197.36
18.5.7 0.00	7.00	120.05	14197.36
18.5.8 0.00	8.00	118.17	14197.36
18.5.9 0.00	9.00	118.15	14197.36
18.5.10	10.00	118.13	14197.36

0.00

18.5.11	11.00	118.15	14197.36
0.00			

18.5.12	12.00	118.25	14197.36
0.00			

18.5.13	13.00	119.85	14197.36
0.00			

18.5.14	14.00	118.14	14197.36
0.00			

18.5.15	15.00	118.14	14197.36
0.00			

18.5.16	16.00	118.17	14197.36
0.00			

18.5.17	17.00	118.15	14197.36
0.00			

18.5.18	18.00	118.13	14197.36
0.00			

18.5.19	19.00	118.15	14197.36
0.00			

18.5.20	20.00	118.25	14197.36
0.00			

18.5.21	21.00	119.85	14197.36
0.00			

18.5.22	22.00	118.14	14197.36
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0.00

18.5.23	23.00	118.14	14197.36
0.00			

18.5.24	24.00	117.58	14197.36
0.00			

18.5.25	25.00	117.68	14197.36
0.00			

18.5.26	26.00	119.44	14197.36
0.00			

18.5.27	27.00	117.88	14197.36
0.00			

18.5.28	28.00	119.50	14197.36
0.00			

18.5.29	29.00	119.49	14197.36
0.00			

18.5.30	30.00	119.94	14197.36
0.00			

18.5.31	31.00	120.05	14197.36
0.00			

18.5.32	32.00	117.58	14197.36
0.00			

18.5.33	33.00	117.68	14197.36
0.00			

18.5.34	34.00	119.44	14197.36
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0.00

18.5.35	35.00	117.88	14197.36
0.00			

18.5.36	36.00	119.50	14197.36
0.00			

18.5.37	37.00	119.49	14197.36
0.00			

18.5.38	38.00	119.94	14197.36
0.00			

18.5.39	39.00	120.05	14197.36
0.00			

18.5.40	40.00	118.17	14197.36
0.00			

18.5.41	41.00	118.14	14197.36
0.00			

18.5.42	42.00	118.13	14197.36
0.00			

18.5.43	43.00	118.14	14197.36
0.00			

18.5.44	44.00	118.24	14197.36
0.00			

18.5.45	45.00	119.85	14197.36
0.00			

18.5.46	46.00	118.14	14197.36
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0.00

18.5.47	47.00	118.14	14197.36
0.00			

18.5.48	48.00	118.17	14197.36
0.00			

18.5.49	49.00	118.14	14197.36
0.00			

18.5.50	50.00	118.13	14197.36
0.00			

18.5.51	51.00	118.14	14197.36
0.00			

18.5.52	52.00	118.24	14197.36
0.00			

18.5.53	53.00	119.85	14197.36
0.00			

18.5.54	54.00	118.14	14197.36
0.00			

18.5.55	55.00	118.14	14197.36
0.00			

18.5.56	56.00	117.58	14197.36
0.00			

18.5.57	57.00	117.68	14197.36
0.00			

18.5.58	58.00	119.43	14197.36
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0.00

18.5.59	59.00	117.88	14197.36
0.00			

18.5.60	60.00	119.50	14197.36
0.00			

18.5.61	61.00	119.49	14197.36
0.00			

18.5.62	62.00	119.94	14197.36
0.00			

18.5.63	63.00	120.05	14197.36
0.00			

18.5.64	64.00	118.17	14197.36
0.00			

18.5.65	65.00	118.14	14197.36
0.00			

18.5.66	66.00	118.13	14197.36
0.00			

18.5.67	67.00	118.14	14197.36
0.00			

18.5.68	68.00	118.24	14197.36
0.00			

18.5.69	69.00	119.85	14197.36
0.00			

18.5.70	70.00	118.14	14197.36
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0.00

18.5.71	71.00	118.14	14197.36
0.00			

18.5.72	72.00	117.58	14197.36
0.00			

18.5.73	73.00	117.68	14197.36
0.00			

18.5.74	74.00	119.44	14197.36
0.00			

18.5.75	75.00	117.88	14197.36
0.00			

18.5.76	76.00	119.50	14197.36
0.00			

18.5.77	77.00	119.49	14197.36
0.00			

18.5.78	78.00	119.94	14197.36
0.00			

18.5.79	79.00	120.05	14197.36
0.00			

18.5.80	80.00	117.58	14197.36
0.00			

18.5.81	81.00	117.68	14197.36
0.00			

18.5.82	82.00	119.44	14197.36
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0.00

18.5.83	83.00	117.88	14197.36
0.00			

18.5.84	84.00	119.50	14197.36
0.00			

18.5.85	85.00	119.49	14197.36
0.00			

18.5.86	86.00	119.94	14197.36
0.00			

18.5.87	87.00	120.05	14197.36
0.00			

18.5.88	88.00	118.17	14197.36
0.00			

18.5.89	89.00	118.14	14197.36
0.00			

18.5.90	90.00	118.13	14197.36
0.00			

18.5.91	91.00	118.14	14197.36
0.00			

18.5.92	92.00	118.24	14197.36
0.00			

18.5.93	93.00	119.85	14197.36
0.00			

18.5.94	94.00	118.14	14197.36
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0.00

18.5.95	95.00	118.14	14197.36
0.00			

18.5.96	96.00	118.20	14197.36
0.00			

18.5.97	97.00	118.18	14197.36
0.00			

18.5.98	98.00	118.17	14197.36
0.00			

18.5.99	99.00	118.18	14197.36
0.00			

18.5.100	100.00	118.27	14197.36
0.00			

18.5.101	101.00	119.89	14197.36
0.00			

18.5.102	102.00	118.16	14197.36
0.00			

18.5.103	103.00	118.16	14197.36
0.00			

18.5.104	104.00	117.71	14197.38
0.00			

18.5.105	105.00	117.85	14197.36
0.00			

18.5.106	106.00	119.58	14197.36
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0.00

18.5.107	107.00	118.03	14197.36
0.00			

18.5.108	108.00	119.64	14197.36
0.00			

18.5.109	109.00	119.64	14197.36
0.00			

18.5.110	110.00	119.97	14197.36
0.00			

18.5.111	111.00	120.08	14197.38
0.00			

18.5.112	112.00	117.59	14197.36
0.00			

18.5.113	113.00	117.69	14197.36
0.00			

18.5.114	114.00	119.44	14197.36
0.00			

18.5.115	115.00	117.89	14197.36
0.00			

18.5.116	116.00	119.50	14197.36
0.00			

18.5.117	117.00	119.49	14197.36
0.00			

18.5.118	118.00	119.94	14197.36
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0.00

18.5.119	119.00	120.05	14197.36
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0.00

18.5.120	120.00	118.17	14197.36
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0.00

18.5.121	121.00	118.15	14197.36
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0.00

18.5.122	122.00	118.13	14197.36
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0.00

18.5.123	123.00	118.15	14197.36
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0.00

18.5.124	124.00	118.25	14197.36
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0.00

18.5.125	125.00	119.85	14197.36
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0.00

18.5.126	126.00	118.14	14197.36
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0.00

18.5.127	127.00	118.14	14197.36
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0.00

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	62.50	476.28	496.98	555.93	1871.00
18.6.1	1.00	0.00	486.32	504.73	580.94	1980.00
18.6.2	2.00	0.00	278.00	489.49	523.05	783.84

18.6.3	3.00	321.00	496.61	519.75	595.71	1852.00
18.6.4	4.00	0.00	0.00	486.82	511.42	1109.75
18.6.5	5.00	0.00	0.00	486.46	511.10	858.48
18.6.6	6.00	0.00	0.00	514.60	542.68	923.66
18.6.7	7.00	0.00	282.25	496.93	525.30	803.64
18.6.8	8.00	184.17	487.45	499.89	566.60	4997.00
18.6.9	9.00	0.00	505.41	511.93	530.34	818.11
18.6.10	10.00	0.00	500.55	511.58	541.60	859.36
18.6.11	11.00	0.00	505.50	517.41	542.94	835.38
18.6.12	12.00	0.00	502.55	516.05	553.71	907.44
18.6.13	13.00	0.00	0.00	495.92	528.90	978.41
18.6.14	14.00	0.00	530.30	540.99	582.89	909.80
18.6.15	15.00	0.00	510.89	521.73	568.72	917.15
18.6.16	16.00	178.67	489.12	500.73	555.21	4992.00
18.6.17	17.00	0.00	505.74	513.31	549.73	872.00
18.6.18	18.00	0.00	497.29	508.56	539.64	961.97
18.6.19	19.00	0.00	505.15	515.49	540.50	993.00
18.6.20	20.00	0.00	501.11	513.55	549.89	990.00
18.6.21	21.00	0.00	0.00	494.95	521.10	924.66
18.6.22	22.00	0.00	530.04	540.12	585.37	962.10
18.6.23	23.00	0.00	510.59	522.10	571.30	1082.00
18.6.24	24.00	61.75	470.98	485.28	555.02	1725.00
18.6.25	25.00	0.00	486.39	506.84	579.73	1894.00
18.6.26	26.00	0.00	278.50	487.01	521.37	788.55

18.6.27	27.00	318.20	495.41	515.36	587.04	1898.00
18.6.28	28.00	0.00	0.00	486.03	512.89	861.33
18.6.29	29.00	0.00	0.00	482.43	506.39	782.45
18.6.30	30.00	0.00	0.00	514.34	540.14	890.13
18.6.31	31.00	0.00	282.75	501.38	523.45	1258.00
18.6.32	32.00	88.50	474.29	496.24	567.55	2644.00
18.6.33	33.00	0.00	498.15	522.33	618.84	2690.00
18.6.34	34.00	0.00	424.25	479.10	519.83	847.15
18.6.35	35.00	249.60	495.12	529.22	609.65	2403.00
18.6.36	36.00	0.00	0.00	485.56	513.55	952.47
18.6.37	37.00	0.00	0.00	483.59	507.63	871.02
18.6.38	38.00	0.00	0.00	503.74	533.92	935.33
18.6.39	39.00	0.00	428.44	491.15	524.08	863.55
18.6.40	40.00	317.80	476.88	492.24	559.43	5442.00
18.6.41	41.00	0.00	488.48	495.13	526.00	850.70
18.6.42	42.00	0.00	495.54	506.35	558.22	945.67
18.6.43	43.00	0.00	496.29	506.54	543.81	1016.31
18.6.44	44.00	0.00	499.13	508.85	538.81	905.00
18.6.45	45.00	0.00	0.00	494.17	519.18	974.39
18.6.46	46.00	0.00	533.87	544.12	583.86	963.59
18.6.47	47.00	0.00	517.25	529.32	577.49	1158.00
18.6.48	48.00	314.60	476.69	492.40	553.56	5565.00
18.6.49	49.00	0.00	488.35	496.05	526.08	845.00
18.6.50	50.00	0.00	496.23	506.21	547.00	880.61

18.6.51	51.00	0.00	496.09	507.69	552.27	970.19
18.6.52	52.00	0.00	497.54	509.66	567.06	931.00
18.6.53	53.00	0.00	0.00	493.67	522.57	870.81
18.6.54	54.00	0.00	532.51	541.90	585.28	927.22
18.6.55	55.00	0.00	514.36	528.71	573.05	1054.00
18.6.56	56.00	88.25	471.70	490.59	544.31	2453.00
18.6.57	57.00	0.00	492.03	515.34	600.98	2470.00
18.6.58	58.00	0.00	421.00	478.41	517.18	779.42
18.6.59	59.00	238.20	493.42	518.19	576.73	2337.00
18.6.60	60.00	0.00	0.00	485.52	513.41	819.15
18.6.61	61.00	0.00	0.00	483.41	514.13	871.44
18.6.62	62.00	0.00	0.00	503.13	531.90	877.38
18.6.63	63.00	0.00	428.00	485.79	513.20	824.43
18.6.64	64.00	234.20	481.11	496.28	558.48	5116.00
18.6.65	65.00	0.00	491.79	500.94	536.19	893.93
18.6.66	66.00	0.00	501.13	510.32	557.68	958.65
18.6.67	67.00	0.00	498.05	509.93	543.34	964.00
18.6.68	68.00	0.00	498.92	512.07	563.77	946.49
18.6.69	69.00	0.00	0.00	495.02	526.63	975.98
18.6.70	70.00	0.00	534.05	550.81	594.82	1099.42
18.6.71	71.00	0.00	514.45	526.65	574.25	1021.51
18.6.72	72.00	148.25	470.84	489.03	556.76	1932.00
18.6.73	73.00	0.00	491.68	516.33	621.57	1917.00
18.6.74	74.00	0.00	291.00	476.88	511.93	885.44

18.6.75	75.00	109.20	492.04	523.78	614.73	1793.00
18.6.76	76.00	0.00	0.00	483.13	515.80	926.20
18.6.77	77.00	0.00	0.00	482.23	515.21	895.40
18.6.78	78.00	0.00	0.00	500.18	535.97	993.83
18.6.79	79.00	0.00	309.50	483.84	521.87	889.95
18.6.80	80.00	147.50	471.53	489.12	559.93	1899.00
18.6.81	81.00	0.00	491.68	516.08	644.66	2122.00
18.6.82	82.00	0.00	298.75	476.59	514.92	940.29
18.6.83	83.00	108.80	493.23	524.79	597.12	1761.00
18.6.84	84.00	0.00	0.00	484.37	510.55	871.97
18.6.85	85.00	0.00	0.00	485.17	509.98	920.33
18.6.86	86.00	0.00	0.00	500.38	538.98	932.04
18.6.87	87.00	0.00	313.94	483.22	507.40	864.98
18.6.88	88.00	233.00	476.66	493.37	569.16	5109.00
18.6.89	89.00	0.00	489.82	498.43	528.05	892.04
18.6.90	90.00	0.00	497.86	508.96	558.13	995.11
18.6.91	91.00	0.00	498.25	509.68	558.88	963.06
18.6.92	92.00	0.00	498.79	510.88	548.53	959.68
18.6.93	93.00	0.00	0.00	497.18	524.90	1054.16
18.6.94	94.00	0.00	533.50	546.33	575.48	964.58
18.6.95	95.00	0.00	513.39	527.49	587.85	996.03
18.6.96	96.00	0.00	488.89	499.17	538.25	4636.00
18.6.97	97.00	0.00	508.45	519.22	539.78	1083.00
18.6.98	98.00	0.00	502.90	517.98	553.42	1520.00

18.6.99	99.00	0.00	508.34	521.89	561.03	959.00
18.6.100	100.00	0.00	506.18	522.68	548.96	1114.00
18.6.101	101.00	0.00	0.00	497.49	527.02	2465.00
18.6.102	102.00	0.00	531.79	543.58	580.08	1379.00
18.6.103	103.00	0.00	512.65	528.24	560.29	1384.00
18.6.104	104.00	0.00	474.88	492.31	546.57	1172.00
18.6.105	105.00	0.00	494.38	513.11	560.28	3186.00
18.6.106	106.00	0.00	146.50	490.39	531.10	825.10
18.6.107	107.00	0.00	503.06	528.39	578.56	1526.00
18.6.108	108.00	0.00	0.00	486.23	514.84	785.07
18.6.109	109.00	0.00	0.00	488.30	514.91	1460.00
18.6.110	110.00	0.00	0.00	516.97	543.08	1050.75
18.6.111	111.00	0.00	161.50	497.71	526.69	2053.50
18.6.112	112.00	0.00	473.90	490.38	548.58	1164.00
18.6.113	113.00	0.00	487.36	507.87	562.73	1474.00
18.6.114	114.00	0.00	153.31	485.51	526.78	921.71
18.6.115	115.00	0.00	498.20	521.00	572.87	1381.00
18.6.116	116.00	0.00	0.00	485.78	515.22	871.21
18.6.117	117.00	0.00	0.00	484.33	509.52	896.85
18.6.118	118.00	0.00	0.00	512.47	547.81	959.32
18.6.119	119.00	0.00	161.80	496.98	524.35	917.79
18.6.120	120.00	124.60	485.31	495.42	541.28	4629.00
18.6.121	121.00	0.00	506.90	515.04	541.37	855.00
18.6.122	122.00	0.00	498.39	509.20	544.99	965.39

18.6.123	123.00	0.00	504.93	517.95	554.22	974.00
18.6.124	124.00	0.00	500.25	514.65	565.63	922.81
18.6.125	125.00	0.00	0.00	495.56	528.06	920.80
18.6.126	126.00	0.00	529.33	542.18	592.39	972.39
18.6.127	127.00	0.00	509.99	522.22	561.40	933.90

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	301.09	334.46	346.69	356.54	489.49
18.7.1	1.00	307.40	329.79	338.06	365.87	497.29
18.7.2	2.00	312.43	346.05	356.61	372.07	533.62
18.7.3	3.00	313.64	344.63	354.45	370.77	529.58
18.7.4	4.00	308.53	350.98	374.71	384.14	535.15
18.7.5	5.00	309.49	350.69	366.54	376.39	520.55
18.7.6	6.00	322.64	371.43	389.29	399.14	560.27
18.7.7	7.00	318.32	365.15	375.92	388.66	563.59
18.7.8	8.00	301.12	343.21	373.49	466.57	666.87
18.7.9	9.00	0.00	362.75	398.24	468.55	666.88
18.7.10	10.00	307.27	350.93	380.16	461.76	667.68
18.7.11	11.00	314.39	365.03	397.95	454.92	668.59
18.7.12	12.00	308.79	353.68	386.24	419.80	625.38
18.7.13	13.00	310.35	355.81	388.40	410.27	618.83
18.7.14	14.00	317.68	358.48	388.10	430.27	605.02
18.7.15	15.00	311.88	361.58	396.61	451.70	665.26
18.7.16	16.00	301.46	343.01	371.15	462.21	664.40

18.7.17	17.00	0.00	362.73	396.88	465.78	663.65
18.7.18	18.00	307.09	350.80	378.14	457.54	662.10
18.7.19	19.00	314.71	364.48	396.00	450.59	662.22
18.7.20	20.00	309.15	354.46	384.38	417.53	612.99
18.7.21	21.00	311.12	356.27	386.72	408.40	611.71
18.7.22	22.00	318.30	358.51	386.12	428.07	600.48
18.7.23	23.00	311.80	361.16	394.71	448.69	661.06
18.7.24	24.00	302.61	334.52	345.80	355.85	499.17
18.7.25	25.00	308.71	328.47	337.24	367.26	504.11
18.7.26	26.00	315.14	346.03	355.72	373.70	508.84
18.7.27	27.00	314.48	343.75	352.75	371.87	506.87
18.7.28	28.00	311.30	356.57	373.23	382.45	534.34
18.7.29	29.00	312.77	352.97	365.70	374.93	514.31
18.7.30	30.00	324.92	374.64	388.13	397.31	551.15
18.7.31	31.00	320.60	364.67	374.82	389.67	541.47
18.7.32	32.00	302.98	343.90	376.55	458.08	661.74
18.7.33	33.00	312.28	361.54	400.05	468.07	658.62
18.7.34	34.00	307.77	348.70	380.56	453.11	659.75
18.7.35	35.00	315.54	363.96	399.99	449.09	635.88
18.7.36	36.00	309.00	353.79	389.06	415.82	600.65
18.7.37	37.00	311.66	355.73	390.44	411.76	612.68
18.7.38	38.00	318.66	366.91	402.19	422.99	641.05
18.7.39	39.00	312.78	354.53	385.83	443.13	658.13
18.7.40	40.00	303.05	335.43	347.42	357.00	499.46

18.7.41	41.00	0.00	331.44	339.80	366.04	505.50
18.7.42	42.00	316.14	344.73	354.68	369.67	515.36
18.7.43	43.00	314.36	343.42	352.60	368.60	511.17
18.7.44	44.00	311.17	356.16	374.26	384.10	547.72
18.7.45	45.00	313.43	353.64	366.64	376.76	541.09
18.7.46	46.00	325.63	374.44	390.17	399.72	558.80
18.7.47	47.00	320.62	360.92	371.03	385.85	539.80
18.7.48	48.00	301.79	330.58	344.19	353.58	502.85
18.7.49	49.00	0.00	327.68	336.74	361.03	523.33
18.7.50	50.00	311.57	340.63	351.03	365.45	502.53
18.7.51	51.00	313.27	340.13	349.67	364.69	501.36
18.7.52	52.00	308.45	347.03	371.63	381.72	538.48
18.7.53	53.00	309.70	347.62	364.11	373.73	526.26
18.7.54	54.00	322.78	364.82	387.55	396.40	551.68
18.7.55	55.00	317.43	356.14	367.96	381.49	523.84
18.7.56	56.00	301.14	341.65	375.02	453.77	657.20
18.7.57	57.00	311.09	358.89	396.96	464.85	654.10
18.7.58	58.00	306.97	347.11	378.45	449.02	658.13
18.7.59	59.00	314.55	361.29	397.87	445.26	628.61
18.7.60	60.00	309.23	351.78	387.54	414.06	587.63
18.7.61	61.00	310.67	353.92	388.30	409.51	588.20
18.7.62	62.00	318.30	364.83	399.93	420.76	629.99
18.7.63	63.00	312.96	352.29	384.23	441.36	654.98
18.7.64	64.00	301.57	339.21	349.84	361.39	512.57

18.7.65	65.00	307.92	332.85	342.17	372.80	539.11
18.7.66	66.00	313.00	347.61	356.74	375.98	515.59
18.7.67	67.00	313.18	346.28	355.24	374.62	515.41
18.7.68	68.00	308.52	359.31	377.66	388.52	540.22
18.7.69	69.00	310.09	356.86	370.31	380.28	516.55
18.7.70	70.00	323.39	377.14	393.09	403.43	549.23
18.7.71	71.00	319.53	362.61	372.78	390.61	542.00
18.7.72	72.00	301.79	352.54	380.88	462.43	657.67
18.7.73	73.00	311.24	371.11	404.49	465.89	663.35
18.7.74	74.00	307.14	357.79	385.00	462.61	664.63
18.7.75	75.00	314.20	372.58	404.94	455.84	661.24
18.7.76	76.00	308.80	362.62	392.80	420.80	644.73
18.7.77	77.00	310.55	365.19	393.89	412.75	630.93
18.7.78	78.00	319.75	377.23	406.39	424.55	618.22
18.7.79	79.00	313.85	362.41	389.72	448.95	656.38
18.7.80	80.00	301.89	350.64	378.96	458.52	653.09
18.7.81	81.00	311.93	369.19	402.96	463.35	659.08
18.7.82	82.00	307.96	355.17	383.01	459.47	659.19
18.7.83	83.00	317.38	370.84	402.77	451.32	655.90
18.7.84	84.00	310.01	361.09	390.91	419.57	646.65
18.7.85	85.00	312.37	363.04	391.91	410.88	626.76
18.7.86	86.00	320.00	374.76	404.54	422.74	622.47
18.7.87	87.00	314.32	359.78	387.54	446.67	652.23
18.7.88	88.00	304.26	339.43	349.57	360.80	497.48

18.7.89	89.00	309.92	331.94	341.79	373.77	515.95
18.7.90	90.00	316.38	347.04	356.84	376.85	511.21
18.7.91	91.00	316.86	345.29	354.58	375.12	507.00
18.7.92	92.00	312.62	362.08	377.18	387.32	545.04
18.7.93	93.00	313.49	357.35	369.33	378.98	526.34
18.7.94	94.00	326.38	378.86	392.34	402.32	554.14
18.7.95	95.00	323.25	362.11	371.88	391.87	529.86
18.7.96	96.00	303.45	345.41	376.46	454.39	659.71
18.7.97	97.00	0.00	363.63	400.96	467.65	660.52
18.7.98	98.00	308.54	352.33	382.38	449.11	663.36
18.7.99	99.00	317.18	365.55	399.88	444.07	665.01
18.7.100	100.00	0.00	354.38	387.64	412.27	614.59
18.7.101	101.00	0.00	357.44	389.36	407.05	587.75
18.7.102	102.00	0.00	359.72	390.18	432.40	632.32
18.7.103	103.00	0.00	362.44	398.73	452.81	651.71
18.7.104	104.00	0.00	333.85	344.43	357.61	485.95
18.7.105	105.00	0.00	328.47	338.65	370.71	481.53
18.7.106	106.00	0.00	345.56	355.11	375.85	1230.00
18.7.107	107.00	0.00	343.00	352.85	374.89	513.50
18.7.108	108.00	0.00	356.28	372.00	382.38	927.00
18.7.109	109.00	0.00	352.16	363.96	375.39	517.34
18.7.110	110.00	0.00	375.47	388.94	397.88	553.94
18.7.111	111.00	0.00	360.45	370.30	392.17	534.29
18.7.112	112.00	302.03	332.57	342.66	356.33	487.21

18.7.113	113.00	0.00	326.65	336.23	369.49	506.92
18.7.114	114.00	314.02	343.37	352.97	374.54	521.82
18.7.115	115.00	0.00	341.70	351.22	373.01	520.13
18.7.116	116.00	308.75	355.01	370.25	380.11	526.76
18.7.117	117.00	0.00	351.69	362.36	373.57	514.97
18.7.118	118.00	323.35	373.87	387.23	396.73	547.14
18.7.119	119.00	0.00	359.58	369.03	390.46	546.01
18.7.120	120.00	301.62	347.09	376.27	450.37	658.55
18.7.121	121.00	0.00	365.71	399.49	464.48	655.24
18.7.122	122.00	307.90	354.03	381.89	445.19	664.29
18.7.123	123.00	315.73	367.66	398.90	440.74	665.33
18.7.124	124.00	308.20	357.02	386.51	409.49	621.06
18.7.125	125.00	0.00	359.86	388.05	404.61	596.37
18.7.126	126.00	309.00	362.02	390.00	429.14	624.87
18.7.127	127.00	0.00	365.32	398.41	450.58	641.71

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00

18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00
18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00

18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00
18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00

18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00
18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00

18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00
18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00

18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00
18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00

18.8.127	127.00	0.00	0.00	0.00	0.00	0.00
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18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.12.0	0.00	0.00	0.00	128.00	128.00	1152.00
18.12.1	1.00	0.00	0.00	128.00	128.00	1152.00
18.12.2	2.00	0.00	0.00	128.00	128.00	1152.00
18.12.3	3.00	0.00	0.00	128.00	128.00	1152.00
18.12.4	4.00	0.00	0.00	128.00	128.00	1152.00
18.12.5	5.00	0.00	0.00	128.00	128.00	1152.00
18.12.6	6.00	0.00	0.00	128.00	128.00	1152.00
18.12.7	7.00	0.00	0.00	128.00	128.00	1152.00
18.12.8	8.00	0.00	0.00	130.00	130.00	1152.00
18.12.9	9.00	0.00	0.00	130.00	130.00	1152.00
18.12.10	10.00	0.00	0.00	130.00	130.00	1152.00
18.12.11	11.00	0.00	0.00	130.00	130.00	1152.00
18.12.12	12.00	0.00	0.00	130.00	130.00	1152.00
18.12.13	13.00	0.00	0.00	130.00	130.00	1152.00
18.12.14	14.00	0.00	0.00	130.00	130.00	1152.00
18.12.15	15.00	0.00	0.00	130.00	130.00	1152.00
18.12.16	16.00	0.00	0.00	130.00	130.00	1152.00
18.12.17	17.00	0.00	0.00	130.00	130.00	1152.00
18.12.18	18.00	0.00	0.00	130.00	130.00	1152.00
18.12.19	19.00	0.00	0.00	130.00	130.00	1152.00
18.12.20	20.00	0.00	0.00	130.00	130.00	1152.00

18.12.21	21.00	0.00	0.00	130.00	130.00	1152.00
18.12.22	22.00	0.00	0.00	130.00	130.00	1152.00
18.12.23	23.00	0.00	0.00	130.00	130.00	1152.00
18.12.24	24.00	0.00	0.00	128.00	128.00	1152.00
18.12.25	25.00	0.00	0.00	128.00	128.00	1152.00
18.12.26	26.00	0.00	0.00	128.00	128.00	1152.00
18.12.27	27.00	0.00	0.00	128.00	128.00	1152.00
18.12.28	28.00	0.00	0.00	128.00	128.00	1152.00
18.12.29	29.00	0.00	0.00	128.00	128.00	1152.00
18.12.30	30.00	0.00	0.00	128.00	128.00	1152.00
18.12.31	31.00	0.00	0.00	128.00	128.00	1152.00
18.12.32	32.00	0.00	0.00	128.00	128.00	1152.00
18.12.33	33.00	0.00	0.00	128.00	128.00	1152.00
18.12.34	34.00	0.00	0.00	128.00	128.00	1152.00
18.12.35	35.00	0.00	0.00	128.00	128.00	1152.00
18.12.36	36.00	0.00	0.00	128.00	128.00	1152.00
18.12.37	37.00	0.00	0.00	128.00	128.00	1152.00
18.12.38	38.00	0.00	0.00	128.00	128.00	1152.00
18.12.39	39.00	0.00	0.00	128.00	128.00	1152.00
18.12.40	40.00	0.00	0.00	130.00	130.00	1152.00
18.12.41	41.00	0.00	0.00	130.00	130.00	1152.00
18.12.42	42.00	0.00	0.00	130.00	130.00	1152.00
18.12.43	43.00	0.00	0.00	130.00	130.00	1152.00
18.12.44	44.00	0.00	0.00	130.00	130.00	1152.00

18.12.45	45.00	0.00	0.00	130.00	130.00	1152.00
18.12.46	46.00	0.00	0.00	130.00	130.00	1152.00
18.12.47	47.00	0.00	0.00	130.00	130.00	1152.00
18.12.48	48.00	0.00	0.00	130.00	130.00	1152.00
18.12.49	49.00	0.00	0.00	130.00	130.00	1152.00
18.12.50	50.00	0.00	0.00	130.00	130.00	1152.00
18.12.51	51.00	0.00	0.00	130.00	130.00	1152.00
18.12.52	52.00	0.00	0.00	130.00	130.00	1152.00
18.12.53	53.00	0.00	0.00	130.00	130.00	1152.00
18.12.54	54.00	0.00	0.00	130.00	130.00	1152.00
18.12.55	55.00	0.00	0.00	130.00	130.00	1152.00
18.12.56	56.00	0.00	0.00	128.00	128.00	1152.00
18.12.57	57.00	0.00	0.00	128.00	128.00	1152.00
18.12.58	58.00	0.00	0.00	128.00	128.00	1152.00
18.12.59	59.00	0.00	0.00	128.00	128.00	1152.00
18.12.60	60.00	0.00	0.00	128.00	128.00	1152.00
18.12.61	61.00	0.00	0.00	128.00	128.00	1152.00
18.12.62	62.00	0.00	0.00	128.00	128.00	1152.00
18.12.63	63.00	0.00	0.00	128.00	128.00	1152.00
18.12.64	64.00	0.00	0.00	130.00	130.00	1152.00
18.12.65	65.00	0.00	0.00	130.00	130.00	1152.00
18.12.66	66.00	0.00	0.00	130.00	130.00	1152.00
18.12.67	67.00	0.00	0.00	130.00	130.00	1152.00
18.12.68	68.00	0.00	0.00	130.00	130.00	1152.00

18.12.69	69.00	0.00	0.00	130.00	130.00	1152.00
18.12.70	70.00	0.00	0.00	130.00	130.00	1152.00
18.12.71	71.00	0.00	0.00	130.00	130.00	1152.00
18.12.72	72.00	0.00	0.00	128.00	128.00	1152.00
18.12.73	73.00	0.00	0.00	128.00	128.00	1152.00
18.12.74	74.00	0.00	0.00	128.00	128.00	1152.00
18.12.75	75.00	0.00	0.00	128.00	128.00	1152.00
18.12.76	76.00	0.00	0.00	128.00	128.00	1152.00
18.12.77	77.00	0.00	0.00	128.00	128.00	1152.00
18.12.78	78.00	0.00	0.00	128.00	128.00	1152.00
18.12.79	79.00	0.00	0.00	128.00	128.00	1152.00
18.12.80	80.00	0.00	0.00	128.00	128.00	1152.00
18.12.81	81.00	0.00	0.00	128.00	128.00	1152.00
18.12.82	82.00	0.00	0.00	128.00	128.00	1152.00
18.12.83	83.00	0.00	0.00	128.00	128.00	1152.00
18.12.84	84.00	0.00	0.00	128.00	128.00	1152.00
18.12.85	85.00	0.00	0.00	128.00	128.00	1152.00
18.12.86	86.00	0.00	0.00	128.00	128.00	1152.00
18.12.87	87.00	0.00	0.00	128.00	128.00	1152.00
18.12.88	88.00	0.00	0.00	130.00	130.00	1152.00
18.12.89	89.00	0.00	0.00	130.00	130.00	1152.00
18.12.90	90.00	0.00	0.00	130.00	130.00	1152.00
18.12.91	91.00	0.00	0.00	130.00	130.00	1152.00
18.12.92	92.00	0.00	0.00	130.00	130.00	1152.00

18.12.93	93.00	0.00	0.00	130.00	130.00	1152.00
18.12.94	94.00	0.00	0.00	130.00	130.00	1152.00
18.12.95	95.00	0.00	0.00	130.00	130.00	1152.00
18.12.96	96.00	0.00	0.00	130.00	130.00	1152.00
18.12.97	97.00	0.00	0.00	130.00	130.00	1152.00
18.12.98	98.00	0.00	0.00	130.00	130.00	1152.00
18.12.99	99.00	0.00	0.00	130.00	130.00	1152.00
18.12.100	100.00	0.00	0.00	130.00	130.00	1152.00
18.12.101	101.00	0.00	0.00	130.00	130.00	1152.00
18.12.102	102.00	0.00	0.00	130.00	130.00	1152.00
18.12.103	103.00	0.00	0.00	130.00	130.00	1152.00
18.12.104	104.00	0.00	0.00	128.00	128.00	1152.00
18.12.105	105.00	0.00	0.00	128.00	128.00	1152.00
18.12.106	106.00	0.00	0.00	128.00	128.00	1152.00
18.12.107	107.00	0.00	0.00	128.00	128.00	1152.00
18.12.108	108.00	0.00	0.00	128.00	128.00	1152.00
18.12.109	109.00	0.00	0.00	128.00	128.00	1152.00
18.12.110	110.00	0.00	0.00	128.00	128.00	1152.00
18.12.111	111.00	0.00	0.00	128.00	128.00	1152.00
18.12.112	112.00	0.00	0.00	128.00	128.00	1152.00
18.12.113	113.00	0.00	0.00	128.00	128.00	1152.00
18.12.114	114.00	0.00	0.00	128.00	128.00	1152.00
18.12.115	115.00	0.00	0.00	128.00	128.00	1152.00
18.12.116	116.00	0.00	0.00	128.00	128.00	1152.00

18.12.117	117.00	0.00	0.00	128.00	128.00	1152.00
18.12.118	118.00	0.00	0.00	128.00	128.00	1152.00
18.12.119	119.00	0.00	0.00	128.00	128.00	1152.00
18.12.120	120.00	0.00	0.00	130.00	130.00	1152.00
18.12.121	121.00	0.00	0.00	130.00	130.00	1152.00
18.12.122	122.00	0.00	0.00	130.00	130.00	1152.00
18.12.123	123.00	0.00	0.00	130.00	130.00	1152.00
18.12.124	124.00	0.00	0.00	130.00	130.00	1152.00
18.12.125	125.00	0.00	0.00	130.00	130.00	1152.00
18.12.126	126.00	0.00	0.00	130.00	130.00	1152.00
18.12.127	127.00	0.00	0.00	130.00	130.00	1152.00

1.4 Fusion MatMulBias Kernel

```
[9]: %%writefile fusion_matmulbias.py
import argparse
import torch

import triton
import triton.language as tl
import triton.profiler as proton

def _matmul_launch_metadata(grid, kernel, args):
    ret = {}
    M, N, K, WS = args["M"], args["N"], args["K"], args.get("WARP_SPECIALIZE",
↪False)
    BM, BN, BK = args["BLOCK_SIZE_M"], args["BLOCK_SIZE_N"],
↪args["BLOCK_SIZE_K"]
    ws_str = "_ws" if WS else ""
    ret["name"] = f"{kernel.name}{ws_str} [M={M}, N={N}, K={K}] [BM={BM},
↪BN={BN}, BK={BK}] "
    if "output_ptr" in args:
        bytes_per_elem = args["output_ptr"].element_size()
```

```

else:
    bytes_per_elem = 2
    ret[f"flops{bytes_per_elem * 8}"] = 2. * M * N * K
    ret["bytes"] = bytes_per_elem * (M * K + N * K + M * N)
    return ret

def matmul_autotune_config(pre_hook=None):
    return [
        triton.Config({'BLOCK_SIZE_M': BM, 'BLOCK_SIZE_N': BN, "BLOCK_SIZE_K": BK, "GROUP_SIZE_M": 8}, num_stages=s,
            num_warps=w, pre_hook=pre_hook)
        for BM in [128]
        for BN in [128, 256]
        for BK in [64, 128]
        for s in ([3, 4, 5])
        for w in [4, 8]
    ]

# MatMulBias Fusion kernel
@triton.autotune(
    configs=matmul_autotune_config(),
    key=['M', 'N', 'K'],
)
@triton.jit(launch_metadata=_matmul_launch_metadata)
def matmulbias_kernel(
    a_ptr, b_ptr, c_ptr,
    bias_ptr,
    M, N, K,
    stride_am, stride_ak,
    stride_bk, stride_bn,
    stride_cm, stride_cn,
    BLOCK_SIZE_M: tl.constexpr, BLOCK_SIZE_N: tl.constexpr, BLOCK_SIZE_K: tl.constexpr,
    GROUP_SIZE_M: tl.constexpr
):
    pid = tl.program_id(axis=0)
    num_pid_m = tl.cdiv(M, BLOCK_SIZE_M)
    num_pid_n = tl.cdiv(N, BLOCK_SIZE_N)
    num_pid_in_group = GROUP_SIZE_M * num_pid_n
    group_id = pid // num_pid_in_group
    first_pid_m = group_id * GROUP_SIZE_M
    group_size_m = min(num_pid_m - first_pid_m, GROUP_SIZE_M)
    pid_m = first_pid_m + ((pid % num_pid_in_group) % group_size_m)
    pid_n = (pid % num_pid_in_group) // group_size_m

    tl.assume(pid_m >= 0)
    tl.assume(pid_n >= 0)

```

```

tl.assume(stride_am > 0)
tl.assume(stride_ak > 0)
tl.assume(stride_bn > 0)
tl.assume(stride_bk > 0)
tl.assume(stride_cm > 0)
tl.assume(stride_cn > 0)

offs_am = (pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)) % M
offs_bn = (pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)) % N
offs_k = tl.arange(0, BLOCK_SIZE_K)
a_ptrs = a_ptr + (offs_am[:, None] * stride_am + offs_k[None, :] *
↪stride_ak)
b_ptrs = b_ptr + (offs_k[:, None] * stride_bk + offs_bn[None, :] *
↪stride_bn)

accumulator = tl.zeros((BLOCK_SIZE_M, BLOCK_SIZE_N), dtype=tl.float32)
for k in range(0, tl.cdiv(K, BLOCK_SIZE_K)):
    a = tl.load(a_ptrs, mask=offs_k[None, :] < K - k * BLOCK_SIZE_K,
↪other=0.0)
    b = tl.load(b_ptrs, mask=offs_k[:, None] < K - k * BLOCK_SIZE_K,
↪other=0.0)
    accumulator = tl.dot(a, b, accumulator)
    a_ptrs += BLOCK_SIZE_K * stride_ak
    b_ptrs += BLOCK_SIZE_K * stride_bk
c = accumulator.to(tl.float16)

bias = tl.load(bias_ptr + offs_bn, mask=offs_bn < N, other=0.0)
output = c + bias[None, :]

offs_cm = pid_m * BLOCK_SIZE_M + tl.arange(0, BLOCK_SIZE_M)
offs_cn = pid_n * BLOCK_SIZE_N + tl.arange(0, BLOCK_SIZE_N)
c_ptrs = c_ptr + stride_cm * offs_cm[:, None] + stride_cn * offs_cn[None, :]
c_mask = (offs_cm[:, None] < M) & (offs_cn[None, :] < N)
tl.store(c_ptrs, output, mask=c_mask)

# MatMulBias kernel wrapper function
def matmulbiasfusion(a: torch.Tensor, b: torch.Tensor, bias: torch.Tensor):
    assert a.shape[1] == b.shape[0], "Incompatible dimensions"
    assert a.is_contiguous(), "Matrix A must be contiguous"
    M, K = a.shape
    K, N = b.shape
    assert bias.shape[0] == N, "BIAS has incompatible dimensions"
    o = torch.empty((M, N), device=a.device, dtype=torch.float16)
    grid = lambda META: (triton.cdiv(M, META['BLOCK_SIZE_M']) * triton.cdiv(N,
↪META['BLOCK_SIZE_N']), )
    matmulbias_kernel[grid](

```

```

        a, b, o,
        bias,
        M, N, K,
        a.stride(0), a.stride(1),
        b.stride(0), b.stride(1),
        o.stride(0), o.stride(1),
    )

    return o

def get_rtol():
    target = triton.runtime.driver.active.get_current_target()
    if target.backend == "hip":
        if target.arch == "gfx90a":
            return 1e-2
        elif target.arch == "gfx942":
            return 1e-2
    else:
        return 0

def show_profile(profile_name):
    import triton.profiler.viewer as proton_viewer
    metric_names = ["time/ms", "tflop16/s"]
    file_name = f"{profile_name}.hatchet"
    tree, metrics = proton_viewer.parse(metric_names, file_name)

    print(f"Proton profile results for {profile_name}")
    proton_viewer.print_tree(tree, metrics)

if __name__ == "__main__":
    parser = argparse.ArgumentParser()
    parser.add_argument("--profile", action="store_true")
    parser.add_argument("--verify", action="store_true")
    args = parser.parse_args()

    # Test matrices
    torch.manual_seed(0)
    M = 1024
    N = 1024
    a = torch.randn((M, N), device='cuda', dtype=torch.float16)
    b = torch.randn((N, M), device='cuda', dtype=torch.float16)
    bias = torch.randn(N, device='cuda', dtype=torch.float16)

    if args.profile:
        print("Profiling the fusion matmulbias kernel ...")

```



```

        proton.start("fusion_matmulbias", hook="triton")
    else:
        print("Running the fusion matmulbias kernel ...")
        fusion_triton_output = matmulbiasfusion(a, b, bias)
    if args.profile:
        proton.finalize()
    print(f"fusion_triton_output: {fusion_triton_output}")

    if args.profile:
        show_profile("fusion_matmulbias")

    if args.verify:
        torch_output = torch.addmm(bias, a, b)
        print(f"torch_output: {torch_output}")
        print("Verifying triton results with torch ...")
        triton.testing.assert_close(fusion_triton_output, torch_output,
        atol=1e-2, rtol=get_rtol())
        print("OK")

```

Overwriting fusion_matmulbias.py

1.4.1 Pre-Build the Fusion MatMulBias Kernel

```
[10]: !python fusion_matmulbias.py --verify
```

```

Running the fusion matmulbias kernel ...
fusion_triton_output: tensor([[ 2.7480, 43.6250, 30.9531, ..., 26.2500,
 8.8203,  6.4609],
 [ 7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],
 [13.8125, -6.4609,  0.5146, ..., 48.0000, 25.1562, -27.5156],
 ...,
 [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
 [ 9.6094, -52.2500, 43.0000, ..., -45.8438, -49.3125, 32.6875],
 [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1250]],
       device='cuda:0', dtype=torch.float16)
torch_output: tensor([[ 2.7480, 43.6562, 30.9531, ..., 26.2500,  8.8203,
 6.4609],
 [ 7.1602, -21.3906, 17.7969, ..., -16.8125, 29.6406, -48.3125],
 [13.8125, -6.4609,  0.5142, ..., 48.0312, 25.1406, -27.5156],
 ...,
 [-54.4375, -1.4375, -0.4214, ...,  5.4297, 17.2812, -39.0625],
 [ 9.6094, -52.2500, 43.0000, ..., -45.8125, -49.3125, 32.7188],
 [-12.8516,  9.5391, 53.3750, ..., -9.3438,  7.6797, 54.1562]],
       device='cuda:0', dtype=torch.float16)
Verifying triton results with torch ...
OK

```

Using Triton Proton to generate the report

Profiling the fusion matmulbias kernel ...

Proton profile results for fusion_matmulbias

```
40.412 89.754 matmulbias_kernel [M=1024, N=1024,
K=1024] [BM=128, BN=128, BK=64]
```

40.41 - 46.07

Only in right graph

```
[12]: !ROCPROF=rocprofiler-sdk rocprof-compute profile -n fusion_matmulbias -- python.
      ↪ fusion_matmulbias.py
```

[illegible]

|_ |

|_ |

WARNING DEPRECATION WARNING: rocm-smi is deprecated in ROCm 7.0 and will be removed from rocprof-compute in ROCm 7.1. Please migrate to amd-smi for compute partition parsing. For migration help, see

<https://github.com/ROCm/amdsmi>

INFO Rocprofiler-Compute version: 3.2.3

INFO Profiler choice: rocprofiler-sdk

INFO Path: /workspace/user/workloads/fusion_matmulbias/MI300X_A1

INFO Target: MI300X_A1

INFO Command: python fusion_matmulbias.py

INFO Kernel Selection: None

INFO Dispatch Selection: None

INFO Hardware Blocks: []

INFO Report Sections: []

INFO

INFO ~~~~~

INFO Collecting Performance Counters

INFO ~~~~~

INFO

INFO [Run 1/12][Approximate profiling time left: pending first measurement...]

INFO [profiling] Current input file: /workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/SQ_IFETCH_LEVEL.txt

INFO |-> [rocprofiler-sdk] W20260112 19:10:42.982423

139694953385536 simple_timer.cpp:55] [rocprofv3] tool initialization ::

0.178938 sec

INFO |-> [rocprofiler-sdk] W20260112 19:10:42.982535

139694953385536 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'

:: 0.000000 sec

INFO |-> [rocprofiler-sdk] W20260112 19:10:44.252752

139694953385536 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO |-> [rocprofiler-sdk] Running the fusion matmulbias kernel ...

INFO |-> [rocprofiler-sdk] fusion_triton_output: tensor([[2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],

INFO |-> [rocprofiler-sdk] [7.1562, -21.3906, 17.7969, ..., -16.8125, 29.6562, -48.3125],

INFO |-> [rocprofiler-sdk] [13.8125, -6.4609, 0.5146, ..., 48.0000, 25.1562, -27.5156],

INFO |-> [rocprofiler-sdk] ...,

INFO |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ..., 5.4297, 17.2812, -39.0625],

```

INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO      |-> [rocprofiler-sdk] W20260112 19:10:45.533803

139694953385536 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.551267 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:10:45.546734

139694953385536 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/634_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:10:45.560455

139694953385536 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/634_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260112 19:10:45.879358

139694953385536 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/634_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:10:45.882235

139694953385536 simple_timer.cpp:55] [rocprofv3] output generation ::
0.343273 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:10:45.883314

139694953385536 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.344416 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 2/12][Approximate profiling time left: 53 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_LDS.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:10:48.413509
139865877130816 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.179497 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:10:48.413617
139865877130816 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:10:49.684479
139865877130816 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:10:50.965728
139865877130816 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.552111 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:10:50.978685
139865877130816 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/654_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:10:50.992469
139865877130816 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/654_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260112 19:10:51.305900
139865877130816 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/654_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:10:51.308772
139865877130816 simple_timer.cpp:55] [rocprofv3] output generation ::
0.337847 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:10:51.309881
139865877130816 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.339024 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 3/12] [Approximate profiling time left: 48 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_SMEM.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:10:53.441526
140005757541952 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.133778 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:10:53.441634
140005757541952 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:10:54.711762
140005757541952 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[

```

```

2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260112 19:10:55.975325

140005757541952 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.533692 sec

INFO    |-> [rocprofiler-sdk] E20260112 19:10:55.988279

140005757541952 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/675_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260112 19:10:56.002198

140005757541952 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/675_counter_collection.c
sv

INFO    |-> [rocprofiler-sdk] E20260112 19:10:56.023621

140005757541952 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/675_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260112 19:10:56.026561

140005757541952 simple_timer.cpp:55] [rocprofv3] output generation ::
0.046062 sec

INFO    |-> [rocprofiler-sdk] W20260112 19:10:56.027530

140005757541952 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.047099 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 4/12][Approximate profiling time left: 39 seconds]...
INFO [profiling] Current input file: /workspace/user/workloads/fusio
n_matmulbias/MI300X_A1/perfmon/SQ_INST_LEVEL_VMEM.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:10:57.508652
140037717196352 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.153469 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:10:57.508764
140037717196352 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:10:58.784006
140037717196352 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```



```

INFO      |-> [rocprofiler-sdk] W20260112 19:11:00.057595
140037717196352 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.548831 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:11:00.070523
140037717196352 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/695_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:00.084310
140037717196352 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/695_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:00.260079
140037717196352 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/695_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:11:00.262908
140037717196352 simple_timer.cpp:55] [rocprofv3] output generation ::
0.200110 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:00.263943
140037717196352 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.201217 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 5/12] [Approximate profiling time left: 33 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/SQ_LEVEL_WAVES.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:11:02.368980
140223726001728 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.178514 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:02.369089
140223726001728 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:03.641553
140223726001728 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[

```

```

2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260112 19:11:04.923924

140223726001728 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.554835 sec

INFO    |-> [rocprofiler-sdk] E20260112 19:11:04.936830
140223726001728 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/716_kernel_trace.csv
INFO    |-> [rocprofiler-sdk] E20260112 19:11:04.950593
140223726001728 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/716_counter_collection.c
sv
INFO    |-> [rocprofiler-sdk] E20260112 19:11:05.262430
140223726001728 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/716_agent_info.csv
INFO    |-> [rocprofiler-sdk] W20260112 19:11:05.265231
140223726001728 simple_timer.cpp:55] [rocprofv3] output generation ::
0.336110 sec
INFO    |-> [rocprofiler-sdk] W20260112 19:11:05.266339
140223726001728 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.337285 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 6/12][Approximate profiling time left: 29 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_0.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:11:07.349460
139807269813824 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127938 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:07.349569
139807269813824 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:08.618438
139807269813824 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:11:09.878896
139807269813824 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.529328 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:11:09.891823
139807269813824 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/736_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:09.905707
139807269813824 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/736_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:09.925880
139807269813824 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/736_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:11:09.928819
139807269813824 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044715 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:09.929752
139807269813824 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045712 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 7/12] [Approximate profiling time left: 23 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_1.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:11:11.238917
140375007215168 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128909 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:11.239027
140375007215168 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:12.504989
140375007215168 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[

```

```

2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260112 19:11:13.770026

140375007215168 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.530999 sec

INFO    |-> [rocprofiler-sdk] E20260112 19:11:13.783227

140375007215168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/757_kernel_trace.csv

INFO    |-> [rocprofiler-sdk] E20260112 19:11:13.797015

140375007215168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/757_counter_collection.c
sv

INFO    |-> [rocprofiler-sdk] E20260112 19:11:13.816801

140375007215168 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/757_agent_info.csv

INFO    |-> [rocprofiler-sdk] W20260112 19:11:13.819701

140375007215168 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044255 sec

INFO    |-> [rocprofiler-sdk] W20260112 19:11:13.820658

140375007215168 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045284 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 8/12][Approximate profiling time left: 18 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_2.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:11:15.126360
140373093432896 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128422 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:15.126465
140373093432896 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:16.392985
140373093432896 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:11:17.655172
140373093432896 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.528708 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:11:17.668124
140373093432896 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/777_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:17.681977
140373093432896 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/777_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:17.701729
140373093432896 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/777_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:11:17.704630
140373093432896 simple_timer.cpp:55] [rocprofv3] output generation ::
0.044223 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:17.705608
140373093432896 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.045271 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 9/12] [Approximate profiling time left: 13 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_3.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:11:19.024762
140183227452992 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128157 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:19.024866
140183227452992 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:20.290368
140183227452992 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[

```

```

2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260112 19:11:21.539067

140183227452992 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.514201 sec

INFO    |-> [rocprofiler-sdk] E20260112 19:11:21.552083
140183227452992 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/797_kernel_trace.csv
INFO    |-> [rocprofiler-sdk] E20260112 19:11:21.566380
140183227452992 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/797_counter_collection.c
sv
INFO    |-> [rocprofiler-sdk] E20260112 19:11:21.587073
140183227452992 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/797_agent_info.csv
INFO    |-> [rocprofiler-sdk] W20260112 19:11:21.589964
140183227452992 simple_timer.cpp:55] [rocprofv3] output generation ::
0.045631 sec
INFO    |-> [rocprofiler-sdk] W20260112 19:11:21.590883
140183227452992 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.046613 sec
INFO    |-> [rocprofiler-sdk]
INFO [Run 10/12] [Approximate profiling time left: 8 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_4.txt

```



```

INFO      |-> [rocprofiler-sdk] W20260112 19:11:22.916446
140124666273344 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.127991 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:22.916547
140124666273344 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:24.181625
140124666273344 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:11:25.414653
140124666273344 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.498106 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:11:25.427668
140124666273344 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/818_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:25.442572
140124666273344 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/818_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:25.466927
140124666273344 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/818_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:11:25.469834
140124666273344 simple_timer.cpp:55] [rocprofv3] output generation ::
0.049927 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:25.470802
140124666273344 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.050962 sec

INFO      |-> [rocprofiler-sdk]
INFO [Run 11/12] [Approximate profiling time left: 4 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_5.txt
INFO      |-> [rocprofiler-sdk] W20260112 19:11:26.803597
140008333934144 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.129104 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:26.803705
140008333934144 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:28.071142
140008333934144 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[

```

```

2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO    |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO    |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO    |-> [rocprofiler-sdk] ...,
INFO    |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO    |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO    |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO    |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)
INFO    |-> [rocprofiler-sdk] W20260112 19:11:29.300006

140008333934144 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.496301 sec

INFO    |-> [rocprofiler-sdk] E20260112 19:11:29.313001
140008333934144 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/838_kernel_trace.csv
INFO    |-> [rocprofiler-sdk] E20260112 19:11:29.327910
140008333934144 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/838_counter_collection.c
sv
INFO    |-> [rocprofiler-sdk] E20260112 19:11:29.352454
140008333934144 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/838_agent_info.csv
INFO    |-> [rocprofiler-sdk] W20260112 19:11:29.355376
140008333934144 simple_timer.cpp:55] [rocprofv3] output generation ::
0.050099 sec
INFO    |-> [rocprofiler-sdk] W20260112 19:11:29.356289
140008333934144 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.051080 sec

INFO    |-> [rocprofiler-sdk]
INFO [Run 12/12] [Approximate profiling time left: 0 seconds]...
INFO [profiling] Current input file:
/workspace/user/workloads/fusion_matmulbias/MI300X_A1/perfmon/pmc_perf_6.txt

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:11:30.690381
140216934547008 simple_timer.cpp:55] [rocprofv3] tool initialization ::
0.128003 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:30.690486
140216934547008 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      0.000000 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:31.957871
140216934547008 tool.cpp:2203] HSA version 8.18.3 initialized (instance=0)

INFO      |-> [rocprofiler-sdk] Running the fusion matmulbias
kernel ...
INFO      |-> [rocprofiler-sdk] fusion_triton_output: tensor([[
2.7480, 43.6250, 30.9531, ..., 26.2500, 8.8203, 6.4609],
INFO      |-> [rocprofiler-sdk] [ 7.1562, -21.3906, 17.7969, ...,
-16.8125, 29.6562, -48.3125],
INFO      |-> [rocprofiler-sdk] [ 13.8125, -6.4609, 0.5146, ...,
48.0000, 25.1562, -27.5156],
INFO      |-> [rocprofiler-sdk] ...,
INFO      |-> [rocprofiler-sdk] [-54.4375, -1.4375, -0.4214, ...,
5.4297, 17.2812, -39.0625],
INFO      |-> [rocprofiler-sdk] [ 9.6094, -52.2500, 43.0000, ...,
-45.8438, -49.3125, 32.6875],
INFO      |-> [rocprofiler-sdk] [-12.8516, 9.5391, 53.3750, ...,
-9.3438, 7.6797, 54.1250]],
INFO      |-> [rocprofiler-sdk] device='cuda:0', dtype=torch.float16)

```

```

INFO      |-> [rocprofiler-sdk] W20260112 19:11:33.170761
140216934547008 simple_timer.cpp:55] [rocprofv3] 'python fusion_matmulbias.py'
::      2.480276 sec

INFO      |-> [rocprofiler-sdk] E20260112 19:11:33.183702
140216934547008 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/858_kernel_trace.csv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:33.199164
140216934547008 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/858_counter_collection.c
sv

INFO      |-> [rocprofiler-sdk] E20260112 19:11:33.216601
140216934547008 output_stream.cpp:111] Opened result file: /workspace/user/workl
oads/fusion_matmulbias/MI300X_A1/out/pmc_1/f57b369e0635/858_agent_info.csv

INFO      |-> [rocprofiler-sdk] W20260112 19:11:33.219509
140216934547008 simple_timer.cpp:55] [rocprofv3] output generation ::
0.043568 sec

INFO      |-> [rocprofiler-sdk] W20260112 19:11:33.220458
140216934547008 simple_timer.cpp:55] [rocprofv3] tool finalization ::
0.044581 sec

INFO      |-> [rocprofiler-sdk]
INFO [roofline] Checking for roofline.csv in
/workspace/user/workloads/fusion_matmulbias/MI300X_A1
INFO [roofline] No roofline data found. Generating...
Empirical Roofline Calculation
Copyright © 2025 Advanced Micro Devices, Inc. All rights reserved.
Total detected GPU devices: 1
GPU Device 0 (gfx942) with 304 CUs: Profiling...
99% [||||| ]
HBM BW, GPU ID: 0, workgroupSize:256, workgroups:6225920, experiments:100,
traffic:25501368320 bytes, duration:6.0 ms, mean:4211.2 GB/sec, stdev=7.7 GB/sec
99% [||||| ]
MALL BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:2611340115968 bytes, duration:415.1 ms, mean:6304.7 GB/sec, stdev=16.9
GB/sec
99% [||||| ]
L2 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:1632087572480 bytes, duration:63.5 ms, mean:25758.7 GB/sec, stdev=40.0
GB/sec

```

```

99% [||||| ]
L1 BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:127506841600 bytes, duration:3.7 ms, mean:34173.9 GB/sec, stdev=174.8
GB/sec
99% [||||| ]
LDS BW, GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100,
traffic:79691776000 bytes, duration:1.3 ms, mean:63345.9 GB/sec, stdev=227.6
GB/sec
99% [||||| ]
Peak FLOPs (FP8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:1081258016768, duration:13.0 ms, mean:83034.7 GFLOPS,
stdev=171.9 GFLOPS
99% [||||| ]
Peak FLOPs (FP16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:13.4 ms, mean:39590.8 GFLOPS,
stdev=47.5 GFLOPS
99% [||||| ]
Peak FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:530428461056, duration:91.534 ms, mean:5792.8 GFLOPS,
stdev=9.9 GFLOPS
99% [||||| ]
Peak FLOPs (FP32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:265214230528, duration:2.040 ms, mean:131205.1 GFLOPS,
stdev=1642.5 GFLOPS
99% [||||| ]
Peak FLOPs (FP64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, FLOP:122406567936, duration:1.1 ms, mean:105952.7 GFLOPS,
stdev=1558.7 GFLOPS
99% [||||| ]
Peak IOPs (INT8), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:1081258016768, duration:13.1 ms, mean:82797.1 GOPS,
stdev=168.5 GOPS
99% [||||| ]
Peak IOPs (INT32), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:265214230528, duration:3.7 ms, mean:72332.7 GOPS,
stdev=528.7 GOPS
99% [||||| ]
Peak IOPs (INT64), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:100, IOP:122406567936, duration:6.2 ms, mean:19818.7 GOPS,
stdev=54.2 GOPS
100% [||||| ]
Peak MFMA FLOPs (F4), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
100% [||||| ]
Peak MFMA FLOPs (F6), GPU ID: 0, workgroupSize:256, workgroups:38912,
experiments:1, FLOP:0, duration:0.0 ms, mean:0.0 GFLOPS, stdev=0.0 GFLOPS
99% [||||| ]
Peak MFMA FLOPs (F8), GPU ID: 0, workgroupSize:256, workgroups:38912,

```

experiments:100, FLOP:10200547328000, duration:5.3 ms, mean:1920381.2 GFLOPS, stdev=16696.4 GFLOPS

99% [||]
Peak MFMA FLOPs (F16), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, FLOP:5100273664000, duration:5.4 ms, mean:940318.3 GFLOPS, stdev=3529.5 GFLOPS

99% [||]
Peak MFMA FLOPs (BF16), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, FLOP:2550136832000, duration:4.8 ms, mean:531152.1 GFLOPS, stdev=2027.0 GFLOPS

99% [||]
Peak MFMA FLOPs (F32), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, FLOP:1275068416000, duration:9.1 ms, mean:140669.2 GFLOPS, stdev=499.1 GFLOPS

99% [||]
Peak MFMA FLOPs (F64), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, FLOP:637534208000, duration:4.4 ms, mean:146309.1 GFLOPS, stdev=692.3 GFLOPS

99% [||]
Peak MFMA IOPs (I8), GPU ID: 0, workgroupSize:256, workgroups:38912, experiments:100, IOP:5100273664000, duration:5.1 ms, mean:1002526.8 GOPS, stdev=5434.5 GOPS

INFO [roofline] Empirical Roofline PDFs saved!

1.4.3 Analyze the Fusion MatMulBias Kernel

Using Triton Proton to analyze the report

```
[13]: !proton-viewer -m flops16,time/s -f full fusion_matmulbias.hatchet
```

```
3627099881472.000 0.097 ROOT
nan 0.057 _ZN2at6native29vectorized_elementwise_kernelILi4EN
SO_11FillFunctorIiEESt5arrayIPcLm1EEEEviT0_T1_
3627099881472.000 0.040 matmulbias_kernel [M=1024,
N=1024, K=1024] [BM=128, BN=128, BK=64]
```

Legend (Metric: flops16 (inc) Min: 3627099881472.00 Max:

3627099881472.00)
3627099881472.00 - 3627099881472.00
3627099881472.00 - 3627099881472.00
3627099881472.00 - 3627099881472.00
3627099881472.00 - 3627099881472.00
3627099881472.00 - 3627099881472.00
3627099881472.00 - 3627099881472.00

name User code Only in left graph
Only in right graph

Using ROCm Systems Compute to analyze the report

```
[14]: !rocprof-compute analyze -p workloads/fusion_matmulbias/MI300X_A1
```

```

----- /_ | ----- _ |_
_/_ / _ \ / _ | ' \ | ' _ / _ \ | _ _ _ _ / _ / _ \ | ' ` _ \ | ' \ | | | _ / _
\
| | | ( ) | ( _ | | ) | | | ( ) | _ | _ _ _ | ( _ | ( ) | | | | | | | ) | | | | |
_/_/
|_ | \_/_ / \_/_ | ._/_ / | | \_/_ / | | \_/_ \_/_ / | | | | | ._/_ /
\_/_ , _ | \_/_ \_/_ |
          | |
          | |

```

```
INFO Analysis mode = cli
```

```
INFO [analysis] deriving rocprofiler-compute metrics...
```

```
WARNING PC sampling: can not detect pc sampling method without /workspa
ce/user/workloads/fusion_matmulbias/MI300X_A1/ps_file_pc_sampling_host_trap.csv
```

```
-----
0. Top Stats
```

```
0.1 Top Kernels
```

	Kernel_Name	Count	Sum(ns)
Mean(ns)	Median(ns)	Pct	
0	void at::native::vectorized_elementwise_43282.56 42377.83 54.76 kernel<4, at::native::FillFunctor<int>, std::array<char*, 1ul> >(int, at::nat...	138.00	5972993.06
1	matmulbias_kernel 26518.38 24777.88 41.82	172.00	4561161.56
2	__amd_rocclr_copyBuffer 3047.06 3010.19 2.49	89.00	271188.56
3	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_ker 6322.33 7346.91 0.17	3.00	18967.00


```

nel<float, 4, at::native::templates::...

4 void at::native::(anonymous namespace)::      7.00      16323.62
2331.95      2225.09      0.15
  CatArrayBatchedCopy_contig<at::native::(
    anonymous namespace)::OpaqueType<2u>,...

5 void at::native::vectorized_elementwise_      3.00      10426.97
3475.66      3481.41      0.10
  kernel<4, at::native::compare_scalar_ker
    nel<double>(at::TensorIteratorBase&, ...

6 void at::native::vectorized_elementwise_      2.00      6224.28
3112.14      3112.14      0.06
  kernel<8, at::native::AUnaryFunctor<c10:
    :Half, c10::Half, bool, at::native::(...

7 void at::native::vectorized_elementwise_      2.00      5245.19
2622.59      2622.59      0.05
  kernel<8, at::native::AbsFunctor<c10::Ha
    lf>, std::array<char*, 2ul> >(int, at...

8 void at::native::vectorized_elementwise_      2.00      5228.25
2614.12      2614.12      0.05
  kernel<8, at::native::BinaryFunctor<c10:
    :Half, c10::Half, bool, at::native::(...

9 void at::native::reduce_kernel<512, 1, a      1.00      4349.59
4349.59      4349.59      0.04
  t::native::ReduceOp<double, at::native::

```

```
func_wrapper_t<double, at::native::Ma...
```

0.2 Dispatch List

Dispatch_ID	Kernel_Name	GPU_ID
0	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
1	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
2	void at::native::(anonymous namespace)::distribution_elementwise_grid_stride_kernel<float, 4, at::native::templates::cuda::normal_and_transform<c10::Half, float, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel<at::CUDAGeneratorImpl*>(at::TensorBase const&, double, double, at::CUDAGeneratorImpl*)::...	0
3	matmulbias_kernel	0
4	void at::native::vectorized_elementwise_kernel<4, at::native::FillFunctor<int>,	0

```

        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
    1>)

5          5  matmulbias_kernel
    0

6          6  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
    1>)

7          7  matmulbias_kernel
    0

8          8  void at::native::vectorized_elementwise_kernel<4,
at::native::FillFunctor<int>,          0
        std::array<char*, 1ul> >(int,
at::native::FillFunctor<int>, std::array<char*, 1u
    1>)

9          9  matmulbias_kernel
    0

```

1. System Info

	Info
workload_name	fusion_matmulbias
command	python fusion_matmulbias.py
ip_blocks	SQ LDS SQC TA TD TCP TCC SPI CPC CPF roofline
timestamp	Mon Jan 12 19:10:39 2026 (UTC)

version	3
hostname	f57b369e0635
cpu_model	Intel(R) Xeon(R) Platinum 8470
sbios	SeaBIOS1.15.0-1
linux_distro	CentOS Stream 10 (Coughlan)
linux_kernel_version	5.15.0-164-generic
amd_gpu_kernel_version	nan
cpu_memory	230893524
gpu_memory	nan
rocm_version	7.0.3
vbios	113-M3000108-103
compute_partition	SPX
memory_partition	nan
gpu_series	MI300
gpu_model	MI300X_A1
gpu_arch	gfx942
gpu_chip_id	29877
gpu_l1	32
gpu_l2	4096
cu_per_gpu	304
simd_per_cu	4
se_per_gpu	32
wave_size	64
workgroup_max_size	1024

max_waves_per_cu	32
max_sclk	2100
max_mclk	1300
cur_sclk	2100
cur_mclk	1300
total_l2_chan	128
lds_banks_per_cu	32
sqc_per_gpu	160
pipes_per_gpu	4
num_xcd	8
num_hbm_channels	128

2. System Speed-of-Light

2.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak
Pct of Peak				
2.1.0	VALU FLOPs	34.68	Gflop/s	
81715.2	0.04			
2.1.1	VALU IOPs	541.6	Giop/s	
81715.2	0.66			
2.1.2	MFMA FLOPs (F8)	0.0	Gflop/s	
2614886.4	0.0			
2.1.3	MFMA FLOPs (BF16)	0.0	Gflop/s	
1307443.2	0.0			

2.1.4	MFMA FLOPs (F16)	32973.27	Gflop/s	
1307443.2	2.52			
2.1.5	MFMA FLOPs (F32)	0.0	Gflop/s	
163430.4	0.0			
2.1.6	MFMA FLOPs (F64)	0.0	Gflop/s	
163430.4	0.0			
2.1.7	MFMA FLOPs (F6F4)		Gflop/s	
2.1.8	MFMA IOPs (Int8)	0.0	Giop/s	
2614886.4	0.0			
2.1.9	Active CUs	101.0	Cus	304.0
33.22				
2.1.10	SALU Utilization	1.3	Pct	100.0
1.3				
2.1.11	VALU Utilization	2.14	Pct	100.0
2.14				
2.1.12	MFMA Utilization	1.94	Pct	100.0
1.94				
2.1.13	VMEM Utilization	0.34	Pct	100.0
0.34				
2.1.14	Branch Utilization	0.25	Pct	100.0
0.25				
2.1.15	VALU Active Threads	62.8	Threads	64.0
98.12				

2.1.16 4.63	IPC	0.23	Instr/cycle	5.0
2.1.17 27.01	Wavefront Occupancy	2627.57	Wavefronts	9728.0
2.1.18 81715.2	Theoretical LDS Bandwidth 2.41	1969.23	Gb/s	
2.1.19 0.0	LDS Bank Conflicts/Access	0.0	Conflicts/access	32.0
2.1.20 26.03	vL1D Cache Hit Rate	26.03	Pct	100.0
2.1.21 81715.2	vL1D Cache BW 6.85	5597.49	Gb/s	
2.1.22 52.03	L2 Cache Hit Rate	52.03	Pct	100.0
2.1.23 34406.4	L2 Cache BW 13.87	4773.08	Gb/s	
2.1.24 5.47	L2-Fabric Read BW	291.24	Gb/s	5324.8
2.1.25 38.02	L2-Fabric Write BW	2024.66	Gb/s	5324.8
2.1.26	L2-Fabric Read Latency	420.69	Cycles	
2.1.27	L2-Fabric Write Latency	479.76	Cycles	

2.1.28	sL1D Cache Hit Rate	86.48	Pct	100.0
86.48				
2.1.29	sL1D Cache BW	34.13	Gb/s	
21504.0	0.16			
2.1.30	L1I Hit Rate	99.36	Pct	100.0
99.36				
2.1.31	L1I BW	463.96	Gb/s	
21504.0	2.16			
2.1.32	L1I Fetch Latency		Cycles	

3. Memory Chart

3.1 Memory Chart

(Normalization: per_kernel)

Instr Buff	Instr Dispatch	Exec
LDS	L2 Cache	

SALU: 350676


```

-----> Active
CUs      Req   : 76033 Util   :    2 %
          xGMI/PCIE

```

```

Wave     0 Instr Buf          10
1: 304    <-----
-----

```

```

          SMEM:
21252          Lat   :
N/A          Hit    :    52
%
-----

```

```

>
~
|
RVGPRseq:  44.0

```

```

      |  |
Wave N-1 Instr Buf          VALU:
509268          Vector L1 Cache
      Request          |
|
-----

```

```

>
|
v
SGPRs :    40

```

```

          MFMA:
52186          Rd    : 887173 Hit    :    26 %    Rd
: 104423    Rd      : 107907

```

```

-----
>    <-----    <-----
-
Fabric

```

```

Wt      : 5464985 Lat   :    N/A    Wr      : 1392186 Wr      :
1392186    Rd      :
59195

```

```

          VMEM: 99310    LDS Alloc:    -
----->

```

```

>    <----- Latency (cycles)    Rd:
59193

```

```

----->
71    Atomic:    0 Coales:    44 %    Atomic:    0    Atomic:

```

```

0      Wr      : 1353046      <----- HBM
                                <
----->      <-----
>      ----->      Rd      :      421      Wr:
1353046
                                LDS:
76033      Stall :      25
cycles      Atomic:      0      Wr      :
480      ----->
                                ----->      Scratch Alloc:

----->      Atomic:      0
                                0

```

```

Wave Occupancy      GWS: 0
Scalar L1D
Cache      Latency (cycles)

26      -----
--->      Rd      :
53

```

```

Wave Life
                                <-----
                                ^
|
23472
BRANCH: 45537      Wavefronts:      Rd      : 21297      Hit      :      79.0
%      Wr      :      0      Rd      :      N/A      |
|
                                ----->      0      <
-----      -----
>      |
|

```

```

Lat      :      N/A      Atomic:      0      Wr      :
N/A      |
v
^
                                <-----
>

```

```

|      Workgroups:

|
0      Instr L1 Cache

```

```

GMI
|
|
|
Fetch :
236908 Hit : 96 % Req
: 3445
-----
<-----
Lat : 2299.0 cycles

```

```
4. Roofline
4.1 Roofline
      Roofline (FP32) -
/workspace/user/workloads/fusion_matmulbias/MI300X_A1

140669.2
140669GFL0P/s
```

36383.2

9410.3

2433.9

629.5 63346 GB/s

34174 GB/s

162.8

42.1 4211 GB/s

0.01

0.18

3.16

56.23

1000.00

Performance (GFLOP/sec) Arithmetic Intensity (FLOP)s/Byte)

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

Metric_ID	Metric	Avg	Min	Max	Unit
5.1.0	CPF Utilization	100.00	100.00	100.00	Pct

5.1.1	CPF Stall	0.61	0.02	4.90	Pct
5.1.2	CPF-L2 Utilization	12.10	1.21	50.46	Pct
5.1.3	CPF-L2 Stall	0.00	0.00	0.00	Pct
5.1.4	CPF-UTCL1 Stall	0.73	0.06	6.93	Pct

5.2 Packet Processor

Metric_ID Unit	Metric	Avg	Min	Max	
5.2.0	CPC SYNC FIFO Full Rate				Pct
5.2.1	CPC CANE Stall Rate				Pct
5.2.2	CPC ADC Utilization				Pct
5.2.3	CPC Utilization	100.0	100.0	100.0	Pct
5.2.4	CPC Stall Rate	24.64	2.96	57.7	Pct
5.2.5	CPC Packet Decoding Utilization	70.25	9.77	363.49	Pct
5.2.6	CPC-Workgroup Manager Utilization	21.79	0.52	69.49	Pct
5.2.7	CPC-L2 Utilization	0.47	0.16	1.32	Pct
5.2.8	CPC-UTCL1 Stall	0.35	0.04	4.86	Pct

5.2.9	CPC-UTCL2 Utilization	0.33	0.1	4.79	Pct
-------	-----------------------	------	-----	------	-----

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

Metric_ID	Metric	Avg	Min	Max	Unit
6.1.0	Schedule-Pipe Wave Occupancy				Wave
6.1.1	Accelerator Utilization	100.73	80.22	367.31	Pct
6.1.2	Scheduler-Pipe Utilization	0.0	0.0	0.0	Pct
6.1.3	Scheduler-Pipe Wave Utilization				Pct
6.1.4	Workgroup Manager Utilization	58.05	5.23	92.75	Pct
6.1.5	Shader Engine Utilization	53.7	0.21	91.23	Pct
6.1.6	SIMD Utilization	30.78	0.01	90.69	Pct
6.1.7	Dispatched Workgroups	0.0	0.0	0.0	

Workgroups

6.1.8	Dispatched Wavefronts	0.0	0.0	0.0
Wavefronts				

6.1.9	VGPR Writes
Cycles/wave	

6.1.10	SGPR Writes
Cycles/wave	

6.2 Workgroup Manager - Resource Allocation

Metric_ID	Metric	Avg	Min	Max
Unit				

6.2.0	Not-scheduled Rate (Workgroup Manager)	0.0	0.0	0.0
Pct				

6.2.1	Not-scheduled Rate (Scheduler-Pipe)	0.0	0.0	0.0
Pct				

6.2.2	Scheduler-Pipe FIFO Full Rate
Pct	

6.2.3	Scheduler-Pipe Stall Rate	0.0	0.0	0.0
Pct				

6.2.4	Scratch Stall Rate	0.0	0.0	0.0
Pct				

6.2.5	Insufficient SIMD Waveslots	0.0	0.0	0.0
Pct				

6.2.6	Insufficient SIMD VGPRs	0.0	0.0	0.0
Pct				

6.2.7 Pct	Insufficient SIMD SGPRs	0.0	0.0	0.0
6.2.8 Pct	Insufficient CU LDS	0.0	0.0	0.0
6.2.9 Pct	Insufficient CU Barriers	0.0	0.0	0.0
6.2.10 Pct	Reached CU Workgroup Limit	0.0	0.0	0.0
6.2.11 Pct	Reached CU Wavefront Limit	0.0	0.0	0.0

7. Wavefront

7.1 Wavefront Launch Stats

Metric_ID	Metric	Avg	Min	Max	Unit
7.1.0 items	Grid Size	1354433.93	32.00	4194304.00	Work
7.1.1 items	Workgroup Size	429.78	32.00	1024.00	Work
7.1.2 Wavefronts	Total Wavefronts	0.00	0.00	0.00	
7.1.3 Wavefronts	Saved Wavefronts	0.00	0.00	0.00	

7.1.4 Wavefronts	Restored Wavefronts	0.00	0.00	0.00	
7.1.5 Registers	VGPRs	43.68	8.00	108.00	
7.1.6 Registers	AGPRs	20.45	0.00	192.00	
7.1.7 Registers	SGPRs	39.56	16.00	112.00	
7.1.8	LDS Allocation	71.11	0.00	29184.00	Bytes
7.1.9 Bytes/workitem	Scratch Allocation	0.00	0.00	0.00	

7.2 Wavefront Runtime Stats

Metric_ID Max Unit	Metric	Avg	Min
7.2.0 65251.50 Ns	Kernel Time (Nanosec)	25248.06	1733.84
7.2.1 322032.62 Cycle	Kernel Time (Cycles)	70292.69	18727.88
7.2.2 3797.00 Instr/wavefront	Instructions per wavefront	1091.01	19.50
7.2.3 949337332.00	Wave Cycles Cycles per kernel	231628220.93	5504.00
7.2.4	Dependency Wait Cycles	53124379.05	4504.00

201746392.00 Cycles per kernel

7.2.5 Issue Wait Cycles 173745800.43 8.00
736339808.00 Cycles per kernel

7.2.6 Active Cycles 4738347.17 296.00
16351996.00 Cycles per kernel

7.2.7 Wavefront Occupancy 2627.57 0.22
35496.48 Wavefronts

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.1.0	VALU	457082.06	20.00	2731520.00	Instr per kernel
10.1.1	VMEM	99310.09	2.00	262144.00	Instr per kernel
10.1.2	LDS	76032.60	0.00	212992.00	Instr per kernel
10.1.3	MFMA	52186.07	0.00	131072.00	Instr per kernel
10.1.4	SALU	350675.87	24.00	924160.00	Instr per kernel
10.1.5	SMEM	21251.87	0.00	68096.00	Instr per kernel
10.1.6	Branch	45536.73	10.00	131072.00	Instr per kernel

10.2 VALU Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.2.0	INT32	240579.02	9.00	393216.00	Instr per kernel
10.2.1	INT64	24626.34	0.00	668160.00	Instr per kernel
10.2.2	F16-ADD	4892.44	0.00	12288.00	Instr per kernel
10.2.3	F16-MUL	0.00	0.00	0.00	Instr per kernel

10.2.4	F16-FMA	75.89	0.00	16384.00	Instr per kernel
10.2.5	F16-Trans	0.00	0.00	0.00	Instr per kernel
10.2.6	F32-ADD	180.30	0.00	38912.00	Instr per kernel
10.2.7	F32-MUL	1292.30	0.00	123392.00	Instr per kernel
10.2.8	F32-FMA	540.89	0.00	116736.00	Instr per kernel
10.2.9	F32-Trans	1021.85	0.00	65024.00	Instr per kernel
10.2.10	F64-ADD	0.00	0.00	0.00	Instr per kernel
10.2.11	F64-MUL	0.00	0.00	1.00	Instr per kernel
10.2.12	F64-FMA	0.01	0.00	6.00	Instr per kernel
10.2.13	F64-Trans	0.00	0.00	1.00	Instr per kernel
10.2.14	Conversion	8234.89	0.00	58368.00	Instr per kernel

10.3 VMEM Instr Mix

Metric_ID Unit	Metric	Avg	Min	Max
10.3.0 Instr per kernel	Global/Generic Instr	83817.35	0.0	262144.0
10.3.1 Instr per kernel	Global/Generic Read	0.51	0.0	6.0
10.3.2 Instr per kernel	Global/Generic Write	83816.84	0.0	262144.0
10.3.3 Instr per kernel	Global/Generic Atomic	0.0	0.0	0.0
10.3.4 Instr per kernel	Spill/Stack Instr	15492.74	0.0	38912.0

10.3.5 Spill/Stack Coalesceable Instr
Instr per kernel

10.3.6 Spill/Stack Read 13861.93 0.0 34816.0
Instr per kernel

10.3.7 Spill/Stack Write 1630.81 0.0 4096.0
Instr per kernel

10.3.8 Spill/Stack Atomic 0.0 0.0 0.0
Instr per kernel

10.4 MFMA Arithmetic Instr Mix

Metric_ID	Metric	Avg	Min	Max	Unit
10.4.0	MFMA-I8	0.0	0.0	0.0	Instr per kernel
10.4.1	MFMA-F8	0.0	0.0	0.0	Instr per kernel
10.4.2	MFMA-F16	52186.07	0.0	131072.0	Instr per kernel
10.4.3	MFMA-BF16	0.0	0.0	0.0	Instr per kernel
10.4.4	MFMA-F32	0.0	0.0	0.0	Instr per kernel
10.4.5	MFMA-F64	0.0	0.0	0.0	Instr per kernel
10.4.6	MFMA-F6F4				Instr per kernel

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit	Peak	Pct of Peak
11.1.0	VALU FLOPs	34.68	Gflop	81715.2	0.04

11.1.1	VALU IOPs	541.6	Giop	81715.2	0.66
11.1.2	MFMA FLOPs (F8)	0.0	Gflop	2614886.4	0.0
11.1.3	MFMA FLOPs (BF16)	0.0	Gflop	1307443.2	0.0
11.1.4	MFMA FLOPs (F16)	32973.27	Gflop	1307443.2	2.52
11.1.5	MFMA FLOPs (F32)	0.0	Gflop	163430.4	0.0
11.1.6	MFMA FLOPs (F64)	0.0	Gflop	163430.4	0.0
11.1.7	MFMA FLOPs (F6F4)		Gflop		
11.1.8	MFMA IOPs (INT8)	0.0	Giop	2614886.4	0.0

11.2 Pipeline Stats

Metric_ID	Metric	Avg	Min	Max	Unit
11.2.0	IPC	0.23	0.04	1.3	Instr/cycle
11.2.1	IPC (Issued)	0.95	0.86	1.0	Instr/cycle

11.2.2	SALU Utilization	1.3	0.0	11.4	Pct
11.2.3	VALU Utilization	2.14	0.0	33.82	Pct
11.2.4	VALU Co-Issue Efficiency				Pct
11.2.5	VMEM Utilization	0.34	0.0	0.85	Pct
11.2.6	Branch Utilization	0.25	0.0	1.56	Pct
11.2.7	VALU Active Threads	62.8	28.46	64.0	Threads
11.2.8	MFMA Utilization	1.94	0.0	5.62	Pct
11.2.9	MFMA Instr Cycles	32.0	32.0	32.0	Cycles/instr
11.2.10	VMEM Latency				Cycles
11.2.11	SMEM Latency				Cycles

11.3 Arithmetic Operations

Metric_ID	Metric	Avg	Min	Max	Unit
-----------	--------	-----	-----	-----	------

11.3.0 kernel	FLOPs (Total)	855568348.15	0.0	2148532224.0	Ops per
11.3.1 kernel	IOPs (Total)	16973142.96	960.0	59179008.0	Ops per
11.3.2 kernel	F8 OPs	0.0	0.0	0.0	Ops per
11.3.3 kernel	F16 OPs	855339467.85	0.0	2148270080.0	Ops per
11.3.4 kernel	BF16 OPs	0.0	0.0	0.0	Ops per
11.3.5 kernel	F32 OPs	228878.22	0.0	29491200.0	Ops per
11.3.6 kernel	F64 OPs	2.07	0.0	896.0	Ops per
11.3.7 kernel	F6F4 OPs				Ops per
11.3.8 kernel	INT8 OPs	0.0	0.0	0.0	Ops per

12. Local Data Share (LDS)

12.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
12.1.0	Utilization	1.85	Pct of peak
12.1.1	Access Rate	1.21	Pct of peak

12.1.2	Theoretical Bandwidth (% of Peak)	2.41	Pct of peak	
12.1.3	Bank Conflict Rate	0.00	Pct of peak	
12.2 LDS Stats				
Metric_ID Unit	Metric	Avg	Min	Max
12.2.0 Instr per kernel	LDS Instrs	76032.6	0.0	212992.0
12.2.1 Instr per kernel	LDS LOAD			
12.2.2 Instr per kernel	LDS STORE			
12.2.3 Instr per kernel	LDS ATOMIC			
12.2.4 Gbps	LDS LOAD Bandwidth			
12.2.5 Gbps	LDS STORE Bandwidth			
12.2.6 Gbps	LDS ATOMIC Bandwidth			
12.2.7 Bytes per kernel	Theoretical Bandwidth	50331929.48	0.0	134217728.0
12.2.8 Cycles	LDS Latency			

12.2.9	Bank Conflicts/Access	0.0	0.0	0.15
Conflicts/access				
12.2.10	Index Accesses	393218.46	0.0	1048576.0
Cycles per kernel				
12.2.11	Atomic Return Cycles	0.0	0.0	0.0
Cycles per kernel				
12.2.12	Bank Conflict	0.26	0.0	112.0
Cycles per kernel				
12.2.13	Addr Conflict	0.06	0.0	24.0
Cycles per kernel				
12.2.14	Unaligned Stall	0.0	0.0	0.0
Cycles per kernel				
12.2.15	Mem Violations	0.0	0.0	0.0
Accesses per kernel				
12.2.16	LDS Command FIFO Full Rate			
Cycles per kernel				
12.2.17	LDS Data FIFO Full Rate			
Cycles per kernel				

13. Instruction Cache

13.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
13.1.0	Bandwidth	2.16	Pct of peak
13.1.1	Cache Hit Rate	95.60	Pct of peak

13.1.2	L1I-L2 Bandwidth	0.02	Pct of peak
--------	------------------	------	-------------

13.2 Instruction Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
13.2.0 per kernel	Req	236908.24	18.0	778240.0	Req
13.2.1 per kernel	Hits	235464.75	14.0	763744.0	Hits
13.2.2 Misses per kernel	Misses - Non Duplicated	101.46	1.0	480.0	
13.2.3 Misses per kernel	Misses - Duplicated	1382.51	1.0	18894.0	
13.2.4	Cache Hit Rate	95.6	77.78	99.66	Pct
13.2.5	Instruction Fetch Latency Cycles				

13.3 Instruction Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
13.3.0 kernel	L1I-L2 Bandwidth	220468.44	1024.00	645120.00	Bytes per

14. Scalar L1 Data Cache

14.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
14.1.0	Bandwidth	0.16	Pct of peak
14.1.1	Cache Hit Rate	78.71	Pct of peak
14.1.2	sL1D-L2 BW	0.00	Pct of peak

14.2 Scalar L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
14.2.0 kernel	Req	21296.96	0.00	77824.00	Req per
14.2.1 kernel	Hits	19726.21	0.00	67904.00	Req per
14.2.2 kernel	Misses - Non Duplicated	52.91	0.00	320.00	Req per
14.2.3 kernel	Misses- Duplicated	1510.37	0.00	9600.00	Req per
14.2.4	Cache Hit Rate	78.71	0.00	93.66	Pct
14.2.5 kernel	Read Req (Total)	21251.87	0.00	68096.00	Req per
14.2.6 kernel	Atomic Req	0.00	0.00	0.00	Req per
14.2.7 kernel	Read Req (1 DWord)	135.92	0.00	29184.00	Req per

14.2.8 kernel	Read Req (2 DWord)	90.49	0.00	19456.00	Req per
14.2.9 kernel	Read Req (4 DWord)	21025.41	0.00	65536.00	Req per
14.2.10 kernel	Read Req (8 DWord)	0.04	0.00	8.00	Req per
14.2.11 kernel	Read Req (16 DWord)	0.00	0.00	0.00	Req per

14.3 Scalar L1D Cache - L2 Interface

Metric_ID	Metric	Avg	Min	Max	Unit
14.3.0	sL1D-L2 BW	3385.93	0.00	20480.00	Bytes per kernel
14.3.1	Read Req	52.91	0.00	320.00	Req per kernel
14.3.2	Write Req	0.00	0.00	0.00	Req per kernel
14.3.3	Atomic Req	0.00	0.00	0.00	Req per kernel
14.3.4	Stall Cycles	0.01	0.00	4.00	Cycles per kernel

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

Metric_ID Max	Metric Unit	Avg	Min
15.1.0 88.45	Address Processing Unit Busy Pct	27.38	0.0
15.1.1 6.75	Address Stall Pct	1.92	0.0

15.1.2	Data Stall	20.48	0.0
67.72	Pct		
15.1.3	Data-Processor → Address Stall	0.0	0.0
0.0	Pct		
15.1.4	Sequencer → TA Address Stall		
	Cycles per kernel		
15.1.5	Sequencer → TA Command Stall		
	Cycles per kernel		
15.1.6	Sequencer → TA Data Stall		
	Cycles per kernel		
15.1.7	Total Instructions	99310.09	2.0
262144.0	Instructions per kernel		
15.1.8	Global/Generic Instructions	83817.35	0.0
262144.0	Instructions per kernel		
15.1.9	Global/Generic Read Instructions	0.51	0.0
6.0	Instructions per kernel		
15.1.10	Global/Generic Read Instructions for LDS		
	Instructions per kernel		
15.1.11	Global/Generic Write Instructions	83816.84	0.0
262144.0	Instructions per kernel		
15.1.12	Global/Generic Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		
15.1.13	Spill/Stack Instructions	15492.74	0.0
38912.0	Instructions per kernel		

15.1.14	Spill/Stack Read Instructions	13861.93	0.0
34816.0	Instructions per kernel		

15.1.15	Spill/Stack Read Instructions for LDS
	Instructions per kernel

15.1.16	Spill/Stack Write Instructions	1630.81	0.0
4096.0	Instructions per kernel		

15.1.17	Spill/Stack Atomic Instructions	0.0	0.0
0.0	Instructions per kernel		

15.1.18	Spill/Stack Total Cycles	247883.85	0.0
622592.0	Cycles per kernel		

15.1.19	Spill/Stack Coalesced Read	0.0	0.0
0.0	Cycles per kernel		

15.1.20	Spill/Stack Coalesced Write	0.0	0.0
0.0	Cycles per kernel		

15.2 Data-Return Path

Metric_ID	Metric	Avg	Min	Max
Unit				
15.2.0	Data-Return Busy	29.35	0.0	79.04
Pct				
15.2.1	Cache RAM → Data-Return Stall	28.09	0.0	88.92
Pct				
15.2.2	Workgroup manager → Data-Return Stall	0.0	0.0	0.0
Pct				

15.2.3	Coalescable Instructions	76.85	0.0	
16384.0	Instructions per kernel			
15.2.4	Read Instructions	13862.44	0.0	
34816.0	Instructions per kernel			
15.2.5	Write Instructions	85447.66	1.0	
262144.0	Instructions per kernel			
15.2.6	Atomic Instructions	0.0	0.0	0.0
	Instructions per kernel			
15.2.7	Write Ack Instructions			
	Instructions per kernel			

INFO Not showing table with empty column(s): 16.6 L1D Addr Translation Stalls

16. Vector L1 Data Cache

16.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
16.1.0	Hit rate	26.03	Pct of peak
16.1.1	Bandwidth	6.85	Pct of peak
16.1.2	Utilization	36.69	Pct of peak
16.1.3	Coalescing	44.33	Pct of peak

16.2 L1D Cache Stalls (%)

Metric_ID	Metric	Min	Q1	Median	Q3	Max
16.2.0	Stalled on L2 Data	0.01	0.27	13.55	92.59	
94.91						

16.2.1 82.98	Stalled on L2 Req	0.0	0.0	0.0	76.66	
16.2.2	Stalled on Address					
16.2.3	Stalled on Data					
16.2.4	Stalled on Latency FIFO					
16.2.5	Stalled on Request FIFO					
16.2.6	Stalled on Read Return					
16.2.7	Tag RAM Stall (Read)	0.0	0.0	0.0	0.0	0.0
16.2.8	Tag RAM Stall (Write)	0.0	0.0	0.0	1.19	1.37
16.2.9	Tag RAM Stall (Atomic)	0.0	0.0	0.0	0.0	0.0

16.3 L1D Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
16.3.0 per kernel	Total Req	6352157.65	32.0	16777216.0	Req

16.3.1 per kernel	Read Req	887173.02	0.0	2228224.0	Req
16.3.2 per kernel	Write Req	5464984.63	16.0	16777216.0	Req
16.3.3 per kernel	Atomic Req	0.0	0.0	0.0	Req
16.3.4 Bytes per kernel	Cache BW	213288314.67	256.0	536870912.0	
16.3.5	Cache Hit Rate	26.03	-29.63	84.09	Pct
16.3.6 per kernel	Cache Accesses	1666314.96	2.0	4194304.0	Req
16.3.7 per kernel	Cache Hits	169705.96	-8.0	425856.0	Req
16.3.8 per kernel	Invalidations	124.31	1.0	304.0	Req
16.3.9 Bytes per kernel	L1-L2 BW	102466076.15	192.0	268435456.0	
16.3.10 per kernel	Tag RAM 0 Req				Req
16.3.11 per kernel	Tag RAM 1 Req				Req
16.3.12 per kernel	Tag RAM 2 Req				Req

16.3.13 per kernel	Tag RAM 3 Req				Req
16.3.14 per kernel	L1-L2 Read	104423.44	0.0	262272.0	Req
16.3.15 per kernel	L1-L2 Write	1392185.56	1.0	4194304.0	Req
16.3.16 per kernel	L1-L2 Atomic	0.0	0.0	0.0	Req
16.3.17 Cycles	L1 Access Latency				
16.3.18 Cycles	L1-L2 Read Latency				
16.3.19 Cycles	L1-L2 Write Latency				

16.4 L1D - L2 Transactions

Metric_ID Max Unit	Metric	Xfer	Coherency	Avg	Min
16.4.0 0.00 Req per kernel	NC - Read	Read	NC	0.00	0.00
16.4.1 0.00 Req per kernel	UC - Read	Read	UC	0.00	0.00
16.4.2 0.00 Req per kernel	CC - Read	Read	CC	0.00	0.00

16.4.3	RW - Read	Read	RW	104423.44	0.00
262272.00	Req per kernel				
16.4.4	RW - Write	Write	RW	1392185.17	0.00
4194304.00	Req per kernel				
16.4.5	NC - Write	Write	NC	0.00	0.00
0.00	Req per kernel				
16.4.6	UC - Write	Write	UC	0.39	0.00
6.00	Req per kernel				
16.4.7	CC - Write	Write	CC	0.00	0.00
0.00	Req per kernel				
16.4.8	NC - Atomic	Atomic	NC	0.00	0.00
0.00	Req per kernel				
16.4.9	UC - Atomic	Atomic	UC	0.00	0.00
0.00	Req per kernel				
16.4.10	CC - Atomic	Atomic	CC	0.00	0.00
0.00	Req per kernel				
16.4.11	RW - Atomic	Atomic	RW	0.00	0.00
0.00	Req per kernel				

16.5 L1D Addr Translation

Metric_ID	Metric	Avg	Min	Max
Units				
16.5.0	Req	1666314.94	2.0	4194304.0
	Req per kernel			
16.5.1	Inflight Req			

Req per kernel

16.5.2	Hit Ratio	82.06	0.0	100.0
Pct				
16.5.3	Hits	1555082.23	0.0	3885174.0
Req per kernel				
16.5.4	Translation Misses	5318.18	0.0	16384.0
Req per kernel				
16.5.5	Misses under Translation Miss			
Req per kernel				
16.5.6	Permission Misses	0.36	0.0	128.0
Req per kernel				

INFO Not showing table with empty column(s): 17.4 L2 Cache Stalls

17. L2 Cache

17.1 Speed-of-Light

Metric_ID	Metric	Avg	Unit
17.1.0	Utilization	48.70	Pct
17.1.1	Bandwidth	13.87	Pct
17.1.2	Hit Rate	52.03	Pct
17.1.3	L2-Fabric Read BW	291.24	Gb/s
17.1.4	L2-Fabric Write and Atomic BW	2024.66	Gb/s
17.1.5	HBM Bandwidth	5324.80	Gb/s

17.2 L2 - Fabric Transactions

Metric_ID	Metric	Avg	Min	Max
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Unit				
17.2.0 18984256.0	Read BW Bytes per kernel	7549135.41	1536.0	
17.2.1 Pct	HBM Read Traffic	99.39	72.41	108.0
17.2.2 Pct	Remote Read Traffic	0.99	0.0	27.59
17.2.3 110.34	Uncached Read Traffic Pct	4.33	0.0	
17.2.4 268435456.0	Write and Atomic BW Bytes per kernel	86594925.19	32.0	
17.2.5 Pct	HBM Write and Atomic Traffic	79.4	0.0	100.0
17.2.6 Pct	Remote Write and Atomic Traffic	20.6	0.0	100.0
17.2.7 Pct	Atomic Traffic	0.0	0.0	0.0
17.2.8 Pct	Uncached Write and Atomic Traffic	20.63	0.0	100.0
17.2.9 709.58	Read Latency Cycles	420.69	160.04	
17.2.10 Cycles	Write and Atomic Latency	479.76	256.0	857.5
17.2.11	Atomic Latency			

Cycles

17.2.12 Read Stall
Pct

17.2.13 Write Stall
Pct

17.3 L2 Cache Accesses

Metric_ID	Metric	Avg	Min	Max	Unit
17.3.0 Bytes per kernel	Bandwidth	192010263.7	3072.0	537408768.0	
17.3.1 Bytes per kernel	Read Bandwidth				
17.3.2 Bytes per kernel	Write Bandwidth				
17.3.3 Bytes per kernel	Atomic Bandwidth				
17.3.4 per kernel	Req	1500080.19	24.0	4198506.0	Req
17.3.5 per kernel	Read Req	107907.17	23.0	272352.0	Req
17.3.6 per kernel	Write Req	1392185.56	1.0	4194304.0	Req
17.3.7 per kernel	Atomic Req	0.0	0.0	0.0	Req

17.3.8 per kernel	Streaming Req	0.01	0.0	3.0	Req
17.3.9 per kernel	Bypasss Req				Req
17.3.10 per kernel	Probe Req	0.04	0.0	4.0	Req
17.3.11 per kernel	Input Buffer Req				Req
17.3.12	Cache Hit	52.03	20.0	61.4	Pct
17.3.13 per kernel	Hits	764596.58	10.0	2101048.0	Hits
17.3.14 Misses per kernel	Misses	735507.39	13.0	2097452.0	
17.3.15 Cachelines per kernel	Writeback	676587.4	6.0	2097208.0	
17.3.16 Cachelines per kernel	Writeback (Internal)	628053.39	0.0	1966093.0	
17.3.17 Cachelines per kernel	Writeback (vL1D Req)	48469.47	0.0	131072.0	
17.3.18 Cachelines per kernel	Evict (Internal)	586218.5	0.0	1835120.0	
17.3.19 Cachelines per kernel	Evict (vL1D Req)	0.0	0.0	0.0	

17.3.20 per kernel	NC Req	0.0	0.0	0.0	Req
17.3.21 per kernel	UC Req	3.3	0.0	195.0	Req
17.3.22 per kernel	CC Req	0.0	0.0	0.0	Req
17.3.23 per kernel	RW Req	1500077.37	24.0	4198304.0	Req

17.5 L2 - Fabric Interface Stalls

Metric_ID Transaction	Metric Avg	Min	Max	Unit	Type	
17.5.0	Read - PCIe Stall Pct				PCIe Stall	Read
17.5.1	Read - Infinity Fabric Stall Pct				Infinity Fabric Stall	Read
17.5.2	Read - HBM Stall Pct				HBM Stall	Read
17.5.3	Write - PCIe Stall Pct				PCIe Stall	Write
17.5.4	Write - Infinity Fabric Stall Pct				Infinity Fabric Stall	Write
17.5.5	Write - HBM Stall Pct				HBM Stall	Write

17.5.6	Write - Credit Starvation	Credit Starvation	Write
0.0	0.0	0.0	Pct

17.6 L2 - Fabric Detailed Transaction Breakdown

Metric_ID	Metric	Avg	Min	Max
Unit				
17.6.0	Read (32B)	0.0	0.0	0.0
Req per kernel				
17.6.1	Read (64B)	434.0	22.0	
1461.0	Req per kernel			
17.6.2	Read (128B)	58760.62	0.0	
147584.0	Req per kernel			
17.6.3	Read (Uncached)	12.18	0.0	408.0
Req per kernel				
17.6.4	HBM Read	59192.66	23.0	
149044.0	Req per kernel			
17.6.5	Remote Read	3.31	0.0	17.0
Req per kernel				
17.6.6	Read Bandwidth - PCIe			
Bytes per kernel				
17.6.7	Read Bandwidth - Infinity Fabric			
Bytes per kernel				
17.6.8	Read Bandwidth - HBM			
Bytes per kernel				

17.6.9	Write and Atomic (32B)	0.48	0.0	6.0
Req per kernel				
17.6.10	Write and Atomic (Uncached)	0.4	0.0	6.0
Req per kernel				
17.6.11	Write and Atomic (64B)	1353045.47	0.0	
4194304.0	Req per kernel			
17.6.12	HBM Write and Atomic	1353045.55	0.0	
4194304.0	Req per kernel			
17.6.13	Remote Write and Atomic	0.39	0.0	6.0
Req per kernel				
17.6.14	Write Bandwidth - PCIe			
Bytes per kernel				
17.6.15	Write Bandwidth - Infinity Fabric			
Bytes per kernel				
17.6.16	Write Bandwidth - HBM			
Bytes per kernel				
17.6.17	Atomic	0.0	0.0	0.0
Req per kernel				
17.6.18	Atomic - HBM			
Req per kernel				
17.6.19	Atomic Bandwidth - PCIe			
Bytes per kernel				
17.6.20	Atomic Bandwidth - Infinity Fabric			
Bytes per kernel				

17.6.21 Atomic Bandwidth - HBM
Bytes per kernel

INFO Not showing table with empty column(s): 18.9 L2-Fabric Read
Stall (Cycles per normUnit)

INFO Not showing table with empty column(s): 18.10 L2-Fabric Write
and Atomic Stall (Cycles per normUnit)

18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

Metric_ID	Metric	Avg	Std Dev	Min	Max	Unit
18.1.0	L2 Cache Hit Rate	52.16	12.61	0.00	61.41	Pct

18.2 L2 Cache Hit Rate (pct)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.2.0	0.00	0.00	50.00	58.52	58.62	61.31
18.2.1	1.00	0.00	50.00	58.54	58.63	59.06
18.2.2	2.00	0.00	50.12	58.54	58.63	65.16
18.2.3	3.00	0.00	50.00	58.54	58.63	65.16
18.2.4	4.00	0.00	50.12	58.52	58.61	61.31
18.2.5	5.00	0.00	50.12	58.54	58.63	61.31
18.2.6	6.00	0.00	50.12	58.54	58.63	61.31
18.2.7	7.00	50.00	50.12	58.54	58.63	61.13
18.2.8	8.00	0.00	50.00	58.50	58.61	61.31
18.2.9	9.00	50.00	50.00	58.70	58.81	61.31
18.2.10	10.00	50.00	50.00	58.70	58.89	61.31
18.2.11	11.00	0.00	50.00	58.70	58.89	61.31
18.2.12	12.00	0.00	50.00	58.70	58.89	61.31

18.2.13	13.00	0.00	50.12	58.70	58.89	61.31
18.2.14	14.00	0.00	50.00	58.70	58.89	61.31
18.2.15	15.00	0.00	50.00	58.60	58.74	61.13
18.2.16	16.00	0.00	50.00	58.50	58.61	61.31
18.2.17	17.00	50.00	50.00	58.70	58.88	61.31
18.2.18	18.00	50.00	50.00	58.70	58.89	61.31
18.2.19	19.00	0.00	50.00	58.70	58.89	61.31
18.2.20	20.00	0.00	50.00	58.70	58.89	61.31
18.2.21	21.00	0.00	50.12	58.70	58.91	61.31
18.2.22	22.00	0.00	50.00	58.70	58.89	61.31
18.2.23	23.00	0.00	50.00	58.60	58.74	61.13
18.2.24	24.00	0.00	50.00	58.52	58.63	61.31
18.2.25	25.00	0.00	50.00	58.54	58.63	58.95
18.2.26	26.00	0.00	50.12	58.54	58.63	65.16
18.2.27	27.00	50.00	50.00	58.54	58.63	65.16
18.2.28	28.00	0.00	50.12	58.54	58.63	61.31
18.2.29	29.00	0.00	50.12	58.54	58.63	61.31
18.2.30	30.00	0.00	50.12	58.54	58.63	61.31
18.2.31	31.00	50.00	50.12	58.54	58.63	61.13
18.2.32	32.00	0.00	50.00	58.52	58.65	61.31
18.2.33	33.00	0.00	50.00	58.54	58.65	58.95
18.2.34	34.00	0.00	50.12	58.54	58.65	65.16
18.2.35	35.00	50.00	50.00	58.54	58.65	65.16
18.2.36	36.00	0.00	50.12	58.54	58.65	61.31

18.2.37	37.00	0.00	50.12	58.54	58.65	61.31
18.2.38	38.00	0.00	50.12	58.54	58.65	61.31
18.2.39	39.00	50.11	50.12	58.54	58.65	61.13
18.2.40	40.00	0.00	50.00	58.50	58.61	61.31
18.2.41	41.00	50.00	50.00	58.70	58.91	61.31
18.2.42	42.00	50.00	50.00	58.70	58.91	61.31
18.2.43	43.00	0.00	50.00	58.70	58.91	61.31
18.2.44	44.00	0.00	50.00	58.70	58.91	61.31
18.2.45	45.00	0.00	50.12	58.70	58.91	61.31
18.2.46	46.00	0.00	50.00	58.70	58.91	61.31
18.2.47	47.00	0.00	50.00	58.60	58.76	61.13
18.2.48	48.00	0.00	50.00	58.50	58.61	61.31
18.2.49	49.00	0.00	50.00	58.70	58.90	61.31
18.2.50	50.00	50.00	50.00	58.70	58.91	61.31
18.2.51	51.00	0.00	50.00	58.70	58.91	61.31
18.2.52	52.00	0.00	50.00	58.70	58.91	61.31
18.2.53	53.00	0.00	50.12	58.70	58.91	61.31
18.2.54	54.00	0.00	50.00	58.70	58.91	61.31
18.2.55	55.00	0.00	50.00	58.60	58.76	61.13
18.2.56	56.00	0.00	50.00	58.52	58.63	61.31
18.2.57	57.00	0.00	50.00	58.54	58.65	59.06
18.2.58	58.00	0.00	50.12	58.54	58.65	65.16
18.2.59	59.00	50.00	50.00	58.54	58.65	65.16
18.2.60	60.00	0.00	50.12	58.54	58.64	61.31

18.2.61	61.00	0.00	50.12	58.54	58.65	61.31
18.2.62	62.00	0.00	50.12	58.54	58.65	61.31
18.2.63	63.00	50.11	50.12	58.54	58.65	61.13
18.2.64	64.00	0.00	50.00	58.50	58.61	61.31
18.2.65	65.00	50.00	50.00	58.70	58.91	61.31
18.2.66	66.00	50.00	50.00	58.70	58.91	61.31
18.2.67	67.00	0.00	50.00	58.70	58.91	61.31
18.2.68	68.00	0.00	50.00	58.70	58.91	61.31
18.2.69	69.00	0.00	50.12	58.70	58.91	61.31
18.2.70	70.00	0.00	50.00	58.70	58.91	61.31
18.2.71	71.00	0.00	50.00	58.60	58.76	61.13
18.2.72	72.00	0.00	50.00	58.54	58.63	61.31
18.2.73	73.00	0.00	50.00	58.54	58.65	59.06
18.2.74	74.00	0.00	50.12	58.54	58.65	65.16
18.2.75	75.00	50.00	50.00	58.54	58.65	65.16
18.2.76	76.00	0.00	50.12	58.54	58.65	61.31
18.2.77	77.00	0.00	50.12	58.54	58.65	61.31
18.2.78	78.00	0.00	50.12	58.54	58.65	61.31
18.2.79	79.00	50.11	50.12	58.54	58.65	61.13
18.2.80	80.00	0.00	50.00	58.54	58.63	61.31
18.2.81	81.00	0.00	50.00	58.54	58.65	59.04
18.2.82	82.00	0.00	50.12	58.54	58.65	65.16
18.2.83	83.00	50.00	50.00	58.54	58.65	65.16
18.2.84	84.00	0.00	50.12	58.54	58.63	61.31

18.2.85	85.00	0.00	50.12	58.54	58.65	61.31
18.2.86	86.00	0.00	50.12	58.54	58.65	61.31
18.2.87	87.00	50.11	50.12	58.54	58.65	61.13
18.2.88	88.00	0.00	50.00	58.50	58.61	61.31
18.2.89	89.00	50.00	50.00	58.70	58.91	61.31
18.2.90	90.00	50.00	50.00	58.70	58.91	61.31
18.2.91	91.00	0.00	50.00	58.70	58.91	61.31
18.2.92	92.00	0.00	50.00	58.70	58.91	61.31
18.2.93	93.00	0.00	50.12	58.70	58.91	61.31
18.2.94	94.00	0.00	50.00	58.70	58.91	61.31
18.2.95	95.00	0.00	50.00	58.60	58.76	61.13
18.2.96	96.00	0.00	50.00	50.00	58.55	61.31
18.2.97	97.00	0.00	50.00	50.00	58.81	61.31
18.2.98	98.00	33.33	50.00	50.00	58.81	61.31
18.2.99	99.00	0.00	50.00	50.00	58.81	61.31
18.2.100	100.00	0.00	50.00	58.60	58.83	61.31
18.2.101	101.00	0.00	50.12	58.60	58.91	61.31
18.2.102	102.00	0.00	50.00	50.00	58.81	61.31
18.2.103	103.00	0.00	50.00	50.00	58.68	61.13
18.2.104	104.00	0.00	50.00	58.47	58.59	61.31
18.2.105	105.00	0.00	50.00	54.25	58.59	59.06
18.2.106	106.00	0.00	50.12	58.48	58.59	65.16
18.2.107	107.00	0.00	50.00	58.48	58.59	65.16
18.2.108	108.00	0.00	50.12	54.30	58.59	62.50

18.2.109	109.00	0.00	50.12	58.48	58.63	87.50
18.2.110	110.00	0.00	50.12	58.48	58.59	61.31
18.2.111	111.00	0.00	50.12	50.12	58.59	61.13
18.2.112	112.00	0.00	50.00	58.52	58.63	61.31
18.2.113	113.00	0.00	50.00	58.54	58.65	59.06
18.2.114	114.00	0.00	50.12	58.54	58.65	65.16
18.2.115	115.00	0.00	50.00	58.54	58.65	65.16
18.2.116	116.00	0.00	50.12	58.52	58.63	61.31
18.2.117	117.00	0.00	50.12	58.54	58.65	61.31
18.2.118	118.00	0.00	50.12	58.54	58.65	61.31
18.2.119	119.00	0.00	50.12	58.54	58.65	61.13
18.2.120	120.00	0.00	50.00	58.50	58.61	61.31
18.2.121	121.00	0.00	50.00	58.70	58.90	61.31
18.2.122	122.00	0.00	50.00	58.70	58.91	61.31
18.2.123	123.00	0.00	50.00	58.70	58.91	61.31
18.2.124	124.00	0.00	50.00	58.70	58.91	61.31
18.2.125	125.00	0.00	50.12	58.70	58.91	61.31
18.2.126	126.00	0.00	50.00	58.70	58.91	61.31
18.2.127	127.00	0.00	50.00	58.60	58.76	61.13

18.3 L2 Requests (per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.3.0	0.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.1	1.00	0.00	0.00	3100.00	32768.00	32788.00
18.3.2	2.00	0.00	0.00	3128.00	32768.00	32788.00

18.3.3	3.00	0.00	0.00	3128.00	32848.00	32868.00
18.3.4	4.00	0.00	0.00	3128.00	32848.00	32869.00
18.3.5	5.00	0.00	0.00	3128.00	32848.00	32868.00
18.3.6	6.00	0.00	0.00	3114.00	32848.00	32868.00
18.3.7	7.00	0.00	0.00	3128.00	32848.00	32869.00
18.3.8	8.00	0.00	1.00	3102.00	32768.00	32790.00
18.3.9	9.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.10	10.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.11	11.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.12	12.00	0.00	0.00	3102.00	32848.00	32868.00
18.3.13	13.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.14	14.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.15	15.00	0.00	0.00	3102.00	32768.00	32794.00
18.3.16	16.00	0.00	1.00	3099.00	32768.00	32790.00
18.3.17	17.00	0.00	0.00	3098.50	32768.00	32789.00
18.3.18	18.00	0.00	0.00	3098.50	32768.00	32789.00
18.3.19	19.00	0.00	0.00	3099.00	32768.00	32789.00
18.3.20	20.00	0.00	0.00	3099.00	32848.00	32868.00
18.3.21	21.00	0.00	0.00	3099.00	32768.00	32788.00
18.3.22	22.00	0.00	0.00	3098.50	32768.00	32789.00
18.3.23	23.00	0.00	0.00	3099.00	32768.00	32795.00
18.3.24	24.00	0.00	0.00	3120.50	32768.00	32788.00
18.3.25	25.00	0.00	0.00	3097.00	32768.00	32788.00
18.3.26	26.00	0.00	0.00	3121.00	32768.00	32788.00

18.3.27	27.00	0.00	0.00	3121.00	32848.00	32868.00
18.3.28	28.00	0.00	0.00	3121.00	32848.00	32869.00
18.3.29	29.00	0.00	0.00	3120.50	32848.00	32868.00
18.3.30	30.00	0.00	0.00	3109.00	32848.00	32868.00
18.3.31	31.00	0.00	0.00	3120.50	32848.00	32869.00
18.3.32	32.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.33	33.00	0.00	0.00	3100.00	32768.00	32788.00
18.3.34	34.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.35	35.00	0.00	0.00	3128.00	32848.00	32868.00
18.3.36	36.00	0.00	0.00	3128.00	32848.00	32869.00
18.3.37	37.00	0.00	0.00	3128.00	32848.00	32868.00
18.3.38	38.00	0.00	0.00	3114.00	32848.00	32868.00
18.3.39	39.00	0.00	0.00	3128.00	32848.00	32869.00
18.3.40	40.00	0.00	1.00	3102.00	32768.00	32790.00
18.3.41	41.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.42	42.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.43	43.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.44	44.00	0.00	0.00	3102.00	32848.00	32868.00
18.3.45	45.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.46	46.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.47	47.00	0.00	0.00	3102.00	32768.00	32794.00
18.3.48	48.00	0.00	1.00	3101.00	32768.00	32790.00
18.3.49	49.00	0.00	0.00	3100.50	32768.00	32789.00
18.3.50	50.00	0.00	0.00	3100.50	32768.00	32789.00

18.3.51	51.00	0.00	0.00	3100.50	32768.00	32789.00
18.3.52	52.00	0.00	0.00	3100.00	32848.00	32868.00
18.3.53	53.00	0.00	0.00	3100.50	32768.00	32788.00
18.3.54	54.00	0.00	0.00	3100.50	32768.00	32789.00
18.3.55	55.00	0.00	0.00	3100.50	32768.00	32793.00
18.3.56	56.00	0.00	0.00	3124.50	32768.00	32788.00
18.3.57	57.00	0.00	0.00	3098.50	32768.00	32788.00
18.3.58	58.00	0.00	0.00	3120.00	32768.00	32788.00
18.3.59	59.00	0.00	0.00	3120.00	32848.00	32868.00
18.3.60	60.00	0.00	0.00	3124.50	32848.00	32869.00
18.3.61	61.00	0.00	0.00	3124.00	32848.00	32868.00
18.3.62	62.00	0.00	0.00	3111.50	32848.00	32868.00
18.3.63	63.00	0.00	0.00	3120.50	32848.00	32869.00
18.3.64	64.00	0.00	1.00	3102.00	32768.00	32790.00
18.3.65	65.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.66	66.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.67	67.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.68	68.00	0.00	0.00	3102.00	32848.00	32868.00
18.3.69	69.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.70	70.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.71	71.00	0.00	0.00	3102.00	32768.00	32794.00
18.3.72	72.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.73	73.00	0.00	0.00	3100.00	32768.00	32788.00
18.3.74	74.00	0.00	0.00	3128.00	32768.00	32788.00

18.3.75	75.00	0.00	0.00	3128.00	32848.00	32868.00
18.3.76	76.00	0.00	0.00	3128.00	32848.00	32869.00
18.3.77	77.00	0.00	0.00	3128.00	32848.00	32868.00
18.3.78	78.00	0.00	0.00	3114.00	32848.00	32868.00
18.3.79	79.00	0.00	0.00	3128.00	32848.00	32869.00
18.3.80	80.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.81	81.00	0.00	0.00	3100.00	32768.00	32788.00
18.3.82	82.00	0.00	0.00	3128.00	32768.00	32788.00
18.3.83	83.00	0.00	0.00	3128.00	32848.00	32868.00
18.3.84	84.00	0.00	0.00	3128.00	32848.00	32869.00
18.3.85	85.00	0.00	0.00	3128.00	32848.00	32868.00
18.3.86	86.00	0.00	0.00	3114.00	32848.00	32868.00
18.3.87	87.00	0.00	0.00	3128.00	32848.00	32869.00
18.3.88	88.00	0.00	1.00	3102.00	32768.00	32790.00
18.3.89	89.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.90	90.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.91	91.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.92	92.00	0.00	0.00	3102.00	32848.00	32868.00
18.3.93	93.00	0.00	0.00	3102.00	32768.00	32788.00
18.3.94	94.00	0.00	0.00	3102.00	32768.00	32789.00
18.3.95	95.00	0.00	0.00	3102.00	32768.00	32795.00
18.3.96	96.00	0.00	4.00	3102.00	32768.00	32790.00
18.3.97	97.00	0.00	4.00	3102.00	32768.00	32789.00
18.3.98	98.00	0.00	4.00	3102.00	32768.00	32789.00

18.3.99	99.00	0.00	4.00	3102.00	32768.00	32789.00
18.3.100	100.00	0.00	4.00	3102.00	32848.00	32868.00
18.3.101	101.00	0.00	4.00	3102.00	32768.00	32788.00
18.3.102	102.00	0.00	4.00	3102.00	32768.00	32789.00
18.3.103	103.00	0.00	4.00	3102.00	32768.00	32794.00
18.3.104	104.00	0.00	4.00	3128.00	32768.00	32788.00
18.3.105	105.00	0.00	5.00	3100.00	32768.00	32788.00
18.3.106	106.00	0.00	4.00	3128.00	32768.00	32788.00
18.3.107	107.00	0.00	4.00	3128.00	32848.00	32868.00
18.3.108	108.00	0.00	4.00	3128.00	32848.00	32869.00
18.3.109	109.00	0.00	5.00	3128.00	32848.00	32868.00
18.3.110	110.00	0.00	5.00	3114.00	32848.00	32868.00
18.3.111	111.00	0.00	6.00	3128.00	32848.00	32869.00
18.3.112	112.00	0.00	0.00	3121.00	32768.00	32788.00
18.3.113	113.00	0.00	0.00	3096.00	32768.00	32788.00
18.3.114	114.00	0.00	0.00	3120.00	32768.00	32788.00
18.3.115	115.00	0.00	0.00	3120.50	32848.00	32868.00
18.3.116	116.00	0.00	0.00	3121.00	32848.00	32869.00
18.3.117	117.00	0.00	0.00	3120.50	32848.00	32868.00
18.3.118	118.00	0.00	0.00	3109.00	32848.00	32868.00
18.3.119	119.00	0.00	0.00	3120.00	32848.00	32869.00
18.3.120	120.00	0.00	1.00	3099.00	32768.00	32790.00
18.3.121	121.00	0.00	0.00	3099.00	32768.00	32789.00
18.3.122	122.00	0.00	0.00	3098.00	32768.00	32789.00

18.3.123	123.00	0.00	0.00	3099.00	32768.00	32789.00
18.3.124	124.00	0.00	0.00	3099.00	32848.00	32868.00
18.3.125	125.00	0.00	0.00	3098.50	32768.00	32788.00
18.3.126	126.00	0.00	0.00	3099.00	32768.00	32789.00
18.3.127	127.00	0.00	0.00	3099.50	32768.00	32793.00

18.4 L2 Requests (per normUnit)

Metric_ID	Channel	L2 Read	L2 Write	L2 Atomic
18.4.0	0.00	831.02	10876.44	0.00
18.4.1	1.00	831.11	10876.44	0.00
18.4.2	2.00	831.08	10876.45	0.00
18.4.3	3.00	856.73	10876.46	0.00
18.4.4	4.00	856.76	10876.44	0.00
18.4.5	5.00	856.71	10876.44	0.00
18.4.6	6.00	856.81	10876.44	0.00
18.4.7	7.00	854.60	10876.44	0.00
18.4.8	8.00	830.36	10876.44	0.00
18.4.9	9.00	837.72	10876.44	0.00
18.4.10	10.00	837.78	10876.44	0.00
18.4.11	11.00	837.72	10876.44	0.00
18.4.12	12.00	861.12	10876.44	0.00
18.4.13	13.00	835.53	10876.44	0.00
18.4.14	14.00	837.75	10876.44	0.00
18.4.15	15.00	834.13	10876.44	0.00
18.4.16	16.00	830.23	10876.44	0.00

18.4.17	17.00	837.32	10876.44	0.00
18.4.18	18.00	837.38	10876.44	0.00
18.4.19	19.00	837.32	10876.44	0.00
18.4.20	20.00	860.67	10876.44	0.00
18.4.21	21.00	835.09	10876.44	0.00
18.4.22	22.00	837.35	10876.44	0.00
18.4.23	23.00	833.87	10876.44	0.00
18.4.24	24.00	830.89	10876.45	0.00
18.4.25	25.00	830.98	10876.45	0.00
18.4.26	26.00	830.95	10876.44	0.00
18.4.27	27.00	856.59	10876.45	0.00
18.4.28	28.00	856.63	10876.44	0.00
18.4.29	29.00	856.58	10876.44	0.00
18.4.30	30.00	856.68	10876.44	0.00
18.4.31	31.00	854.42	10876.44	0.00
18.4.32	32.00	830.65	10876.44	0.00
18.4.33	33.00	830.74	10876.44	0.00
18.4.34	34.00	830.71	10876.44	0.00
18.4.35	35.00	856.35	10876.45	0.00
18.4.36	36.00	856.39	10876.44	0.00
18.4.37	37.00	856.34	10876.44	0.00
18.4.38	38.00	856.44	10876.44	0.00
18.4.39	39.00	854.23	10876.44	0.00
18.4.40	40.00	829.99	10876.44	0.00

18.4.41	41.00	836.99	10876.44	0.00
18.4.42	42.00	837.05	10876.44	0.00
18.4.43	43.00	836.99	10876.44	0.00
18.4.44	44.00	860.39	10876.44	0.00
18.4.45	45.00	834.81	10876.44	0.00
18.4.46	46.00	837.02	10876.44	0.00
18.4.47	47.00	833.58	10876.44	0.00
18.4.48	48.00	830.31	10876.44	0.00
18.4.49	49.00	837.40	10876.44	0.00
18.4.50	50.00	837.45	10876.44	0.00
18.4.51	51.00	837.39	10876.44	0.00
18.4.52	52.00	860.72	10876.44	0.00
18.4.53	53.00	835.13	10876.44	0.00
18.4.54	54.00	837.42	10876.44	0.00
18.4.55	55.00	833.94	10876.44	0.00
18.4.56	56.00	830.97	10876.44	0.00
18.4.57	57.00	831.05	10876.44	0.00
18.4.58	58.00	831.02	10876.44	0.00
18.4.59	59.00	856.67	10876.45	0.00
18.4.60	60.00	856.70	10876.44	0.00
18.4.61	61.00	856.66	10876.44	0.00
18.4.62	62.00	856.75	10876.44	0.00
18.4.63	63.00	854.47	10876.44	0.00
18.4.64	64.00	830.15	10876.44	0.00

18.4.65	65.00	837.49	10876.44	0.00
18.4.66	66.00	837.54	10876.44	0.00
18.4.67	67.00	837.49	10876.44	0.00
18.4.68	68.00	860.94	10876.44	0.00
18.4.69	69.00	835.36	10876.44	0.00
18.4.70	70.00	837.51	10876.44	0.00
18.4.71	71.00	833.90	10876.44	0.00
18.4.72	72.00	830.82	10876.44	0.00
18.4.73	73.00	830.90	10876.44	0.00
18.4.74	74.00	830.87	10876.44	0.00
18.4.75	75.00	856.52	10876.44	0.00
18.4.76	76.00	856.55	10876.44	0.00
18.4.77	77.00	856.51	10876.44	0.00
18.4.78	78.00	856.60	10876.44	0.00
18.4.79	79.00	854.45	10876.44	0.00
18.4.80	80.00	831.31	10876.44	0.00
18.4.81	81.00	831.39	10876.44	0.00
18.4.82	82.00	831.36	10876.44	0.00
18.4.83	83.00	857.01	10876.44	0.00
18.4.84	84.00	857.04	10876.44	0.00
18.4.85	85.00	857.00	10876.44	0.00
18.4.86	86.00	857.09	10876.44	0.00
18.4.87	87.00	854.78	10876.44	0.00
18.4.88	88.00	830.64	10876.44	0.00

18.4.89	89.00	838.00	10876.44	0.00
18.4.90	90.00	838.05	10876.44	0.00
18.4.91	91.00	838.00	10876.44	0.00
18.4.92	92.00	861.29	10876.44	0.00
18.4.93	93.00	835.71	10876.44	0.00
18.4.94	94.00	838.02	10876.44	0.00
18.4.95	95.00	834.41	10876.44	0.00
18.4.96	96.00	831.60	10876.47	0.00
18.4.97	97.00	839.14	10876.53	0.00
18.4.98	98.00	838.38	10876.63	0.00
18.4.99	99.00	838.40	10876.47	0.00
18.4.100	100.00	862.53	10876.46	0.00
18.4.101	101.00	836.73	10876.47	0.00
18.4.102	102.00	839.23	10876.47	0.00
18.4.103	103.00	835.52	10876.47	0.00
18.4.104	104.00	831.55	10876.46	0.00
18.4.105	105.00	831.65	10876.45	0.00
18.4.106	106.00	831.60	10876.46	0.00
18.4.107	107.00	857.24	10876.45	0.00
18.4.108	108.00	857.24	10876.44	0.00
18.4.109	109.00	857.16	10876.45	0.00
18.4.110	110.00	857.24	10876.45	0.00
18.4.111	111.00	855.15	10876.56	0.00
18.4.112	112.00	831.57	10876.44	0.00

18.4.113	113.00	831.66	10876.44	0.00
18.4.114	114.00	831.63	10876.44	0.00
18.4.115	115.00	857.27	10876.45	0.00
18.4.116	116.00	857.32	10876.45	0.00
18.4.117	117.00	857.26	10876.46	0.00
18.4.118	118.00	857.37	10876.44	0.00
18.4.119	119.00	855.03	10876.45	0.00
18.4.120	120.00	830.91	10876.44	0.00
18.4.121	121.00	838.48	10876.45	0.00
18.4.122	122.00	838.55	10876.44	0.00
18.4.123	123.00	838.48	10876.45	0.00
18.4.124	124.00	861.78	10876.44	0.00
18.4.125	125.00	836.19	10876.45	0.00
18.4.126	126.00	838.58	10876.44	0.00
18.4.127	127.00	834.84	10876.45	0.00

18.5 L2-Fabric Requests (per normUnit)

Metric_ID	Channel	L2-Fabric Read	L2-Fabric Write and Atomic
18.5.0 0.00	0.00	461.93	10570.67
18.5.1 0.00	1.00	461.33	10570.67
18.5.2 0.00	2.00	463.21	10570.67

18.5.3 0.00	3.00	461.93	10570.67
18.5.4 0.00	4.00	463.39	10570.67
18.5.5 0.00	5.00	463.48	10570.67
18.5.6 0.00	6.00	462.11	10570.67
18.5.7 0.00	7.00	463.22	10570.67
18.5.8 0.00	8.00	462.30	10570.67
18.5.9 0.00	9.00	462.25	10570.67
18.5.10 0.00	10.00	462.40	10570.67
18.5.11 0.00	11.00	462.28	10570.67
18.5.12 0.00	12.00	461.91	10570.67
18.5.13 0.00	13.00	463.18	10570.67
18.5.14 0.00	14.00	462.41	10570.68

18.5.15 0.00	15.00	461.91	10570.68
18.5.16 0.00	16.00	462.29	10570.67
18.5.17 0.00	17.00	462.24	10570.67
18.5.18 0.00	18.00	462.40	10570.67
18.5.19 0.00	19.00	462.28	10570.67
18.5.20 0.00	20.00	461.91	10570.68
18.5.21 0.00	21.00	463.18	10570.68
18.5.22 0.00	22.00	462.41	10570.67
18.5.23 0.00	23.00	461.91	10570.67
18.5.24 0.00	24.00	461.93	10570.67
18.5.25 0.00	25.00	461.33	10570.67
18.5.26 0.00	26.00	463.21	10570.67

18.5.27 0.00	27.00	461.93	10570.67
18.5.28 0.00	28.00	463.38	10570.67
18.5.29 0.00	29.00	463.48	10570.67
18.5.30 0.00	30.00	462.11	10570.67
18.5.31 0.00	31.00	463.22	10570.67
18.5.32 0.00	32.00	461.92	10570.67
18.5.33 0.00	33.00	461.32	10570.67
18.5.34 0.00	34.00	463.20	10570.67
18.5.35 0.00	35.00	461.92	10570.67
18.5.36 0.00	36.00	463.37	10570.67
18.5.37 0.00	37.00	463.47	10570.67
18.5.38 0.00	38.00	462.10	10570.67

18.5.39 0.00	39.00	463.21	10570.67
18.5.40 0.00	40.00	462.29	10570.67
18.5.41 0.00	41.00	462.23	10570.67
18.5.42 0.00	42.00	462.39	10570.67
18.5.43 0.00	43.00	462.27	10570.67
18.5.44 0.00	44.00	461.90	10570.67
18.5.45 0.00	45.00	463.17	10570.67
18.5.46 0.00	46.00	462.41	10570.67
18.5.47 0.00	47.00	461.90	10570.67
18.5.48 0.00	48.00	462.28	10570.67
18.5.49 0.00	49.00	462.23	10570.67
18.5.50 0.00	50.00	462.39	10570.67

18.5.51 0.00	51.00	462.27	10570.67
18.5.52 0.00	52.00	461.90	10570.67
18.5.53 0.00	53.00	463.17	10570.67
18.5.54 0.00	54.00	462.41	10570.67
18.5.55 0.00	55.00	461.90	10570.67
18.5.56 0.00	56.00	461.92	10570.67
18.5.57 0.00	57.00	461.32	10570.67
18.5.58 0.00	58.00	463.20	10570.67
18.5.59 0.00	59.00	461.92	10570.67
18.5.60 0.00	60.00	463.37	10570.67
18.5.61 0.00	61.00	463.47	10570.67
18.5.62 0.00	62.00	462.10	10570.67

18.5.63 0.00	63.00	463.21	10570.67
18.5.64 0.00	64.00	462.28	10570.67
18.5.65 0.00	65.00	462.23	10570.67
18.5.66 0.00	66.00	462.39	10570.67
18.5.67 0.00	67.00	462.27	10570.67
18.5.68 0.00	68.00	461.90	10570.67
18.5.69 0.00	69.00	463.17	10570.67
18.5.70 0.00	70.00	462.40	10570.67
18.5.71 0.00	71.00	461.89	10570.67
18.5.72 0.00	72.00	461.92	10570.67
18.5.73 0.00	73.00	461.32	10570.67
18.5.74 0.00	74.00	463.19	10570.67

18.5.75 0.00	75.00	461.92	10570.67
18.5.76 0.00	76.00	463.37	10570.67
18.5.77 0.00	77.00	463.47	10570.67
18.5.78 0.00	78.00	462.10	10570.67
18.5.79 0.00	79.00	463.21	10570.67
18.5.80 0.00	80.00	461.92	10570.67
18.5.81 0.00	81.00	461.32	10570.67
18.5.82 0.00	82.00	463.19	10570.67
18.5.83 0.00	83.00	461.92	10570.67
18.5.84 0.00	84.00	463.37	10570.67
18.5.85 0.00	85.00	463.46	10570.67
18.5.86 0.00	86.00	462.10	10570.67

18.5.87 0.00	87.00	463.22	10570.67
18.5.88 0.00	88.00	462.28	10570.67
18.5.89 0.00	89.00	462.23	10570.67
18.5.90 0.00	90.00	462.39	10570.67
18.5.91 0.00	91.00	462.27	10570.67
18.5.92 0.00	92.00	461.90	10570.67
18.5.93 0.00	93.00	463.17	10570.67
18.5.94 0.00	94.00	462.40	10570.67
18.5.95 0.00	95.00	461.89	10570.67
18.5.96 0.00	96.00	463.33	10570.68
18.5.97 0.00	97.00	463.27	10570.85
18.5.98 0.00	98.00	463.25	10570.69

18.5.99 0.00	99.00	463.34	10570.70
18.5.100 0.00	100.00	462.11	10570.67
18.5.101 0.00	101.00	463.47	10570.67
18.5.102 0.00	102.00	463.49	10570.67
18.5.103 0.00	103.00	462.97	10570.67
18.5.104 0.00	104.00	462.22	10570.68
18.5.105 0.00	105.00	461.72	10570.79
18.5.106 0.00	106.00	463.48	10570.69
18.5.107 0.00	107.00	462.23	10570.70
18.5.108 0.00	108.00	463.60	10570.68
18.5.109 0.00	109.00	463.70	10570.69
18.5.110 0.00	110.00	462.38	10570.68

18.5.111 0.00	111.00	463.45	10570.69
18.5.112 0.00	112.00	461.94	10570.67
18.5.113 0.00	113.00	461.34	10570.67
18.5.114 0.00	114.00	463.21	10570.67
18.5.115 0.00	115.00	461.93	10570.67
18.5.116 0.00	116.00	463.39	10570.67
18.5.117 0.00	117.00	463.49	10570.67
18.5.118 0.00	118.00	462.12	10570.67
18.5.119 0.00	119.00	463.23	10570.67
18.5.120 0.00	120.00	462.30	10570.68
18.5.121 0.00	121.00	462.24	10570.68
18.5.122 0.00	122.00	462.40	10570.67

18.5.123 0.00	123.00	462.28	10570.67
18.5.124 0.00	124.00	461.92	10570.67
18.5.125 0.00	125.00	463.18	10570.67
18.5.126 0.00	126.00	462.48	10570.67
18.5.127 0.00	127.00	461.97	10570.67

18.6 L2-Fabric Read Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.6.0	0.00	120.00	492.62	503.19	513.92	696.78
18.6.1	1.00	125.00	506.23	516.25	528.15	717.33
18.6.2	2.00	122.67	278.62	506.95	531.64	246796.50
18.6.3	3.00	132.00	516.76	526.30	538.10	725.67
18.6.4	4.00	125.00	262.62	504.17	531.26	246789.50
18.6.5	5.00	128.00	264.50	506.73	527.17	246858.50
18.6.6	6.00	179.75	288.12	538.87	564.85	61939.50
18.6.7	7.00	123.00	283.50	524.76	546.72	246791.75
18.6.8	8.00	116.00	494.27	505.50	517.92	691.43
18.6.9	9.00	127.00	516.77	527.80	539.95	726.17
18.6.10	10.00	123.00	500.15	521.05	532.56	719.16
18.6.11	11.00	121.00	517.17	529.16	540.08	720.87

18.6.12	12.00	125.00	516.50	527.64	541.50	719.28
18.6.13	13.00	160.83	265.50	505.21	527.69	725.60
18.6.14	14.00	143.00	526.97	548.10	559.18	751.20
18.6.15	15.00	124.00	521.46	534.97	545.91	728.91
18.6.16	16.00	129.00	504.54	518.89	528.09	662.73
18.6.17	17.00	285.40	521.67	537.22	548.25	682.16
18.6.18	18.00	228.00	507.00	527.13	536.06	675.99
18.6.19	19.00	122.00	526.31	535.87	548.72	734.00
18.6.20	20.00	126.00	522.62	534.97	544.80	682.60
18.6.21	21.00	162.00	268.00	509.39	534.59	687.67
18.6.22	22.00	142.00	532.18	555.18	566.12	710.47
18.6.23	23.00	174.50	526.99	537.25	550.49	691.71
18.6.24	24.00	120.00	495.81	507.61	516.78	649.40
18.6.25	25.00	126.00	510.48	524.43	535.38	679.65
18.6.26	26.00	123.33	279.50	512.77	537.04	686.32
18.6.27	27.00	152.75	521.40	531.94	544.65	686.78
18.6.28	28.00	125.00	263.50	508.42	534.85	678.38
18.6.29	29.00	127.00	265.50	510.80	531.97	724.40
18.6.30	30.00	180.75	289.38	545.59	569.07	718.10
18.6.31	31.00	126.00	283.50	527.81	549.15	1221.00
18.6.32	32.00	216.00	501.29	512.50	524.22	686.46
18.6.33	33.00	238.00	520.11	531.84	543.69	709.15
18.6.34	34.00	228.33	447.12	517.98	534.12	247016.75
18.6.35	35.00	288.50	528.15	537.76	547.35	722.87

18.6.36	36.00	231.00	411.31	516.76	539.18	246989.00
18.6.37	37.00	239.00	421.25	517.56	540.41	247044.00
18.6.38	38.00	138.00	392.06	537.39	561.13	62060.75
18.6.39	39.00	232.00	432.56	529.09	548.07	247005.75
18.6.40	40.00	152.75	497.49	508.19	522.78	671.38
18.6.41	41.00	159.75	516.59	529.89	543.70	689.71
18.6.42	42.00	122.00	511.19	530.61	543.21	707.26
18.6.43	43.00	123.00	525.31	535.51	547.54	761.00
18.6.44	44.00	191.00	522.49	534.49	548.61	702.42
18.6.45	45.00	321.00	382.50	518.04	539.02	1199.00
18.6.46	46.00	136.00	545.05	566.62	576.42	745.15
18.6.47	47.00	124.00	540.92	555.55	565.91	748.62
18.6.48	48.00	152.00	500.84	512.15	524.10	696.13
18.6.49	49.00	159.00	518.16	528.58	541.24	690.52
18.6.50	50.00	122.00	512.94	531.74	542.66	709.80
18.6.51	51.00	128.00	524.38	534.40	543.99	718.34
18.6.52	52.00	196.00	524.79	539.06	550.60	721.12
18.6.53	53.00	323.75	375.41	521.53	540.90	700.58
18.6.54	54.00	135.00	547.29	569.19	580.42	750.94
18.6.55	55.00	124.00	544.84	555.79	565.87	733.22
18.6.56	56.00	220.00	503.69	514.92	526.91	687.08
18.6.57	57.00	237.00	524.84	535.81	546.92	702.75
18.6.58	58.00	237.33	443.25	519.99	538.48	246732.50
18.6.59	59.00	282.50	531.63	541.21	550.38	723.04

18.6.60	60.00	232.00	410.69	519.91	541.19	246675.75
18.6.61	61.00	240.00	417.62	518.53	539.07	246726.75
18.6.62	62.00	154.00	385.81	542.47	561.62	61967.25
18.6.63	63.00	231.00	431.44	528.78	545.90	246624.50
18.6.64	64.00	212.00	507.12	516.11	525.13	691.47
18.6.65	65.00	154.00	525.56	532.76	542.08	707.95
18.6.66	66.00	228.00	522.18	535.58	546.51	719.47
18.6.67	67.00	236.00	529.42	538.29	549.26	768.00
18.6.68	68.00	225.00	526.92	536.72	547.28	707.94
18.6.69	69.00	237.25	259.00	518.88	538.68	710.54
18.6.70	70.00	247.00	550.84	566.16	576.94	750.89
18.6.71	71.00	168.00	545.14	556.46	566.10	733.80
18.6.72	72.00	129.00	500.65	510.09	517.51	681.69
18.6.73	73.00	293.00	522.94	531.31	538.81	718.69
18.6.74	74.00	233.25	318.75	516.67	532.70	247162.50
18.6.75	75.00	299.00	527.97	534.94	546.64	722.36
18.6.76	76.00	235.50	307.60	521.15	536.45	247132.25
18.6.77	77.00	243.25	313.20	520.54	533.77	247155.50
18.6.78	78.00	240.25	270.31	542.75	558.83	61959.00
18.6.79	79.00	246.50	314.34	529.50	544.64	247051.25
18.6.80	80.00	130.00	500.85	509.31	519.58	697.18
18.6.81	81.00	299.00	519.43	527.20	537.32	718.54
18.6.82	82.00	233.75	317.75	514.88	529.56	246959.75
18.6.83	83.00	298.00	523.19	531.59	542.30	709.71

18.6.84	84.00	234.25	314.60	518.31	532.78	246924.25
18.6.85	85.00	242.75	318.25	515.45	530.95	246951.50
18.6.86	86.00	234.00	270.62	540.23	556.48	61917.25
18.6.87	87.00	247.25	314.62	522.55	538.45	246899.75
18.6.88	88.00	212.00	500.74	509.74	519.40	693.18
18.6.89	89.00	146.00	518.35	527.31	536.97	696.23
18.6.90	90.00	228.00	516.47	528.86	537.77	712.14
18.6.91	91.00	228.00	522.68	530.38	541.53	711.39
18.6.92	92.00	226.00	523.86	531.56	544.00	718.26
18.6.93	93.00	223.60	261.39	517.25	532.99	699.48
18.6.94	94.00	247.00	547.93	559.21	570.12	752.63
18.6.95	95.00	125.00	537.73	548.29	558.07	728.60
18.6.96	96.00	117.00	221.81	502.53	517.76	791.86
18.6.97	97.00	133.00	228.75	525.03	541.94	694.13
18.6.98	98.00	124.00	231.17	515.03	532.38	687.15
18.6.99	99.00	123.50	232.00	526.08	540.66	711.76
18.6.100	100.00	138.00	519.31	533.28	542.15	877.00
18.6.101	101.00	119.00	146.06	339.25	533.65	693.60
18.6.102	102.00	135.25	241.56	538.35	560.63	911.00
18.6.103	103.00	126.25	234.56	527.61	545.21	712.49
18.6.104	104.00	119.00	498.49	510.46	519.00	668.72
18.6.105	105.00	133.00	477.00	523.87	536.39	914.00
18.6.106	106.00	122.75	167.75	515.17	542.27	246732.75
18.6.107	107.00	121.25	523.61	537.56	546.82	704.54

18.6.108	108.00	125.00	163.69	509.48	538.20	246738.75
18.6.109	109.00	126.50	164.19	514.13	535.45	246765.75
18.6.110	110.00	138.50	162.38	545.60	572.15	61774.75
18.6.111	111.00	126.50	165.62	526.07	556.34	246654.50
18.6.112	112.00	122.00	501.90	511.89	520.84	673.59
18.6.113	113.00	265.00	519.41	530.14	541.08	741.00
18.6.114	114.00	132.00	170.44	518.71	539.19	246246.25
18.6.115	115.00	297.00	527.95	537.62	545.21	922.00
18.6.116	116.00	124.75	161.85	513.95	533.32	246232.25
18.6.117	117.00	127.00	163.50	514.72	532.42	246273.25
18.6.118	118.00	138.75	158.31	546.90	568.52	61660.50
18.6.119	119.00	134.00	166.38	530.92	551.84	246196.75
18.6.120	120.00	117.00	504.27	513.61	524.12	761.40
18.6.121	121.00	349.00	526.99	536.83	547.70	700.35
18.6.122	122.00	229.50	505.92	526.36	538.09	733.56
18.6.123	123.00	230.00	527.68	536.37	546.09	708.14
18.6.124	124.00	126.00	522.43	530.79	542.24	693.60
18.6.125	125.00	120.50	137.50	512.82	533.85	701.17
18.6.126	126.00	136.00	529.57	550.68	562.52	763.34
18.6.127	127.00	229.00	527.32	539.55	547.71	712.76

18.7 L2-Fabric Write and Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.7.0	0.00	302.99	308.23	341.26	353.22	497.03
18.7.1	1.00	308.57	314.49	333.39	342.07	471.23

18.7.2	2.00	313.42	320.09	351.59	361.56	505.83
18.7.3	3.00	314.40	320.63	349.27	359.20	499.12
18.7.4	4.00	310.28	316.02	369.24	384.42	535.15
18.7.5	5.00	311.79	317.93	361.97	374.69	519.88
18.7.6	6.00	324.07	330.82	385.22	397.13	544.04
18.7.7	7.00	321.16	326.49	370.47	382.61	524.99
18.7.8	8.00	302.78	308.28	370.09	462.51	661.52
18.7.9	9.00	311.63	317.81	396.12	466.49	656.99
18.7.10	10.00	308.49	314.34	378.29	459.45	652.32
18.7.11	11.00	315.31	321.44	396.29	450.94	659.36
18.7.12	12.00	309.93	315.45	381.63	425.35	594.79
18.7.13	13.00	312.33	317.49	386.80	412.28	580.54
18.7.14	14.00	0.00	324.92	386.01	434.51	637.75
18.7.15	15.00	0.00	319.67	394.58	455.83	666.55
18.7.16	16.00	303.86	310.36	366.25	456.48	651.59
18.7.17	17.00	313.53	320.19	392.77	463.83	651.98
18.7.18	18.00	309.75	316.15	374.59	455.81	649.36
18.7.19	19.00	317.92	323.99	393.97	446.57	654.39
18.7.20	20.00	0.00	317.99	378.09	420.20	597.62
18.7.21	21.00	0.00	319.98	383.74	409.56	584.56
18.7.22	22.00	320.54	327.34	382.11	432.43	641.88
18.7.23	23.00	0.00	321.01	391.12	453.83	665.35
18.7.24	24.00	304.22	309.97	340.41	351.62	485.80
18.7.25	25.00	310.13	316.01	333.03	341.65	468.91

18.7.26	26.00	318.23	323.81	350.77	361.14	499.24
18.7.27	27.00	317.68	323.04	348.07	358.47	496.14
18.7.28	28.00	314.11	319.92	366.43	382.79	542.27
18.7.29	29.00	315.66	321.68	358.80	372.23	512.90
18.7.30	30.00	327.69	334.42	382.02	395.63	538.91
18.7.31	31.00	323.20	329.45	368.78	379.55	517.64
18.7.32	32.00	305.60	309.32	369.46	455.00	669.34
18.7.33	33.00	314.66	319.35	396.48	466.48	661.55
18.7.34	34.00	309.74	315.18	376.85	452.53	647.09
18.7.35	35.00	318.28	323.04	397.29	448.42	634.47
18.7.36	36.00	310.84	316.33	382.47	419.34	592.59
18.7.37	37.00	314.13	319.00	384.27	412.83	580.73
18.7.38	38.00	320.62	326.07	397.23	426.18	625.82
18.7.39	39.00	315.17	320.25	381.43	445.34	636.23
18.7.40	40.00	305.13	309.74	339.11	349.80	487.96
18.7.41	41.00	311.54	315.58	331.71	340.01	469.83
18.7.42	42.00	318.59	322.36	345.90	355.79	494.79
18.7.43	43.00	318.12	322.08	343.82	353.12	490.82
18.7.44	44.00	314.43	318.47	366.03	379.97	542.57
18.7.45	45.00	316.10	321.20	358.27	369.44	534.19
18.7.46	46.00	328.31	333.01	382.81	394.94	557.16
18.7.47	47.00	0.00	328.20	362.27	372.00	526.01
18.7.48	48.00	305.23	309.15	336.68	347.74	476.95
18.7.49	49.00	311.60	315.51	327.86	337.95	474.86

18.7.50	50.00	315.87	320.51	342.12	353.69	486.38
18.7.51	51.00	317.23	321.23	341.17	352.06	488.47
18.7.52	52.00	312.31	316.74	364.39	376.02	534.68
18.7.53	53.00	313.62	319.06	356.77	367.52	527.40
18.7.54	54.00	326.43	331.30	379.60	392.11	550.64
18.7.55	55.00	0.00	326.30	359.37	370.95	517.93
18.7.56	56.00	304.68	308.82	367.72	448.32	660.48
18.7.57	57.00	314.53	318.30	394.29	463.67	656.15
18.7.58	58.00	310.83	315.38	375.58	446.97	641.72
18.7.59	59.00	318.29	321.80	395.84	444.77	630.12
18.7.60	60.00	311.91	315.51	378.75	415.83	583.47
18.7.61	61.00	314.46	318.12	382.40	409.62	574.70
18.7.62	62.00	321.21	325.67	394.43	421.93	612.34
18.7.63	63.00	314.89	319.66	380.99	443.46	630.21
18.7.64	64.00	304.39	311.28	341.72	354.49	482.99
18.7.65	65.00	311.74	317.88	332.48	343.28	461.21
18.7.66	66.00	316.85	323.11	348.04	359.68	483.44
18.7.67	67.00	317.53	323.71	346.20	357.75	480.86
18.7.68	68.00	311.86	318.16	370.04	384.50	547.33
18.7.69	69.00	314.46	320.27	361.66	374.07	516.54
18.7.70	70.00	327.54	333.32	385.79	397.61	550.72
18.7.71	71.00	323.53	328.79	363.54	374.87	511.53
18.7.72	72.00	303.55	310.03	370.23	461.99	654.93
18.7.73	73.00	313.79	319.86	397.11	467.41	654.59

18.7.74	74.00	310.80	317.17	377.35	461.21	650.47
18.7.75	75.00	318.31	323.85	398.91	456.15	647.27
18.7.76	76.00	310.73	316.95	383.54	421.17	586.77
18.7.77	77.00	313.12	319.52	385.94	414.07	560.44
18.7.78	78.00	321.53	328.36	397.74	421.23	585.49
18.7.79	79.00	316.29	322.29	383.10	445.25	639.19
18.7.80	80.00	305.09	310.58	371.29	456.79	652.71
18.7.81	81.00	315.36	320.90	398.86	464.63	651.46
18.7.82	82.00	310.86	316.80	378.38	458.27	648.08
18.7.83	83.00	320.12	325.20	399.14	450.83	643.88
18.7.84	84.00	312.41	318.00	384.05	419.98	589.83
18.7.85	85.00	314.76	320.39	389.13	411.67	559.42
18.7.86	86.00	322.74	328.61	401.71	421.61	577.18
18.7.87	87.00	316.12	322.21	382.67	443.38	619.25
18.7.88	88.00	307.38	312.16	341.86	353.07	487.93
18.7.89	89.00	313.09	317.76	332.01	344.48	472.43
18.7.90	90.00	320.61	324.72	348.54	359.09	493.95
18.7.91	91.00	319.68	324.73	345.90	356.89	488.64
18.7.92	92.00	315.45	320.26	368.14	383.58	547.28
18.7.93	93.00	316.91	321.93	359.88	373.75	521.66
18.7.94	94.00	329.97	335.36	385.12	397.43	551.98
18.7.95	95.00	326.04	330.77	364.36	374.22	510.99
18.7.96	96.00	0.00	309.79	365.37	458.77	650.00
18.7.97	97.00	0.00	318.82	347.93	468.34	653.54

18.7.98	98.00	0.00	316.00	368.88	450.30	644.43
18.7.99	99.00	0.00	324.49	381.89	443.86	640.33
18.7.100	100.00	312.28	317.08	377.14	419.75	591.45
18.7.101	101.00	315.29	319.50	381.14	411.27	566.41
18.7.102	102.00	321.39	326.88	381.06	434.12	606.79
18.7.103	103.00	0.00	321.40	386.21	455.59	646.25
18.7.104	104.00	0.00	311.58	336.27	348.87	510.50
18.7.105	105.00	0.00	315.32	323.27	337.23	550.50
18.7.106	106.00	0.00	324.26	347.93	358.07	535.94
18.7.107	107.00	0.00	323.88	344.12	355.08	534.75
18.7.108	108.00	0.00	318.93	359.65	378.22	549.34
18.7.109	109.00	0.00	320.75	353.27	369.13	535.11
18.7.110	110.00	0.00	333.44	379.29	395.66	572.85
18.7.111	111.00	0.00	328.85	362.68	374.00	559.93
18.7.112	112.00	303.56	307.85	335.71	346.76	493.06
18.7.113	113.00	0.00	313.80	328.53	338.69	512.25
18.7.114	114.00	316.75	320.20	347.30	356.41	510.59
18.7.115	115.00	0.00	320.39	345.28	354.08	508.61
18.7.116	116.00	310.42	313.90	361.95	377.32	535.20
18.7.117	117.00	0.00	315.83	354.78	367.51	515.32
18.7.118	118.00	325.15	328.96	380.49	392.97	554.77
18.7.119	119.00	0.00	325.20	362.37	372.81	540.68
18.7.120	120.00	0.00	307.47	364.46	453.09	650.42
18.7.121	121.00	0.00	317.21	388.22	464.99	652.00

18.7.122	122.00	310.51	314.31	371.95	447.66	625.83
18.7.123	123.00	0.00	321.31	388.82	439.41	630.55
18.7.124	124.00	309.75	314.03	378.40	418.60	572.64
18.7.125	125.00	313.30	316.84	381.61	409.24	556.17
18.7.126	126.00	320.39	324.38	382.02	433.17	607.84
18.7.127	127.00	0.00	319.04	386.52	453.86	641.72

18.8 L2-Fabric Atomic Latency (Cycles)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.8.0	0.00	0.00	0.00	0.00	0.00	0.00
18.8.1	1.00	0.00	0.00	0.00	0.00	0.00
18.8.2	2.00	0.00	0.00	0.00	0.00	0.00
18.8.3	3.00	0.00	0.00	0.00	0.00	0.00
18.8.4	4.00	0.00	0.00	0.00	0.00	0.00
18.8.5	5.00	0.00	0.00	0.00	0.00	0.00
18.8.6	6.00	0.00	0.00	0.00	0.00	0.00
18.8.7	7.00	0.00	0.00	0.00	0.00	0.00
18.8.8	8.00	0.00	0.00	0.00	0.00	0.00
18.8.9	9.00	0.00	0.00	0.00	0.00	0.00
18.8.10	10.00	0.00	0.00	0.00	0.00	0.00
18.8.11	11.00	0.00	0.00	0.00	0.00	0.00
18.8.12	12.00	0.00	0.00	0.00	0.00	0.00
18.8.13	13.00	0.00	0.00	0.00	0.00	0.00
18.8.14	14.00	0.00	0.00	0.00	0.00	0.00
18.8.15	15.00	0.00	0.00	0.00	0.00	0.00

18.8.16	16.00	0.00	0.00	0.00	0.00	0.00
18.8.17	17.00	0.00	0.00	0.00	0.00	0.00
18.8.18	18.00	0.00	0.00	0.00	0.00	0.00
18.8.19	19.00	0.00	0.00	0.00	0.00	0.00
18.8.20	20.00	0.00	0.00	0.00	0.00	0.00
18.8.21	21.00	0.00	0.00	0.00	0.00	0.00
18.8.22	22.00	0.00	0.00	0.00	0.00	0.00
18.8.23	23.00	0.00	0.00	0.00	0.00	0.00
18.8.24	24.00	0.00	0.00	0.00	0.00	0.00
18.8.25	25.00	0.00	0.00	0.00	0.00	0.00
18.8.26	26.00	0.00	0.00	0.00	0.00	0.00
18.8.27	27.00	0.00	0.00	0.00	0.00	0.00
18.8.28	28.00	0.00	0.00	0.00	0.00	0.00
18.8.29	29.00	0.00	0.00	0.00	0.00	0.00
18.8.30	30.00	0.00	0.00	0.00	0.00	0.00
18.8.31	31.00	0.00	0.00	0.00	0.00	0.00
18.8.32	32.00	0.00	0.00	0.00	0.00	0.00
18.8.33	33.00	0.00	0.00	0.00	0.00	0.00
18.8.34	34.00	0.00	0.00	0.00	0.00	0.00
18.8.35	35.00	0.00	0.00	0.00	0.00	0.00
18.8.36	36.00	0.00	0.00	0.00	0.00	0.00
18.8.37	37.00	0.00	0.00	0.00	0.00	0.00
18.8.38	38.00	0.00	0.00	0.00	0.00	0.00
18.8.39	39.00	0.00	0.00	0.00	0.00	0.00

18.8.40	40.00	0.00	0.00	0.00	0.00	0.00
18.8.41	41.00	0.00	0.00	0.00	0.00	0.00
18.8.42	42.00	0.00	0.00	0.00	0.00	0.00
18.8.43	43.00	0.00	0.00	0.00	0.00	0.00
18.8.44	44.00	0.00	0.00	0.00	0.00	0.00
18.8.45	45.00	0.00	0.00	0.00	0.00	0.00
18.8.46	46.00	0.00	0.00	0.00	0.00	0.00
18.8.47	47.00	0.00	0.00	0.00	0.00	0.00
18.8.48	48.00	0.00	0.00	0.00	0.00	0.00
18.8.49	49.00	0.00	0.00	0.00	0.00	0.00
18.8.50	50.00	0.00	0.00	0.00	0.00	0.00
18.8.51	51.00	0.00	0.00	0.00	0.00	0.00
18.8.52	52.00	0.00	0.00	0.00	0.00	0.00
18.8.53	53.00	0.00	0.00	0.00	0.00	0.00
18.8.54	54.00	0.00	0.00	0.00	0.00	0.00
18.8.55	55.00	0.00	0.00	0.00	0.00	0.00
18.8.56	56.00	0.00	0.00	0.00	0.00	0.00
18.8.57	57.00	0.00	0.00	0.00	0.00	0.00
18.8.58	58.00	0.00	0.00	0.00	0.00	0.00
18.8.59	59.00	0.00	0.00	0.00	0.00	0.00
18.8.60	60.00	0.00	0.00	0.00	0.00	0.00
18.8.61	61.00	0.00	0.00	0.00	0.00	0.00
18.8.62	62.00	0.00	0.00	0.00	0.00	0.00
18.8.63	63.00	0.00	0.00	0.00	0.00	0.00

18.8.64	64.00	0.00	0.00	0.00	0.00	0.00
18.8.65	65.00	0.00	0.00	0.00	0.00	0.00
18.8.66	66.00	0.00	0.00	0.00	0.00	0.00
18.8.67	67.00	0.00	0.00	0.00	0.00	0.00
18.8.68	68.00	0.00	0.00	0.00	0.00	0.00
18.8.69	69.00	0.00	0.00	0.00	0.00	0.00
18.8.70	70.00	0.00	0.00	0.00	0.00	0.00
18.8.71	71.00	0.00	0.00	0.00	0.00	0.00
18.8.72	72.00	0.00	0.00	0.00	0.00	0.00
18.8.73	73.00	0.00	0.00	0.00	0.00	0.00
18.8.74	74.00	0.00	0.00	0.00	0.00	0.00
18.8.75	75.00	0.00	0.00	0.00	0.00	0.00
18.8.76	76.00	0.00	0.00	0.00	0.00	0.00
18.8.77	77.00	0.00	0.00	0.00	0.00	0.00
18.8.78	78.00	0.00	0.00	0.00	0.00	0.00
18.8.79	79.00	0.00	0.00	0.00	0.00	0.00
18.8.80	80.00	0.00	0.00	0.00	0.00	0.00
18.8.81	81.00	0.00	0.00	0.00	0.00	0.00
18.8.82	82.00	0.00	0.00	0.00	0.00	0.00
18.8.83	83.00	0.00	0.00	0.00	0.00	0.00
18.8.84	84.00	0.00	0.00	0.00	0.00	0.00
18.8.85	85.00	0.00	0.00	0.00	0.00	0.00
18.8.86	86.00	0.00	0.00	0.00	0.00	0.00
18.8.87	87.00	0.00	0.00	0.00	0.00	0.00

18.8.88	88.00	0.00	0.00	0.00	0.00	0.00
18.8.89	89.00	0.00	0.00	0.00	0.00	0.00
18.8.90	90.00	0.00	0.00	0.00	0.00	0.00
18.8.91	91.00	0.00	0.00	0.00	0.00	0.00
18.8.92	92.00	0.00	0.00	0.00	0.00	0.00
18.8.93	93.00	0.00	0.00	0.00	0.00	0.00
18.8.94	94.00	0.00	0.00	0.00	0.00	0.00
18.8.95	95.00	0.00	0.00	0.00	0.00	0.00
18.8.96	96.00	0.00	0.00	0.00	0.00	0.00
18.8.97	97.00	0.00	0.00	0.00	0.00	0.00
18.8.98	98.00	0.00	0.00	0.00	0.00	0.00
18.8.99	99.00	0.00	0.00	0.00	0.00	0.00
18.8.100	100.00	0.00	0.00	0.00	0.00	0.00
18.8.101	101.00	0.00	0.00	0.00	0.00	0.00
18.8.102	102.00	0.00	0.00	0.00	0.00	0.00
18.8.103	103.00	0.00	0.00	0.00	0.00	0.00
18.8.104	104.00	0.00	0.00	0.00	0.00	0.00
18.8.105	105.00	0.00	0.00	0.00	0.00	0.00
18.8.106	106.00	0.00	0.00	0.00	0.00	0.00
18.8.107	107.00	0.00	0.00	0.00	0.00	0.00
18.8.108	108.00	0.00	0.00	0.00	0.00	0.00
18.8.109	109.00	0.00	0.00	0.00	0.00	0.00
18.8.110	110.00	0.00	0.00	0.00	0.00	0.00
18.8.111	111.00	0.00	0.00	0.00	0.00	0.00

18.8.112	112.00	0.00	0.00	0.00	0.00	0.00
18.8.113	113.00	0.00	0.00	0.00	0.00	0.00
18.8.114	114.00	0.00	0.00	0.00	0.00	0.00
18.8.115	115.00	0.00	0.00	0.00	0.00	0.00
18.8.116	116.00	0.00	0.00	0.00	0.00	0.00
18.8.117	117.00	0.00	0.00	0.00	0.00	0.00
18.8.118	118.00	0.00	0.00	0.00	0.00	0.00
18.8.119	119.00	0.00	0.00	0.00	0.00	0.00
18.8.120	120.00	0.00	0.00	0.00	0.00	0.00
18.8.121	121.00	0.00	0.00	0.00	0.00	0.00
18.8.122	122.00	0.00	0.00	0.00	0.00	0.00
18.8.123	123.00	0.00	0.00	0.00	0.00	0.00
18.8.124	124.00	0.00	0.00	0.00	0.00	0.00
18.8.125	125.00	0.00	0.00	0.00	0.00	0.00
18.8.126	126.00	0.00	0.00	0.00	0.00	0.00
18.8.127	127.00	0.00	0.00	0.00	0.00	0.00

18.12 L2-Fabric (128B read requests per normUnit)

Metric_ID	Channel	Min	Q1	Median	Q3	Max
18.12.0	0.00	0.00	0.00	0.00	1152.00	1152.00
18.12.1	1.00	0.00	0.00	0.00	1152.00	1152.00
18.12.2	2.00	0.00	0.00	0.00	1152.00	1152.00
18.12.3	3.00	0.00	0.00	0.00	1152.00	1152.00
18.12.4	4.00	0.00	0.00	0.00	1152.00	1152.00
18.12.5	5.00	0.00	0.00	0.00	1152.00	1152.00

18.12.6	6.00	0.00	0.00	0.00	1152.00	1152.00
18.12.7	7.00	0.00	0.00	0.00	1152.00	1152.00
18.12.8	8.00	0.00	0.00	0.00	1154.00	1154.00
18.12.9	9.00	0.00	0.00	0.00	1154.00	1154.00
18.12.10	10.00	0.00	0.00	0.00	1154.00	1154.00
18.12.11	11.00	0.00	0.00	0.00	1154.00	1154.00
18.12.12	12.00	0.00	0.00	0.00	1154.00	1154.00
18.12.13	13.00	0.00	0.00	0.00	1154.00	1154.00
18.12.14	14.00	0.00	0.00	0.00	1154.00	1154.00
18.12.15	15.00	0.00	0.00	0.00	1154.00	1154.00
18.12.16	16.00	0.00	0.00	0.00	1154.00	1154.00
18.12.17	17.00	0.00	0.00	0.00	1154.00	1154.00
18.12.18	18.00	0.00	0.00	0.00	1154.00	1154.00
18.12.19	19.00	0.00	0.00	0.00	1154.00	1154.00
18.12.20	20.00	0.00	0.00	0.00	1154.00	1154.00
18.12.21	21.00	0.00	0.00	0.00	1154.00	1154.00
18.12.22	22.00	0.00	0.00	0.00	1154.00	1154.00
18.12.23	23.00	0.00	0.00	0.00	1154.00	1154.00
18.12.24	24.00	0.00	0.00	0.00	1152.00	1152.00
18.12.25	25.00	0.00	0.00	0.00	1152.00	1152.00
18.12.26	26.00	0.00	0.00	0.00	1152.00	1152.00
18.12.27	27.00	0.00	0.00	0.00	1152.00	1152.00
18.12.28	28.00	0.00	0.00	0.00	1152.00	1152.00
18.12.29	29.00	0.00	0.00	0.00	1152.00	1152.00

18.12.30	30.00	0.00	0.00	0.00	1152.00	1152.00
18.12.31	31.00	0.00	0.00	0.00	1152.00	1152.00
18.12.32	32.00	0.00	0.00	0.00	1152.00	1152.00
18.12.33	33.00	0.00	0.00	0.00	1152.00	1152.00
18.12.34	34.00	0.00	0.00	0.00	1152.00	1152.00
18.12.35	35.00	0.00	0.00	0.00	1152.00	1152.00
18.12.36	36.00	0.00	0.00	0.00	1152.00	1152.00
18.12.37	37.00	0.00	0.00	0.00	1152.00	1152.00
18.12.38	38.00	0.00	0.00	0.00	1152.00	1152.00
18.12.39	39.00	0.00	0.00	0.00	1152.00	1152.00
18.12.40	40.00	0.00	0.00	0.00	1154.00	1154.00
18.12.41	41.00	0.00	0.00	0.00	1154.00	1154.00
18.12.42	42.00	0.00	0.00	0.00	1154.00	1154.00
18.12.43	43.00	0.00	0.00	0.00	1154.00	1154.00
18.12.44	44.00	0.00	0.00	0.00	1154.00	1154.00
18.12.45	45.00	0.00	0.00	0.00	1154.00	1154.00
18.12.46	46.00	0.00	0.00	0.00	1154.00	1154.00
18.12.47	47.00	0.00	0.00	0.00	1154.00	1154.00
18.12.48	48.00	0.00	0.00	0.00	1154.00	1154.00
18.12.49	49.00	0.00	0.00	0.00	1154.00	1154.00
18.12.50	50.00	0.00	0.00	0.00	1154.00	1154.00
18.12.51	51.00	0.00	0.00	0.00	1154.00	1154.00
18.12.52	52.00	0.00	0.00	0.00	1154.00	1154.00
18.12.53	53.00	0.00	0.00	0.00	1154.00	1154.00

18.12.54	54.00	0.00	0.00	0.00	1154.00	1154.00
18.12.55	55.00	0.00	0.00	0.00	1154.00	1154.00
18.12.56	56.00	0.00	0.00	0.00	1152.00	1152.00
18.12.57	57.00	0.00	0.00	0.00	1152.00	1152.00
18.12.58	58.00	0.00	0.00	0.00	1152.00	1152.00
18.12.59	59.00	0.00	0.00	0.00	1152.00	1152.00
18.12.60	60.00	0.00	0.00	0.00	1152.00	1152.00
18.12.61	61.00	0.00	0.00	0.00	1152.00	1152.00
18.12.62	62.00	0.00	0.00	0.00	1152.00	1152.00
18.12.63	63.00	0.00	0.00	0.00	1152.00	1152.00
18.12.64	64.00	0.00	0.00	0.00	1154.00	1154.00
18.12.65	65.00	0.00	0.00	0.00	1154.00	1154.00
18.12.66	66.00	0.00	0.00	0.00	1154.00	1154.00
18.12.67	67.00	0.00	0.00	0.00	1154.00	1154.00
18.12.68	68.00	0.00	0.00	0.00	1154.00	1154.00
18.12.69	69.00	0.00	0.00	0.00	1154.00	1154.00
18.12.70	70.00	0.00	0.00	0.00	1154.00	1154.00
18.12.71	71.00	0.00	0.00	0.00	1154.00	1154.00
18.12.72	72.00	0.00	0.00	0.00	1152.00	1152.00
18.12.73	73.00	0.00	0.00	0.00	1152.00	1152.00
18.12.74	74.00	0.00	0.00	0.00	1152.00	1152.00
18.12.75	75.00	0.00	0.00	0.00	1152.00	1152.00
18.12.76	76.00	0.00	0.00	0.00	1152.00	1152.00
18.12.77	77.00	0.00	0.00	0.00	1152.00	1152.00

18.12.78	78.00	0.00	0.00	0.00	1152.00	1152.00
18.12.79	79.00	0.00	0.00	0.00	1152.00	1152.00
18.12.80	80.00	0.00	0.00	0.00	1152.00	1152.00
18.12.81	81.00	0.00	0.00	0.00	1152.00	1152.00
18.12.82	82.00	0.00	0.00	0.00	1152.00	1152.00
18.12.83	83.00	0.00	0.00	0.00	1152.00	1152.00
18.12.84	84.00	0.00	0.00	0.00	1152.00	1152.00
18.12.85	85.00	0.00	0.00	0.00	1152.00	1152.00
18.12.86	86.00	0.00	0.00	0.00	1152.00	1152.00
18.12.87	87.00	0.00	0.00	0.00	1152.00	1152.00
18.12.88	88.00	0.00	0.00	0.00	1154.00	1154.00
18.12.89	89.00	0.00	0.00	0.00	1154.00	1154.00
18.12.90	90.00	0.00	0.00	0.00	1154.00	1154.00
18.12.91	91.00	0.00	0.00	0.00	1154.00	1154.00
18.12.92	92.00	0.00	0.00	0.00	1154.00	1154.00
18.12.93	93.00	0.00	0.00	0.00	1154.00	1154.00
18.12.94	94.00	0.00	0.00	0.00	1154.00	1154.00
18.12.95	95.00	0.00	0.00	0.00	1154.00	1154.00
18.12.96	96.00	0.00	0.00	0.00	1154.00	1154.00
18.12.97	97.00	0.00	0.00	0.00	1154.00	1154.00
18.12.98	98.00	0.00	0.00	0.00	1154.00	1154.00
18.12.99	99.00	0.00	0.00	0.00	1154.00	1154.00
18.12.100	100.00	0.00	0.00	0.00	1154.00	1154.00
18.12.101	101.00	0.00	0.00	0.00	1154.00	1154.00

18.12.102	102.00	0.00	0.00	0.00	1154.00	1154.00
18.12.103	103.00	0.00	0.00	0.00	1154.00	1154.00
18.12.104	104.00	0.00	0.00	0.00	1152.00	1152.00
18.12.105	105.00	0.00	0.00	0.00	1152.00	1152.00
18.12.106	106.00	0.00	0.00	0.00	1152.00	1152.00
18.12.107	107.00	0.00	0.00	0.00	1152.00	1152.00
18.12.108	108.00	0.00	0.00	0.00	1152.00	1152.00
18.12.109	109.00	0.00	0.00	0.00	1152.00	1152.00
18.12.110	110.00	0.00	0.00	0.00	1152.00	1152.00
18.12.111	111.00	0.00	0.00	0.00	1152.00	1152.00
18.12.112	112.00	0.00	0.00	0.00	1152.00	1152.00
18.12.113	113.00	0.00	0.00	0.00	1152.00	1152.00
18.12.114	114.00	0.00	0.00	0.00	1152.00	1152.00
18.12.115	115.00	0.00	0.00	0.00	1152.00	1152.00
18.12.116	116.00	0.00	0.00	0.00	1152.00	1152.00
18.12.117	117.00	0.00	0.00	0.00	1152.00	1152.00
18.12.118	118.00	0.00	0.00	0.00	1152.00	1152.00
18.12.119	119.00	0.00	0.00	0.00	1152.00	1152.00
18.12.120	120.00	0.00	0.00	0.00	1154.00	1154.00
18.12.121	121.00	0.00	0.00	0.00	1154.00	1154.00
18.12.122	122.00	0.00	0.00	0.00	1154.00	1154.00
18.12.123	123.00	0.00	0.00	0.00	1154.00	1154.00
18.12.124	124.00	0.00	0.00	0.00	1154.00	1154.00
18.12.125	125.00	0.00	0.00	0.00	1154.00	1154.00

18.12.126	126.00	0.00	0.00	0.00	1154.00	1154.00
18.12.127	127.00	0.00	0.00	0.00	1154.00	1154.00

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